FACULTY OF
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Optimization of high-resolution drone-based geophysical prospection

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That's all Folks!

Admission to loan

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Optimization of high-resolution drone-based geophysical prospection

by
Eduard Ene

Master's dissertation submitted in order to obtain the academic degree of Master of Science in Bioscience engineering

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Abstract - Drone-based magnetometry is becoming a reliable alternative to ground surveys. Drones are flexible, less costly, and enable faster coverage. However, noise originating from electronics, vibrations, and flight parameters can obscure small or low-contrast anomalies. This thesis focuses on optimizing high-resolution UAV fluxgate magnetometry by characterizing noise sources, modelling their behaviour, and developing processing methods to improve anomaly detection and visualisation. Field experiments were conducted using the Sensys R4 setup on a DJI Matrice 350 RTK quadcopter. The campaign combined flights at different altitudes, headings and velocities. Data processing steps involved track detection, sensor geolocalisation, filtering, and tieline correction. Noise was investigated by analyzing persistent harmonic frequency, transient fluctuations, and vibrational traits from the drone platform. Signal processing approaches such as Short-Time Fourier Transform, synchrosqueezing, wavelet matching, and ARMA-based modelling were employed. Results demonstrate that interference is not random but structured, with distinct frequency bands linked to propeller behaviour and electronic components. Modelling these effects could enable better noise prediction and filtering, significantly improving the Signal-To-Noise Ratio. Anomaly detection tests confirmed that ferromagnetic features can be identified reliably by adjusting flight parameters (e.g., flying slower or lower Above Ground Level) and signal compensation. Comparison to pre-existing literature confirmed the findings and effectiveness of the proposed methods.

Disclaimer: Generative AI was used in this thesis to assist with literature searches and to rephrase texts written by the author. It was not used to generate original scientific content, perform analysis, or introduce unsourced material. The author assumes full responsibility for all content presented in this thesis.

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List of Acronyms

AGL Above Ground Level

ARMA Autoregressive Moving Average

BLDC Brushless DC Motors

CSV Comma-Separated Value

ESC Electronics Speed Controller

FEM Finite Element Method

FT Fourier Transform

FWHM Full-Width at Half-Maximum

GE Geological Exploration

GMF Geomagnetic Field

GNSS Global Navigation Satellite System

GPS Global Positioning System

IMU Inertial Measurement Unit

IRM Isothermal Remanent Magnetization

LiPo Lithium Polymer

MDDT MagDrone DataTool

MRA Multi Resolution Analysis

PPS Pulse Per Second

PSS Proximal Soil Sensing

RS Remote Sensing

RPM Revolutions Per Minute

SNR Signal-To-Noise ratio

STFT Short-Time Fourier Transform

SSQ Synchrosqueezed

TRM Thermoremanent Magnetization

SQUID Superconducting Quantum Interference Device

UAV Unmanned Aerial Vehicle

UXO Unexploded Ordnance

1

Introduction

1.1 Brief history of subsurface exploration

The shallow subsurface is challenging to investigate without disturbing its natural structure. Accurate soil characterization is helpful for optimization of soil processes and agriculture [78], archaeological prospection [73, 2], ore exploration [27], and hazard monitoring [55]. Since the second half of the 20th century, researchers have made progress in remote sensing (RS) and geological exploration (GE).

RS offers broad spatial surface coverage, but is limited in depth and temporal resolution. For example, satellites allow for coarse temporal measurements (2-3 days) of dielectric properties by low-frequency signals penetrating within the upper centimeters [52]. In contrast, GE allows for the exploration of the deeper Earth's crust by using physical properties. For example, employing magnetic susceptibility analysis to locate large mineral clusters for mining [27].

Proximal soil sensing (PSS) bridges the gap between surface-level RS and deep GE through geophysical and geochemical methods. It utilizes field-based sensors to obtain signals from the soil when in contact or close to the soil [60]. This study focuses on a specific geophysical subfield of PSS, namely magnetometry.

Magnetometry efficiently characterizes soil magnetic properties using magnetometers. These devices are sensitive and capable of capturing variations in the Geomagnetic Field caused by magnetic features, [65]. Applications include detecting pollution hotspots [81], archeological structures [68], and unexploded ordnance or UXO [63]. It is a passive type of measurement with a field variation sensitivity ranging from 10^{-4} nT in Superconducting Quantum Interference Device or SQUID magnetometers, [25], to 1 nT in fluxgate magnetometers [57]. The latter excels at rapid data acquisition in field environments [3].

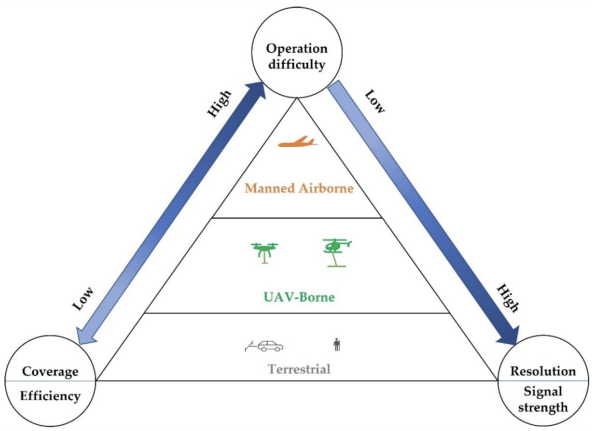

Figure 1.1: Comparison of resolution, coverage, efficiency, operation difficulty, and signal strength of manned airborne and ground magnetic surveys compared with UAV-magnetic surveys. Currently, UAV magnetometry occupies the middle ground across all previously mentioned performance metrics [83].

Traditionally, magnetometry surveys involved walking in a grid pattern with a handheld magnetometer [3, 59, 75]. More recently, hand-pulled carts replaced this survey method, which allows for different magnetometer configurations [7], fixed sensor altitude, and fixed geometry. New setups implement a Global Navigation Satellite System (GNSS) for precise positioning and attitude of the sensors [35]. GNSS also supports non-gridded survey configurations, which grant leniency, particularly useful for efficient data collection in irregular terrains or when obstacles prevent uniform line spacing [53, 38].

While ground-based methods deliver high resolution with high Signal-To-Noise ratio (SNR) data, they are laborious, require operational knowledge, and are limited in survey coverage [83]. Rugged terrains often remain inaccessible due to hazards being present or the inability to cover them on foot, such as volcanic areas [41]. The quality of the measurements depends on the expertise of the researcher. Sensor height, positioning, or environmental factors are all variables that can potentially mask small or low-contrast features, which render the data unusable [3].

Automating surveying benefits researchers greatly, as Unmanned Aerial Vehicles or UAVs can guarantee consistency. UAVs allow for consistent Above Ground Level (AGL) control and velocity tracking while utilizing high-performance magnetometers [68]. UAVs collect data more efficiently and faster [66] on large-scale coverage [80], while operational costs become cheaper [76], Figure 1.1. In addition, UAV-based magnetometry provides comparable results to ground-based surveys [28]. Despite its potential, different factors limit drone-based magnetometry, such as interference caused by the onboard electronics, calibration [74], and inconsistent flight parameters such as AGL, speed, and direction.

1.2 Research objectives

The goal of a magnetometry survey is to maximize the likelihood of detecting small, widely spaced, low-contrast features, to maximize the reliability with which the site is mapped, and to assess the degree of preservation of the buried resource [3]. Drone-induced interference can mask features, which can impact data quality. Thus, the objective of this study is to mitigate the limitations of drone-based magnetometry [68, 28]. Optimizing involves adapting a precise protocol concerning the data acquisition and processing of drone-based triaxial fluxgate magnetometry. The following research questions guide the study:

1. What are the characteristics and sources of UAV-induced noise in fluxgate magnetometry?
2. How can noise be modelled to enable its identification in future datasets?
3. How can magnetic features be accurately and reliably extracted and modelled after noise filtering?

The following chapter reviews the principles of magnetism, the geomagnetic field, and the magnetic dipole behaviour. It also provides a technical explanation about magnetometers, different possible setups, and what features this instrument can measure, followed by the possible noise sources that can be present. Lastly, the chapter will cover the different analysis tools and models to process the data.

2

Literature review

This literature review chapter outlines the scientific and technical background required to support high-resolution drone-based magnetometry. The physical principles behind magnetism are explored, followed by the origin of the Geomagnetic Field and its interaction with magnetic objects in the ground. This interaction forms the core of the signal recorded by fluxgate magnetometers. It then expands on how subsurface features affect magnetic signals based on their shape, depth, and susceptibility. The review continues by covering the different types of noise sources, primarily originating from the quadcopter and onboard electronics, that degrade the magnetic signal. Lastly, the chapter concludes with data processing techniques used to identify and model noise and anomalies.

2.1 Magnetism

Magnetic fields originate from the displacement of electric charges, such as electrons with an intrinsic magnetic moment, which are paired or unpaired. Paired spins cancel each other, while unpaired spins produce a net magnetic moment. In the presence of an external field $\vec{\mathbf{H}}$, magnetic components align accordingly, a phenomenon known as induced magnetization $\vec{\mathbf{M}}$. In most particles, including soil particles, $\vec{\mathbf{M}}$ is proportional to a continuously applied field $\vec{\mathbf{H}}$. This is described by Equation 2.1:

$$\mathbf{M} = \chi_m \cdot \mathbf{H} \tag{2.1}$$

Equation 2.1: The induced magnetic field of the material $\mathbf{M}$ equals the product of the magnetic susceptibility χ and the applied external magnetic field $\mathbf{H}$.

The proportionality factor χ refers to the nature and intensity of the response of the material towards $\vec{\mathbf{H}}$ [32]. The behaviour is categorized as dia-, para-, or ferromagnetic [32, 30]. Diamagnetic materials (e.g., copper) exhibit a negative χ and weakly oppose the applied field $\vec{\mathbf{H}}$, Equation 2.2. The induced $\vec{\mathbf{M}}$ is an order of 10^-5 weaker than ferromagnetic materials. Paramagnetic materials, align to $\vec{\mathbf{H}}$ with a weak net positive $\vec{\mathbf{M}}$. The strength is similar to that of diamagnets.

$$\mathbf{B} = \mu_0 \cdot (\mathbf{H} + \mathbf{M}) = \mu_0 \cdot (\mathbf{H} + \chi_m \cdot \mathbf{H}) \tag{2.2}$$

Equation 2.2: The total magnetic field $\mathbf{B}$ equals the sum of the induced and external magnetic field $\mathbf{M}$ and $\mathbf{H}$ multiplied by the permeability of free space μ_0 [32].

Ferromagnetic materials exhibit a stronger response due to their internal domain structure, which is multiple magnetic components aligned into regions [33]. Two physical processes align these domains to a particular orientation: Thermoremanent Magnetization (TRM) and Isothermal Remanent Magnetization (IRM). Both are essential for understanding anomaly detection.

TRM occurs when a material heats up above a specific threshold temperature, referred to as the Curie Temperature (CT). The material behaves paramagnetically and thus magnetic components orient randomly resulting in no net magnetization. As the temperature decreases, domains align to the constant $\vec{\mathbf{H}}$ and permanently become magnetized. For most materials, the CT is the order of 1000 K, e.g., iron has a CT of 1063 K. TRM can be observed in archaeological features such as clay walls [62].

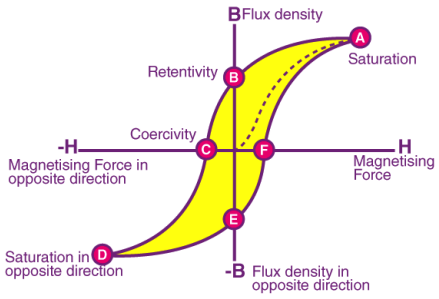

Figure 2.1: Hysteresis loop [10].

A phenomenon called hysteresis aligns domains within IRM. This involves an external field $\vec{\mathbf{H}}$ which exerts a magnetic torque $\vec{\mathbf{N}}$ equivalent to the net magnetic component $\vec{\mu} \times \vec{\mathbf{H}}$. Figure 2.1 illustrates the relationship between the total flux density $\vec{\mathbf{B}}$ and the external field $\vec{\mathbf{H}}$. An increase in $\vec{\mathbf{H}}$, shrinks oppositely oriented domains, while favourably oriented ones grow until achieving magnetic saturation. Removing the field does not return the material to its original state: domains remain aligned and create remanent magnetization. The loop also passes through coercivity points, which refer to the magnitude of the reversed field $\vec{\mathbf{H}}$ required to reduce the net magnetization $\vec{\mathbf{H}}$ to zero [32].

2.2 The Geomagnetic Field

The remanent signals are preserved after alignment with the external magnetic field $\vec{\mathbf{H}}$ called the Geomagnetic Field (GMF). The field provides the background against which magnetic anomalies appear. Understanding the origin, characteristics, and variations of the GMF is crucial for understanding anomaly signals.

2.2.1 Geodynamo theory

According to the geodynamo theory, the GMF $\vec{\mathbf{H}}$ (**B** Figure 2.2) is produced by a self-generated current **I** [51, 9, 8]. These currents, within an electrically conducting fluid in the Earth's core, induce a magnetic field. Thermal instability, caused by heat gradients, drives fluid motion, thus creating buoyancy forces. Additionally, these are also created by non-thermal convection. It contributes to fluid motion by settling of the heavier elements around the Earth's inner core, excluding lighter elements.

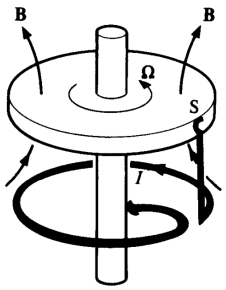

Figure 2.2: Geodynamo disk [51].

2.2.2 Inclination and declination

The Geomagnetic field is a dipolar field with magnetic flux lines emerging at the magnetic South pole and re-entering at the magnetic North pole. At any point on Earth, the GMF has a specific intensity, declination, and inclination, Figure 2.3. The intensity can be decomposed into an X-, Y-, and Z-component. The declination is defined as the angle between magnetic North and geographic North Pole. Lastly, the inclination represents the angle between the magnetic field and the Earth's surface [3, 21]. These characteristics relate to directional magnetic gradients and how anomaly signals should be interpreted.

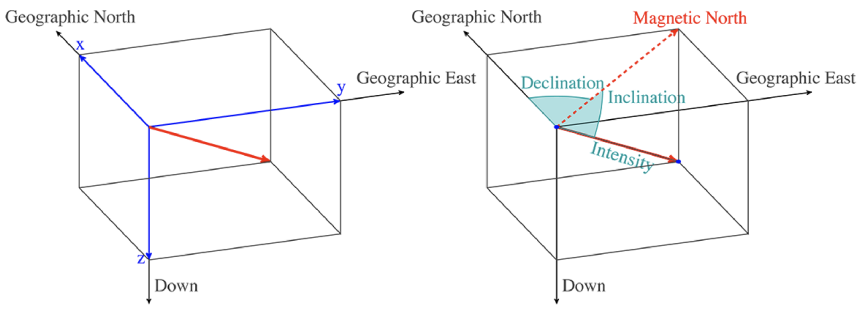

Figure 2.3: Magnetic fluxline portrayed with its characteristics: intensity, declination, and inclination [21].

2.3 Magnetic Dipole

Buried features produce magnetic anomalies with a characteristic signature determined by their susceptibility, shape, and burial depth. Anomalies arise from the interaction between the GMF and the feature, which creates magnetic poles at the boundaries of the feature. Consider a rectangular and ferromagnetic object. As the GMF intersects the object with a local inclination angle, field lines converge at one boundary, thus creating a South pole. At the other boundary, the field lines diverge, resulting in a North pole, Figure 2.4. Each pole of the object has a strength of ± p, creating a dipole moment $\vec{\mathbf{m}}$. The magnetic field for a dipole [32]:

$$\mathbf{B}_{\text{dip}}(\mathbf{r}) = \frac{\mu_0\, m}{4\pi r^3}\left(2\cos\theta\,\hat{\mathbf{r}} + \sin\theta\,\hat{\boldsymbol{\theta}}\right) \tag{2.3}$$

Equation 2.3: m is the magnetic dipole moment, r is the distance from the measurement point to the dipole center, and θ refers to the angle between the direction of r and dipole axis [32].

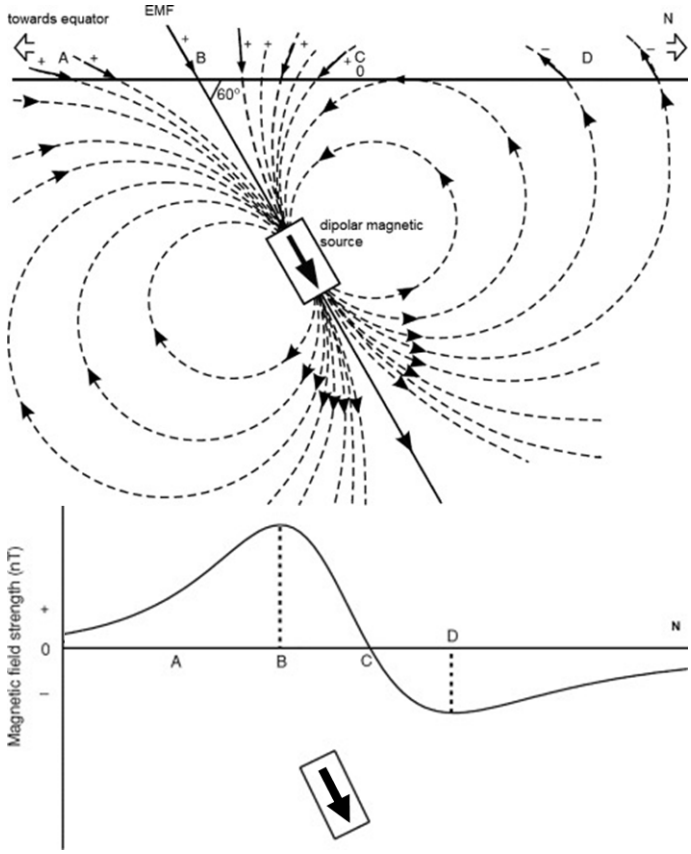

Figure 2.4: The magnetic dipole. The first figure illustrates the induced magnetic field within the anomaly by the Geomagnetic field, in this example, within the Northern Hemisphere. The second figure illustrates the measured total field by a passing sensor over the anomaly. Adapted from [22].

As the sensor approaches a feature, it will measure the total field. The magnetic total field $\vec{\mathbf{B}}_{tot}$ measured is a superposition of the GMF $\vec{\mathbf{B}}_{GMF}$ and the induced field $\vec{\mathbf{B}}_{dip}$. When a sensor approaches along the regional field, the total field increases due to the addition of the induced field to the GMF. Past the object, the total field decreases due to the induced field opposing the GMF, Figure 2.4. Decomposition of the total anomaly field shows that the horizontal $\mathbf{B}_h$-component initially experiences an increase, followed by a decrease due to the reversal of the induced field after passing the feature.

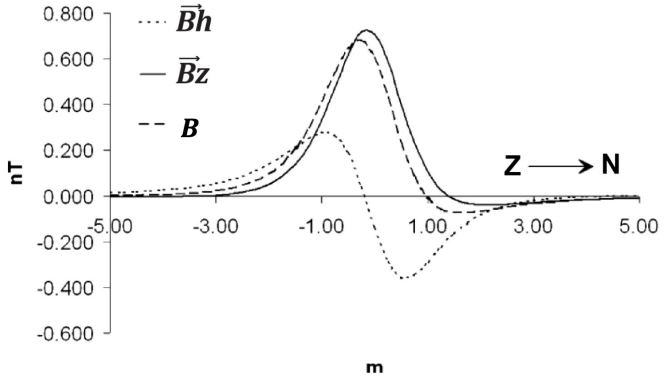

Figure 2.5: Evolution of the total field $\mathbf{B}$, and the horizontal $\mathbf{B}_h$ and vertical $\mathbf{B}_z$ components for a magnetic dipole [3].

This decline continues until the $\mathbf{B}_h$-component reaches zero, at which point the induced horizontal component completely cancels out the GMF horizontal component. As the sensor moves further, the $\mathbf{B}_h$-component reaches a trough which levels out. The $\mathbf{B}_z$-component exhibits a similar peak-trough pattern. However, peaks develop differently because with the object's inclination, one pole lies close to the surface, which creates a stronger signal than the deeper opposing pole.

2.3.1 Depth and shape

Depth

Magnetic anomalies suggest the presence of a buried object, but do not identify the exact depth on their own. In general, though, deeper buried anomalies result in a weaker and broader signal, while shallow anomalies produce narrow and intense signals. The field strength decreases proportionally with $\frac{1}{r^3}$, thus reducing the distance between the sensor and object greatly amplifies the measured field, Equation 2.3. Therefore, positioning a sensor closer to the ground significantly enhances the Signal-To-Noise Ratio.

Figure 2.6 illustrates how the anomaly broadens and weakens with increasing depth. For simply shaped anomalies (e.g., symmetric) and isolated bodies, the depth is often approximated by the Full-Width at Half-Maximum (FWHM) of the magnetic signal. The FWHM is the horizontal distance between the principal maximum of the anomaly and the point where the value is exactly one-half the maximum value [6, 3].

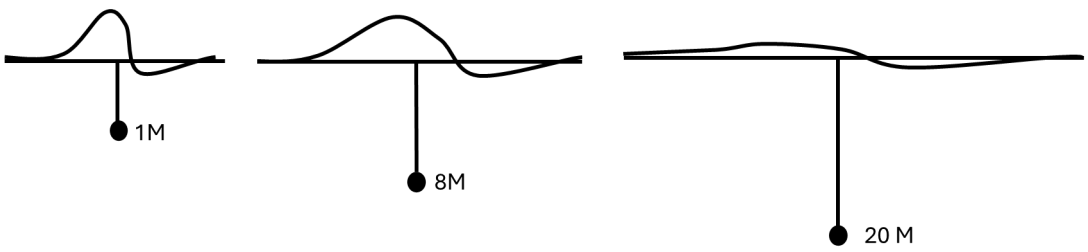

Figure 2.6: Effect of burial depth on the shape and amplitude of a magnetic anomaly. Adapted from [6].

Shape

The geometry of a buried object strongly influences the resulting magnetic response, thus determining the shape is crucial for the interpretation of the signal. In general, the response depends on the angle between the GMF and the object's axis. Depending on orientation, anomalies may exhibit a dipolar (Figure 2.4) or monopolar (Figure 2.7) response. Ferromagnetic objects with a high length-to-diameter ratio, such as iron pipes, exhibit an apparent monopole or a dipole response depending on their alignment to the GMF.

For example, a vertically oriented object often produces a monopole response, since one pole lies near the surface while the other pole lies deeper. The asymmetry arises from the stronger signal of the shallower pole, while the deeper pole decays with $\frac{1}{r^3}$. This monopole causes the total field signal to be recorded as weaker and broader. On the contrary, a horizontally oriented object results in two or more dipole signatures given that multiple edges of the structure are shallowly positioned (Figure 2.7). For elongated or compact objects, demagnetization effects can occur and minimize the strength of the anomaly field, thus complicating interpretation.

These characteristics of a magnetic anomaly are used later to identify the detection limit of the sensors and to distinguish UAV noise from anomalies.

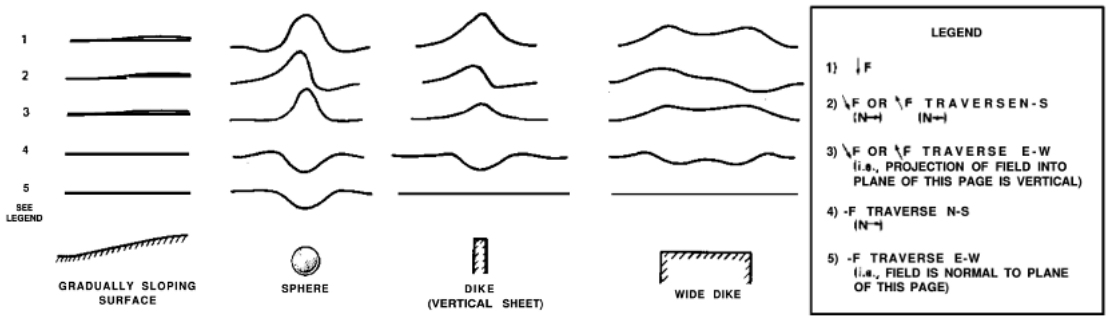

Figure 2.7: Magnetic anomaly responses originated by different geometries and orientations, with different incoming angles of the Geomagnetic field [6].

2.4 Magnetometers

2.4.1 Scalar vs vector magnetometers

Variations in the GMF are measured with magnetometers. Magnetometers can be divided into two groups: scalar and vector. Scalar magnetometers, often referred to as total-field magnetometers, measure the total intensity of a magnetic field but cannot pinpoint the orientation of the magnetic vector. In contrast to scalar magnetometers, vector magnetometers can measure the intensity in every direction. Multiple vector magnetometers record the spatial variations of the total magnetic field by decomposing it into X-, Y-, and Z-directional components [3].

2.4.2 Physical principles of magnetometers

The proprietary design of the Sensys magnetometer system is not disclosed, it likely operates on the following principle. Faraday's law describes the induced voltage within a coil by electromagnetic induction due to a varying magnetic flux ϕ [32, 30].

$$V_{ind} = \frac{d\phi}{dt} = \frac{d(N \cdot A \cdot \mu_0 \cdot \mu_r(t) \cdot B(t))}{dt} \tag{2.4}$$

Equation 2.4: The induced voltage V_{ind} is proportional to the rate of change of the magnetic flux, where N is the number of turns in the coil, A is the cross-sectional area of the core, μ_0 the permeability of free space, $\mu_r(t)$ the relative permeability of the core and $B(t)$ the magnetic field.

The magnetic flux can change due to variations in the magnetic field over time, changes in the permeability of the core material, or alterations in the effective area through which the magnetic field passes. Equation 2.5 shows the complete derivative of Equation 2.4, which is also called the general equation for induction sensors:

$$V_{ind} = N \cdot A \cdot \mu_0 \cdot \mu_r \cdot \frac{dB(t)}{dt} + N \cdot \mu_0 \cdot \mu_r(t) \cdot B(t) \cdot \frac{dA(t)}{dt} + N \cdot A \cdot \mu_0 \cdot B(t) \cdot \frac{d\mu_r(t)}{dt} \tag{2.5}$$

Equation 2.5: The induced voltage V_{ind} is proportional to the rate of change of the magnetic flux, where n represents the number of turns in the coil, and A is the cross-sectional area of the core [57, 57].

The three terms correspond to different induction sensors. The first term describes basic induction or "search" sensors, the second applies to rotating coil sensors. The last term captures sensors which detect variations of the cores' permeability in function of the spatially varying GMF, which apply to fluxgate magnetometers [57, 58].

2.4.3 Fluxgate magnetometer mechanism

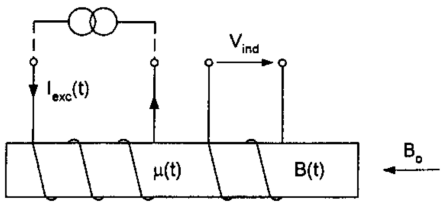

Figure 2.8: Fluxgate magnetometer schematic [58].

A ferromagnetic core is wrapped with a coil carrying an alternating current I_{exc} with a frequency f. This current produces an alternating excitation field, causing the sensor core to periodically traverse the hysteresis loop and reach both saturation points at a frequency $2f$ (Figure 2.1). The external field B_0 changes the core flux, which in turn induces a voltage V_{ind} [58]. The core is typically composed of soft magnetic material (low coercivity $< 1000\,A/m$), allowing fast demagnetization and high magnetic permeability. A standard material, Mu-metal, saturates at levels comparable to the GMF [57, 3].

2.4.4 Gradiometry

In gradiometry, the field gradient is measured with the aid of two sensors, whose respective fields are subtracted. The sensors are separated by a gradiometric baseline, which refers to the distance between the sensors. A vertical gradiometric configuration positions one sensor at a certain baseline x under the other sensor at baseline x plus a distance y. This setup helps isolate the GMF measured by the upper sensor, while the lower sensor captures variations caused by buried features.

Another option is to have a stationary base, which registers the GMF completely isolated from the survey, Figure 2.9. However, this configuration limits the ability to observe spatial variations in the GMF across the survey area [3, 58]. This setup could help eliminate the GMF from the signal to obtain a better estimate of a field anomaly.

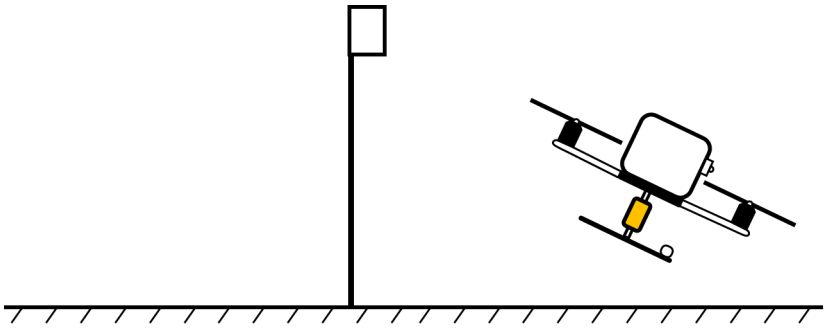

Figure 2.9: The schematic illustrates a simplistic gradiometric setup where the pole-shaped structure represents the stationary base, while the drone surveys above a field.

2.5 Soil variations

Magnetic properties in the upper 20 to 30 cm of soil are exhibited by rocks containing ferromagnetic minerals [46]. The most important and widespread minerals are Fe-oxides available in 3 forms of oxides: magnetites Fe_3O_4 ($\chi = 5 \cdot 10^7$ (-)), maghemites γ - Fe_2O_3 ($\chi = 4 \cdot 10^7$(-)) and hematites α - Fe_2O_3 ($\chi = 60$(-)) [44].

Fassbinder [24] summarized the transformation from one mineral form to another, being driven by reduction-oxidation processes, or temperature (e.g., bush fires), Table 2.1. Maghemite can originate due to oxidation of siderite $FeCO_3$ or magnetite from parent rock or sediment, lepidocrocite γ^- FeOOH dehydrating at 300 $°C$, or goethite burning in the presence of organic matter. Magnetite can be obtained due to inorganic oxidation of ferrous iron or through biologically controlled mineralization by bacteria.

α–Fe_2O_3 hematite	$\Rightarrow$	Fe_3O_4 magnetite	$\Rightarrow$	γ–Fe_2O_3 maghemite
	Reduction		Oxidation	

Table 2.1: Overview of the reduction–oxidation processes of iron oxides [24].

The presence of magnetite leads to weak magnetic properties in naturally occurring rocks and soils. When heated, such materials exhibit much higher magnetism upon cooling from a high temperature, for example, igneous rocks [6].

These soil variations contribute to the signal as background noise in magnetic surveys, potentially masking the anomalies caused by features of interest.

2.6 Drones

Drones are increasingly used for magnetic surveys, offering advantages such as high mobility, flexibility, and the ability to cover large areas quickly [66, 80]. Magnetometers can be mounted on these drones to perform magnetic surveys. This section will shortly cover the mechanisms behind a drone, specifically a quadcopter.

Drones are aircraft equipped with a control system, which includes a flight controller, an Electronics Speed Controller (ESC) for RPM regulation, and a telemetry module for recording flight data. Additionally, drones also carry an Inertial Measurement Unit (IMU) composed of accelerometers, gyroscopes, and magnetometers, to determine attitude, orientation, and movement [72, 39].

A proximity camera is also present for the integrated anti-collision system. Power is usually provided by a Lithium Polymer (LiPo) battery, composed of 2 or more voltaic cells connected in series with a typical autonomy of 15 minutes. The controller monitors battery condition and reduces RPM when the power drops below a safe threshold [39].

Quadcopters are one of the most common types of drones. These are multi-rotor helicopters with the most common configuration, the X-structure. It consists of four arms, each equipped with a propeller [39]. Two propellers rotate clockwise while the other two rotate counterclockwise, which counteract the rotating forces [31].

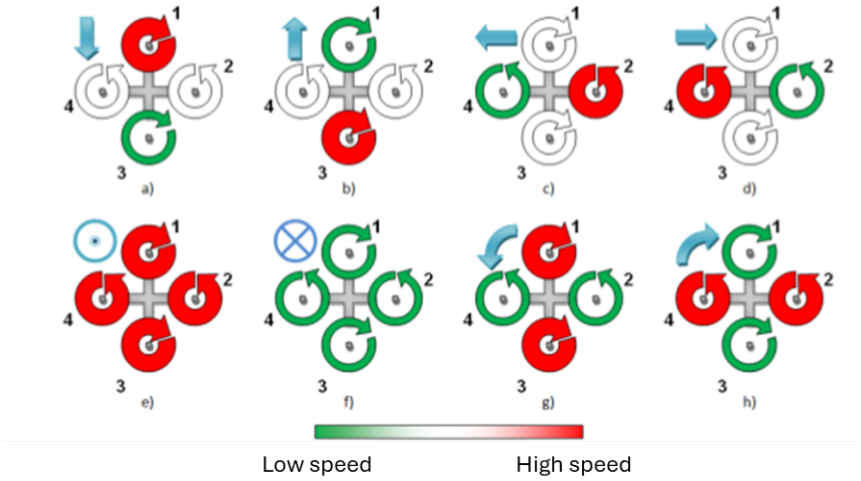

Figure 2.10: Propeller speed and direction combinations during UAV flight. Each configuration illustrates changes in lift or torque to induce different ways of movement. Green indicates decreased speed, red indicates increased speed. Adapted from [77].

Movement in the horizontal frame is achieved by tilting the platform: two propellers increase RPM while two others slow down. Vertical movement consists of increasing total thrust in all motors [31]. The orientation of a drone is defined by a specific roll, pitch, and yaw, Figure 2.11. These are controlled by adjusting RPM, which creates differential thrust.

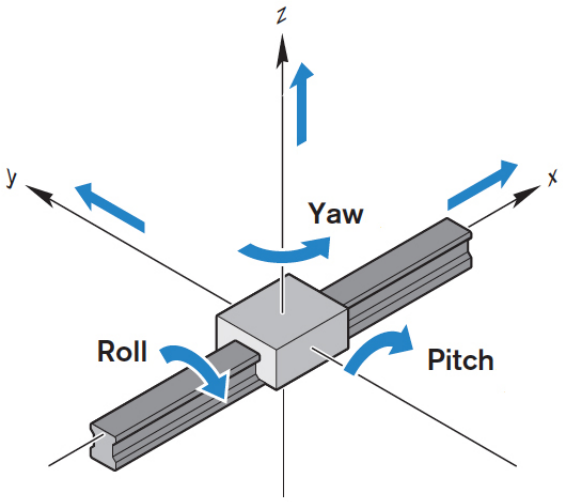

Figure 2.11: The three rotational degrees of freedom for a drone: roll (rotation around the x-axis), pitch (rotation around the y-axis), and yaw (rotation around the z-axis). Adapted from [47].

Drones are driven by either brushed DC motors or brushless DC motors (BLDC), the most commonly used in UAV applications [72, 31]. It consists of a stator (stationary part) and a rotor (rotating part), separated by an air gap. The stator contains copper windings placed in slots, which guide the magnetic flux. The rotor is composed of permanent magnets. When DC electricity flows through the stator windings, a rotating magnetic field is generated [11]. This magnetic field interacts with the rotor magnets, causing the rotor to spin, Figure 2.12.

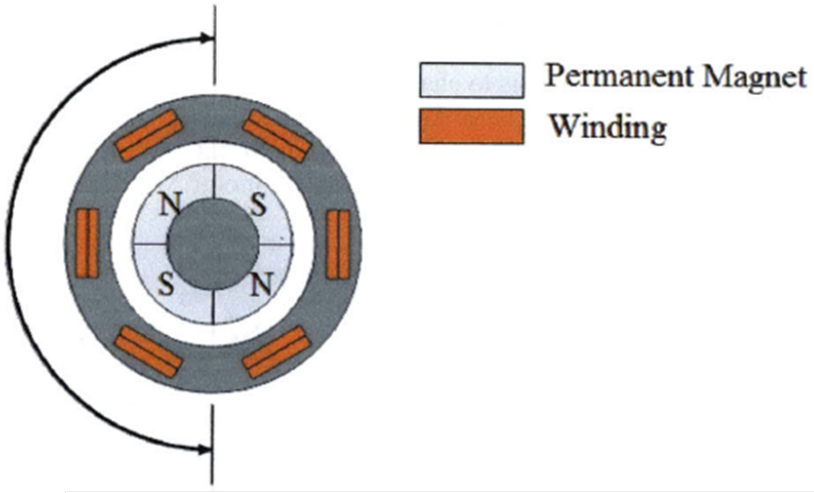

Figure 2.12: Schematic of a Brushless DC Motor (BLDC). The stator contains windings (orange) that produce a rotating magnetic field. This field interacts with the permanent magnets (white) on the rotor, generating motion. Adapted from [11].

In general, a quadcopter should provide twice the weight of the drone in thrust to maintain stability. The size of the propellers influences the amount of aerodynamic lift generated. Larger propellers generate more lift but also exert more pressure on the hub. Most frames are composed of a plastic compound, which introduces flexibility that bends the frame, thus creating vibrations within the setup. Additional payload weight can introduce vibrations by imbalancing [72].

2.7 Offset and Noise Sources

In addition to the measurement of the Geomagnetic Field (B_{GMF}) and magnetic anomalies (B_{dip}), the magnetometer signal is inevitably contaminated by various sources of noise. The overall measured the magnetic field can be expressed as the sum of contributions from signal and noise sources:

$$B_{meas} = B_{GMF} + B_{dip} + B_{env} + B_{elec} + B_{oper} + B_{sensor} + B_{offset} \tag{2.6}$$

Equation 2.6. The total measured magnetic field is the superposition of the GMF, the anomaly signal, and multiple noise components.

The sensor offset (B_{offset}) refers to the discrepancy between the sensor's output at zero input and the expected output value. This offset can originate from residual remanent magnetization, thermal drift ($0.1\ \frac{nT}{^\circ C}$ in fluxgate magnetometers), or perming, which refers to hysteresis caused by exposure to a magnetic shock. Perming can impact the sensor's ability for proper measurements, but can be solved by demagnetization with a periodic high-amplitude current [58].

Sensor noise (B_{sensor}) arises from material imperfections, such as dislocations, inclusions, and surface roughness in the ferromagnetic core. These factors cause uneven magnetization, especially in regions that are not fully saturated during each excitation cycle. Studies have shown that increasing the excitation current, as well as improving core quality through annealing or polishing, can reduce this noise [58].

Noise coming from drone electronics (B_{elec}) is typically white noise. The sources are the internal components such as brushless motors (BLDC), circuits, and batteries. Motors can generate high-frequency interference, which is beyond the Nyquist density of any magnetometer. Magnetic anomalies of up to 800 nT from engines have been recorded, and 300 nT from onboard ferromagnetic components [70].

Environmental noise (B_{env}) can originate from anthropogenic sources such as power lines (50 Hz in Belgium [45]), vehicles, and buildings. Natural sources also contribute to environmental noise. Soil variations can also cause noise, e.g., igneous rocks contain strong magnetic properties, which can generate intense background noise due to high magnetic susceptibility and remanent magnetization. Careful site selection or a gradiometry setup can reduce this noise [3].

Operator-induced noise B_{oper} typically results from deviation from standard survey protocols, such as uneven walking speed or an uneven gait [3]. However, in drones, operator noise refers to variations in above-ground level (AGL) height or vibrations originating from the drone frame (B_{vibr}).

Fluctuations in the B_{GMF} are primarily driven by thermal gradient, Coriolis forces [9, 8], and solar winds, which distort the magnetosphere substantially within seconds, minutes, or hours [6].

Other sources of noise involve magnetic cleanliness. Procedures which refer to maintaining a high signal-to-noise by keeping the sensor clean, avoiding perming, and avoiding ferromagnetic components near the setup, as it can corrupt the signal [3].

The most important sources are listed in the equation and will be discussed in more detail in the discussion.

2.8 Short Time Fourier Transform, Synchrosqueezing and ridge extraction

Time-frequency representation of the magnetometer signal allows for the separation of UAV-induced noise from the low-frequency geomagnetic signal. The Fourier Transform (FT) portrays which frequencies are present but does not show when these occur. This is due to the Heisenberg-Gabor uncertainty principle, which does not allow simultaneous time and frequency representation of a signal [5, 16]. Short Time Fourier Transform (STFT) allows time localization by partitioning the signal into overlapping, windowed segments with duration τ [17, 18, 19, 20].

$$g(t) = \begin{cases} 1, & \text{if } t \in [0, \tau] \\ 0, & \text{otherwise} \end{cases} \tag{2.7}$$

Equation 2.7 Example of a rectangular window which equals 1 on [0, τ] and 0 elsewhere.

Sliding the window along the signal creates signal segments, to which the FT is applied, thus yielding the STFT, Equation 2.8.

$$V_f^g(t, \omega) = \int_{\mathbb{R}} f(\tau) g^*(\tau - t) e^{-i\omega\tau} d\tau \tag{2.8}$$

Equation 2.8 Short Time Fourier Transform of a given signal $f \in L^1(\mathbb{R})$ with the complex conjugate of the window function $g^*(t)$ [5, 16].

A spectrogram is the squared magnitude of the STFT or the $|V_f^g(t, \omega)|^2$ of a given signal. It shows energy as a function of time and frequency. STFT is a widely used robust tool that provides consistent information, however, it has notable limitations.

Firstly, sharp window edges can introduce discontinuities. E.g., at time step τ the parsed signal is nonzero, while at time step $\tau + 1$ it is valued zero. This rapid change in values can introduce frequencies which are not necessarily present within the original signal, also called spectral leakage or artefacts [17]. Tapering the windows or overlapping segments helps mitigate the artefacts, which is the case in practice.

Secondly, the most essential parameter within the spectrogram is the window size. If τ increases, the larger the time-frame, meaning more elements are included within the FT, leading to a finer frequency resolution. The opposite holds true when τ decreases, which concludes the trade-off between time-frequency [16]. This time-frequency uncertainty appears as blurred resolution within a spectrogram, Figure 2.13. This is particularly true for non-stationary signals.

To sharpen the signal, a technique called the synchrosqueeze transform (SST) is applied to the STFT. SST reduces the uncertainty originating from first computing the phase oscillation of the signal. The reconstruction of the signal $f(t)$ equals the Inverse STFT given by Equation 2.9.

$$f(t) = \frac{1}{2\pi g(0)^*} \int_{\mathbb{R}} V_f^g(t, \omega) e^{i\omega t} d\omega \tag{2.9}$$

Equation 2.9 Inverse Short Time Fourier Transform of a given signal $f \in L^1(R)$ with the complex conjugate of the window function $g^*(t)$ [5, 16].

After reconstruction, SST sharpens the time-frequency reconstruction by localizing energy to a meaningful time-frequency location with a Dirac-delta function δ, according to 2.10. It concentrates the energy into localized ridges.

$$T_f^g(t, \omega) = \frac{1}{2\pi g(0)^*} \int_{\mathbb{R}} V_f^g(t, \nu) e^{i\omega t} \delta(\omega - \hat{\omega}(t, \nu)) d\nu \tag{2.10}$$

Equation 2.10 SST of a given signal $f \in L^1(\mathbb{R})$ with the complex conjugate of the window function $g^*(t)$ [5, 16].

This localization of the energy can be extracted by ridge extraction. A greedy algorithm extracts the most energetic frequency ridges over time. Isolation of the ridges can be improved by penalising ridge jumps with a specific penalty factor [54]. Ridge-extraction makes it possible to isolate features of interest, such as propeller frequency fluctuations, which in turn can be linked to variables such as velocity or internal drone voltages.

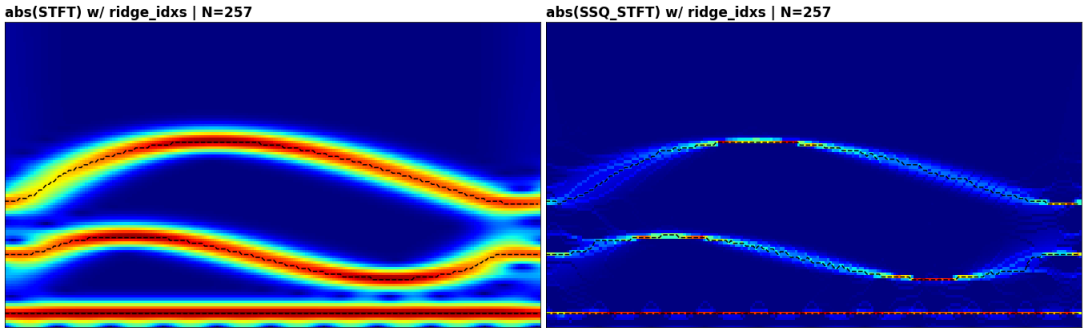

Figure 2.13: Left: Short Time Fourier Transform (STFT) with ridges plotted over the most energetic frequencies. Right: Synchrosqueezed STFT with ridges plotted over the same energetic frequencies showing clearer separation of frequency components [54].

2.9 Modelling

2.9.1 Multiplicative and Auto-Regressive Moving Average model

Onboard electronics generate periodic magnetic interference, each component exhibiting distinct characteristics. Electronics can be assumed to be (quasi)-sinusoidal oscillators, producing sharp ridges in the time-frequency representation. Ideally, the signals can be represented by a multiplicative model whose spectrum consists of impulses at $\pm\omega_0$, assuming random fluctuations do not occur, Equation 2.11.

$$V_{out} = A \cdot \cos(2\pi f_0 \cdot t + \phi) \tag{2.11}$$

Equation 2.11 represents the multiplicative model for oscillators without random fluctuations, V_{out} is the output voltage, A the amplitude and f the frequency and ϕ the phase-shift of the noise source [34].

However, in practice, oscillators portray time-dependency and thus fluctuate [34], Equation 2.12. For example, internal voltages of the drone can fluctuate due to battery charge, temperature, or payload.

$$V_{out} = A(t) \cdot f(2\pi f_0 \cdot t + \phi(t)) \tag{2.12}$$

Equation 2.12 extends the ideal model by incorporating time-varying $A(t)$ and phase $\phi(t)$, thus capturing random fluctuations [34].

When telemetric data is unavailable (e.g., internal voltages or RPM of the propellers), the noise signals can be modeled with an Auto-Regressive Moving Average model (ARMA). ARMA models excel at extracting time-dependent characteristics from a given time series [4]. They consist of 2 components, an autoregressive (AR) and a Moving Average (MA) component, Equation 2.13.

$$ARMA(p,q) : x_t = \varepsilon_t + \sum_{i=1}^{p} \Phi_i x_{t-i} + \sum_{i=0}^{q} \varphi_i \varepsilon_{t-i} \tag{2.13}$$

Equation 2.13: Autoregressive Moving Average (ARMA) model combining AR(p) and MA(q) components.

The AR component models the current value x_t at time stamp t as a weighted sum of its previous values, with weights Φ_t, Equation 2.14.

$$AR(p) : x_t = c + \sum_{i=1}^{p} \Phi_i x_{t-i} + \varepsilon_t \tag{2.14}$$

Equation 2.14: Autoregressive (AR) model of order p, where x_t is the current value, x_{t-i} are previous values weighted by coefficients Φ_i, c is a constant, and ε_t is white noise.

The MA component incorporates the influence of past errors ε_t, weighted by a ϕ_t Equation 2.15. These errors are typically assumed to be white Gaussian noise, which is applicable for electronics noise.

$$MA(q) : x_t = \mu + \varepsilon_t + \sum_{i=0}^{q} \varphi_i \varepsilon_{t-i} \tag{2.15}$$

Equation 2.15: Moving Average (MA) model of order q, where x_t is expressed as the mean μ (usually assumed to be 0) plus the current and past white noise terms ε_{t-i} weighted by φ_i.

2.9.2 Natural Neighbour Bathymetry

Despite high sampling frequencies, magnetometers cannot provide continuous coverage across the entire survey area. Data is acquired at discrete locations based on the setup configurations, such as even spacing between sensors. This results in gaps that require interpolation between the samples. Natural Neighbor Bathymetry (NNI) uses area-based weights from Voronoi tessellation surrounding each query point. NNI produces smooth and visually coherent surfaces while preserving the original data values at sample points. Additionally, it is sufficiently fast for geospatial data [49].

$$\hat{z} = \frac{1}{\sum_{i=1}^{n} CapturedArea_i} \sum_{k=1}^{n} CapturedArea_i \cdot z_k \tag{2.16}$$

Equation 2.16: Natural Neighbour Bathymetry (NNI). The interpolated value $\hat{z}$ is calculated as a weighted average of neighbouring values z_k. The weights are proportional to the area of the new Voronoi cell that overlaps with each existing cell when inserting the query point.

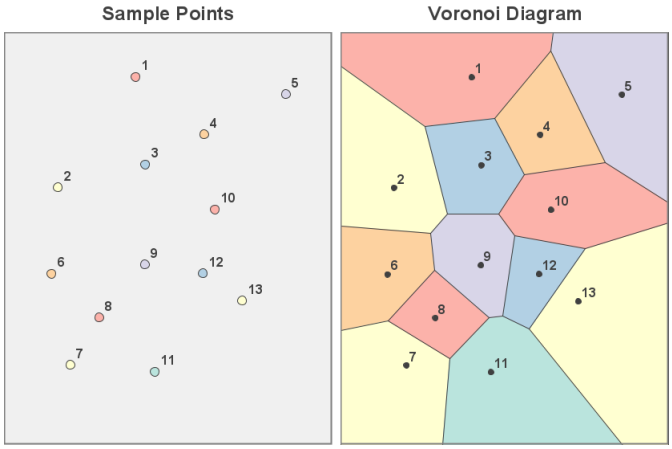

Figure 2.14: Illustration of Voronoi tessellation. Left: Original sample points. Right: Corresponding Voronoi diagram, where each cell contains the region closest to a given sample. Adapted from [49].

3

Materials and Methods

Field experiments were performed on January 21, 2025, between 11:00 and 14:00 at the field site in Vijverhoek, Bottelare, see Figure 3.1. The following section describes the environmental and geological context of the testing field.

3.1 Testing Field

The site provided an open, unobstructed area with minimal magnetic environmental interferences. Weather conditions at the start of the testing included mist and low temperatures, but improved gradually. These conditions ensured stable flying conditions and guaranteed consistent drone stability without the need for roll readjustments, which windy conditions would have required.

The diamagnetic quartz, feldspar, and mica within the sandy loam of the topsoil created a relatively weak magnetic signal. As a consequence, anomalies would only occur if the soil contained ferromagnetic or significantly paramagnetic materials.

Although the field's agricultural use allowed for the possibility of anthropogenic anomalies, the only ferromagnetic materials present were two spherical anomalies planted at a depth 0.5 m. The coordinates are (50.95866029°, 3.76733557°) and (50.95865358°, 3.76734453°). Overall, the site and weather conditions proved to be suitable for conducting analyses aimed at distinguishing noise from features and soil variations.

According to the Geological 3-D model obtained from Databank Ondergrond Vlaanderen (DOV), the first 10.7 m belong to the Formation of Ghent, formed during the Holocene up to the Pleistocene. The investigated field (green square Figure 3.1) consists entirely of the Ldp soil type.

Ldp is the sole soil type present in the area according to the model, which indicates a moderately wet sandy loam soil without profile development [15]. The model does contain inaccuracies, as in the southeast part of the field, the soil contains a clay part of the Gentbrugge Formation at shallow depth [79]. Overall, the area is diamagnetic mainly and spans approximately 1950 m^2.

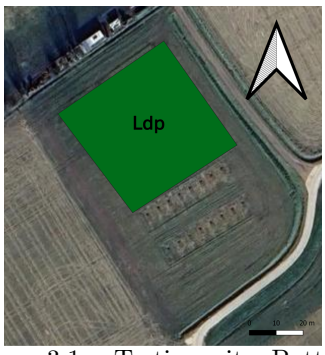

Figure 3.1: Testing site Bottelare (50.957954° N, 3.767978° E) [15].

3.2 Setup

3.2.1 UAV Configuration

The setup utilizes a DJI Matrice 350 RTK with a cylindrical frame attached to the legs, which houses the Sensys FGM3D magnetometers (Figure 3.2). When unfolded, the drone has dimensions of 810x670x430 mm (LxWxH) with an estimated payload of approximately 6.5 kg during the performed experiments. It is a quadcopter with four propellers along its diagonal axes, each with a blade diameter of approximately 50 cm. Figure A.1 displays the distances between each propeller and sensor.

The survey routes were preprogrammed in the UgCS application and transmitted to the Skyhub mounted on the drone. Preprogramming enabled faster and more efficient surveying by dividing a larger field into multiple flight plans. This approach was crucial because the drone required battery swaps every 20-30 minutes of actual flight time, a duration that may shorten under harsh weather conditions. The altimeter, mounted beneath the Skyhub, maintained a constant Above-Ground Level (AGL) during flight. Additional procedural details are available on the website of SPH engineering [67].

Figure 3.2: The DJI Matrice 350 RTK with the magnetometers within the cylindrical frame. Sensor 1 is positioned to the outer right, while sensor 5 is to the outer left. The Skyhub and the altimeter are both mounted on the drone's right leg.

3.2.2 Sensys FGM3D

Figure 3.3: FGM3D from Sensys [64].

The Sensys FGM3D/75 is a triaxial fluxgate magnetometer capable of measuring the X-, Y-, and Z-components of the magnetic field vector. The Magdrone R4 comprises five such sensors, each spaced 0.5 m apart, forming a total frame length of 2 m. The triaxial fluxgate measures the X-component in the flight direction, the Y-component horizontally perpendicular to the left, and the Z-component vertically perpendicular to the ground. Each magnetometer operates over a range of $\pm$ 75000 nT with a resolution of less than 150 pT. The manufacturer specifies a thermal drift of $\leq \pm$ 0.3 nT/K, a zero error or offset of $\leq \pm$ 5 nT, and a relative error of measurement around $\pm$ 0.1 %. Data sampling of all sensors occurs at a frequency of 200 Hz [64]. No magnetic calibration preceded the flight.

3.3 Experiments

Twelve surveys investigated the influence of flight heading, altitude, and propeller activity on signal-to-noise ratio, signal quality, and noise characteristics. All surveys flew at a yaw angle of 323 °, except survey 4, which flew at an angle of 233 °. An overview of the surveys is provided in Table 3.1. Each survey serves a specific purpose:

Surveys 1, 2, and 3 examined the effects of flight direction and Above Ground Level (AGL) on the signal. Survey 2 was the baseline survey flown with a constant heading and 0.87 m AGL. Survey 1 tested a higher AGL at 1.17 m with a constant heading, while survey 3 tested a variable heading at 0.87 m AGL.

Note: the AGL refers to the onboard altimeter, which was 17 cm mounted above the magnetometer housing. The magnetometer height above ground was at 1 m instead of 1.17 m during survey 1, at 0.7 m instead of 0.87 m for the rest of the surveys. For consistency, the altimeter AGL will always be mentioned.

Survey 4 investigated the tieline correction by only applying it to survey 2. The tieline correction was not applied to survey 1 due to its higher AGL and unknown pitch angle during the flight.

Surveys 5 and 6 investigated geostationary noise sources. Survey 5 recorded the frequency bands during propeller activity, while survey 6 recorded with propellers inactive. Comparison between these datasets enabled distinction of noise originating from the internal electronics versus the motors and propellers.

Survey 7 assessed whether the findings of surveys 5 and 6 persisted in a higher altitude environment where the sensors would primarily measure the Geomagnetic Field. It also compares the stationary and the non-stationary behaviour of these frequency bands.

Survey 8-12 investigated the influence of the velocity on the frequency bands. Velocities were increased with increments of 0.5 m/s, from 1 m/s to 3.0 m/s. The drone flew along the same straight line with constant heading during each survey.

Table 3.1: Overview of survey configurations.

Survey #	**Above Ground Level [m]**	**Heading Style**	**Propeller Activity**	**Velocity [m/s]**	**Description**
1	1.17	Constant	Yes	1.5	Height variation experiment
2	0.87	Constant	Yes	1.5	Baseline experiment
3	0.87	Variable	Yes	1.5	Heading variation experiment
4	0.87	Constant	Yes	1.5	Tieline dataset
5	0.87	Constant	Yes	0.0	Propellers-on dataset (stationary)
6	0.87	Constant	No	0.0	Propeller-off dataset (stationary)
7	30.00	Constant	Yes	1.5	Altitude test
8	0.87	Constant	Yes	0.5	Velocity test
9	0.87	Constant	Yes	1.0	Velocity test
10	0.87	Constant	Yes	2.0	Velocity test
11	0.87	Constant	Yes	2.5	Velocity test
12	0.87	Constant	Yes	3.0	Velocity test

3.4 Data processing and preparation

Although the data processing and preparation steps described in this section do not directly address the research questions mentioned in the introduction, they remain essential for the subsequent noise analyses. Due to the lack of documentation of Sensys' MagDrone DataTool (MDDT) software, this section aims to provide insight into the processing steps. The goal is to maintain scientific transparency and reproducibility.

Flights are encoded in the PMXLF format, which can only be accessed using the MDDT software. It offers a limited toolset to process data: a track detection and selection algorithm, frequency filtering (low-, high-, band- and notch-filters), smoothing filters and moving mean filters, and correction of spatial and temporal offsets. The tool provides a quick, reliable way to apply filters and visualize modifications on the collected data.

However, performance issues are present during the application of these tools and the data export. Modifying parameters in the tool requires a few seconds to process, while exporting can take up to a minute, depending on the dataset size. The toolbox remains under development and requires further optimization. It was programmed in Python and is available on GitHub [23].

3.4.1 Pmxlf translation

The files are encrypted by Sensys within the PMXLF format, meaning that a mandatory import and export within the MDDT is required. The export from the PMXLF format files to unfiltered comma-separated (CSV) files require only a few seconds and generate a total of 5 files per survey in the NMEA-0183 standard [43]:

1. A Pulse Per Second (PPS) file, commonly used for time synchronization of the acquired data by providing a high-precision pulse at 1 Hz frequency. The first recorded pulse by the system defines the configured start time.
2. The configuration file provides the X-, Y-, and Z-offset of each sensor relative to the centrally positioned third sensor within the setup. For this case study, the X- and Z-offsets are equal to zero, while the Y-offsets range from -1 to 1 m with a 0.5 m interval between each sensor. Sensor 1 sits at 1 m, while sensor 5 at 1 m offset.
3. A magdata file which contains the X-, Y-, and Z-values of each magnetometer and the timestamp of the signal recording, the sampling occurs at a 200 Hz frequency. Table 3.3 shows how the data format for a given timestamp and specific sensor. The same column format is present for all sensors 1 to 5.
4. A ZDA file which contains the timestamp [μs] of the signal recording and the GPS UTC time [ms].
5. A GGA file which includes the UTC and coordinates of the GPS, recorded at a frequency of approximately 5 Hz. Table 3.2 provides the GGA data format. The file only contains the coordinates of the GPS receiver. The coordinates are attributed to sensor 3, which is positioned centrally beneath the GPS receiver.

Table 3.2: Example of the GGA data for a single datapoint.

Timestamp [µs]	**GPS Time [ms]**	**Latitude [°]**	**Longitude [°]**	**Mean Sea Level [m]**
1122080706	38997987	50.95888779	3.76730828	-7.80

Table 3.3: Example of the magdata from an arbitrary sensor for a single datapoint. The '.' is a placeholder for sensors (1-5).

Timestamp [µs]	$\mathbf{B}_{.,x}$ **[nT]**	$\mathbf{B}_{.,y}$ **[nT]**	$\mathbf{B}_{.,z}$ **[nT]**
1122804722	24467.31	10917.06	41926.86

3.4.2 Track detection

The track detection is based on the methodology of Dalitz et al. [14]. The algorithm allows detection and separation of curves in 3D point clouds without making a priori assumptions about their parametric shape. Its use is well-suited for this study, given that the curves represent flight tracks and the sequential characteristic of the data. Each data point location is dependent on the previous data point position.

The original GGA data, recorded as degree-based coordinates, requires a conversion to longitudinal and latitudinal meter coordinates. The coordinates were converted from EPSG:4326 to EPSG:32631 using the Transformer function within the pyproj package. After conversion, the track detection function will iterate through the whole dataset and apply the following steps:

1. The coordinates are smoothed by applying a centered rolling mean window to the latitude $[m]$ and longitude $[m]$ coordinates. Perpendicular point spread can result in adjacent triplets being mislabelled as rogue datapoints, while in reality belonging to the same track. The original coordinates remain.

2. Primary detection scan:

   (a) The smoothing window is selected based on the minimal triplet length (MTL) provided by the user. MTL is element-based, which computes faster than distance-based tracking. The MTL essentially determines how long the minimal track is, which is required to detect a track. In Figure 3.4, the window size is equal to twice the MTL. While the choice is arbitrary, the window should preferably correspond to a certain distance threshold r_{smooth} between the points.

   (b) The algorithm iterates over triplets starting from i = 1 in the dataset up to the n^{th}-MTL datapoint: point A corresponds to i = 1, midpoint B to i + $\frac{MTL}{2}$ and point C is located at i + MTL.

   (c) Compute the segment vectors $\vec{AB}$, $\vec{BC}$.

   (d) A triplet cluster belongs together if it does not exceed a certain threshold α. The angle is computed via the dot product of $\cos\alpha = \frac{\langle \vec{AB}, \vec{BC} \rangle}{||\vec{AB}|| \cdot ||\vec{BC}||}$. A threshold α equaling 14.07 °is sufficient. Triplets exceeding α are labelled as "off_{track}", otherwise "on_{track}".

3. Secondary detection scan:

   (a) Surveys proceed along two dominant directions: θ and $\theta + 180$ °. The angle is user-defined.

   (b) The angle at which the drone flies γ is determined by calculating the direction between two subsequent points (at timestamp t and timestamp $t + 1$).

   (c) On-track datapoints may deviate slightly from the main directions, whereas the off-track datapoints will deviate substantially. Therefore, the introduction of tolerance angle β allows for a certain deviation within the proximity of the principal direction of the drone, Figure 3.5.

4. As a final step, the correct triplets are labelled as "on_{track}" or "off_{track}" only if they comply with the previous conditions.

3.4.3 Geolocalisation of the sensors

The GGA file provides only the coordinates of the GPS tracker. Thus, the positions of the remaining individual sensors require manual calculation. This calculation introduces a potential source of error that likely also affects the MDDT software. The cylindrical frame housing the magnetometers can be positioned arbitrarily within the UAV chassis. As a result, an initial horizontal offset x between the GPS receiver (Skyhub) and the frame may be present at each setup. Ideally, this offset is measured before surveying.

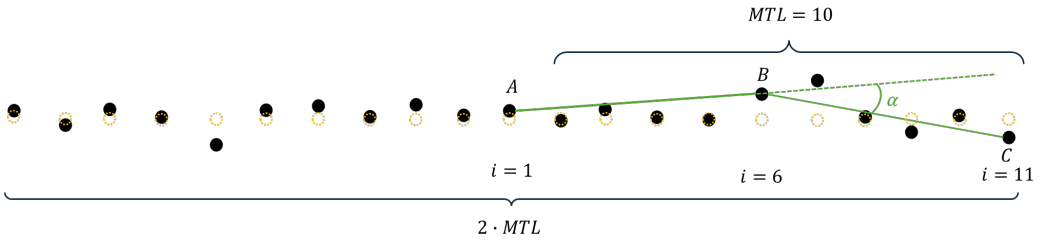

Figure 3.4: Primary detection: triplet selection and angle thresholding. Black dots represent the unfiltered data, orange the smoothed coordinates, α the triplet angle threshold, and MTL, the minimal triplet length.

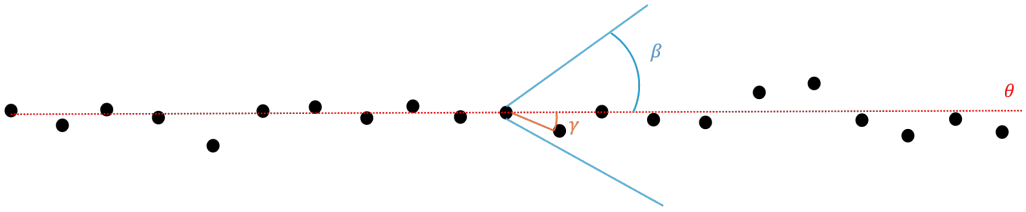

Figure 3.5: Secondary detection: principal direction and deviation tolerance. γ represents the direction of the drone movement, θ represents the direction angle at which the survey takes place and β is the maximal deviation allowed from the angle θ.

During flight, the drone has a specific pitch (inclination angle), which can shift the frame's position relative to the GPS. The frame might align itself with the GPS tracker when flying forward or fall behind it when flying backward. Thus, without knowing the pitch, the exact sensor positions cannot be determined, as shown in Figure 3.6. Each uncharacterized offset allows for errors to propagate into the georeferencing of the signal, eventually distorting the anomaly signal shape.

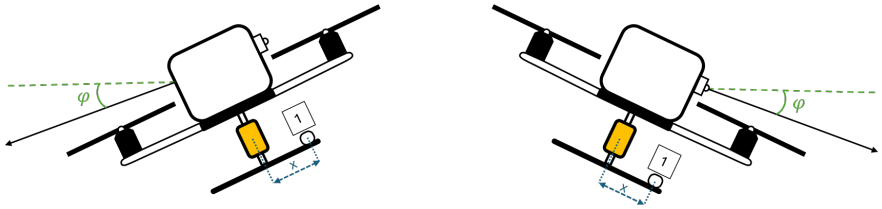

Figure 3.6: Right side of the drone portrayed while flying backwards (left) and forwards (right) at a certain pitch ϕ. The orange box represents the GPS receiver, which has a certain offset x from sensor 1.

The calculation of the individual sensor coordinates happens based on the previously calculated γ. This angle, combined with the offset available in the configuration file, is used to adjust the GPS coordinates according to Equation 3.1. The equation is generalized and not completely accurate when the drone flies to the left or right. Complete logic statements are available in the GitHub code [23]. Additionally, it does not include the pitch of the drone, as it was unknown during the time of the thesis development. The pitch is assumed to be zero. Both the GPS offset and pitch uncertainty induce a systematic error in the final georeferencing of the data.

$$\begin{aligned} x_{sensor} &= x_{GPS} + x_{offset} \cdot \cos(\gamma) \\ y_{sensor} &= y_{GPS} + y_{offset} \cdot \sin(\gamma) \end{aligned} \tag{3.1}$$

Equation 3.1: GPS coordinate offset adjustment formula where (x_{sensor}, y_{sensor}) represent the adjusted sensor coordinates, (x_{GPS}, y_{GPS}) the GPS tracker coordinates γ the angle between subsequent datapoints.

3.4.4 Output file

The import function incorporates these algorithms and extracts the correct survey track from each flight, storing the results in a column $trajectory_{status}$. It requires the specific flight number, the deviation angle β, the MTL, and the survey angle θ. Additionally, the different sampling frequencies of PPS and GGA files require linear interpolation to match the sampling rate of the magdata file. Linear interpolation was chosen because the time variable is continuous and the survey datasets exhibit linear spatial progression.

3.5 Noise characterization

This section addresses Research Question 1: Identifying the characteristics and sources of UAV-induced noise during surveys.

3.5.1 Persistent frequency bands analysis

The Geomagnetic field (GMF) is known to be highly variable and unpredictable, indicating that measurements at a specific timestamp t may differ significantly from those at timestamp $t + \Delta t$. The use of a fifth-order high-pass filter with a cutoff frequency of 2 Hz minimizes the influence of GMF variations while characterizing the noise from various sources.

Most GMF variations occur in the frequency range of 0.001 to 1 Hz [50]. Based on this, frequencies below 2 Hz are primarily attributed to natural GMF variability, with minimal contribution from other noise sources (e.g., sensor white noise).

The stationary dataset (Survey 6) provides substantial information concerning the electronic and environmental noise. Each of the five sensors continuously measured the magnetic field at its fixed location. Any variations would be attributed to environmental, electronic noise, or GMF variability.

The initial analysis involved plotting the time series of the X-, Y-, and Z- components, as well as the total magnetic field, for each sensor. The Euclidean norm of the three components computes the total field. This analysis helped assess the presence of systematic offsets between sensors. The offsets were estimated by computing the difference between each sensor pair at every timestamp and averaging it over the whole dataset.

Spectrograms were generated using window sizes varying from 512 to 4096 samples, with no overlap and a Nyquist density of 100 Hz. These spectrograms were examined for the presence of persistent high-frequency bands.

To isolate and quantify this high-frequency noise, a fifth-order high-pass filter was applied to characterize the strength of the noise. Additionally, the signal's high-frequency variance within aeromagnetic data can be examined by using the 4th difference [13, 36, 37]. It is computed by taking four samples adjacent to a selected sample ΔT_0, Equation 3.2:

$$\Delta = \frac{\Delta T_{-2} - 4\Delta T_{-1} + 6\Delta T_0 - 4\Delta T_1 + \Delta T_2}{16} \tag{3.2}$$

Equation 3.2: Fourth difference.

3.5.2 Harmonics

Narrow-bandpass filters were designed to isolate individual frequency bands of interest. The passband for each filter was $[f - 0.5, f + 0.5]$ Hz with a Kaiser window, Figure 3.7, which minimizes frequency leakage [71]. The stopband attenuation is –60 dB and the transition bandwidth is 0.4 Hz.

After signal extraction, a Hilbert transform was subsequently applied to extract the instantaneous frequency and envelope of the filtered signal. The instantaneous frequency was examined to assess whether the identified components represented real signals or spectrogram artefacts. The amplitudes from the Hilbert transform were then plotted as a histogram and fit to a Gaussian distribution truncated at 0 nT.

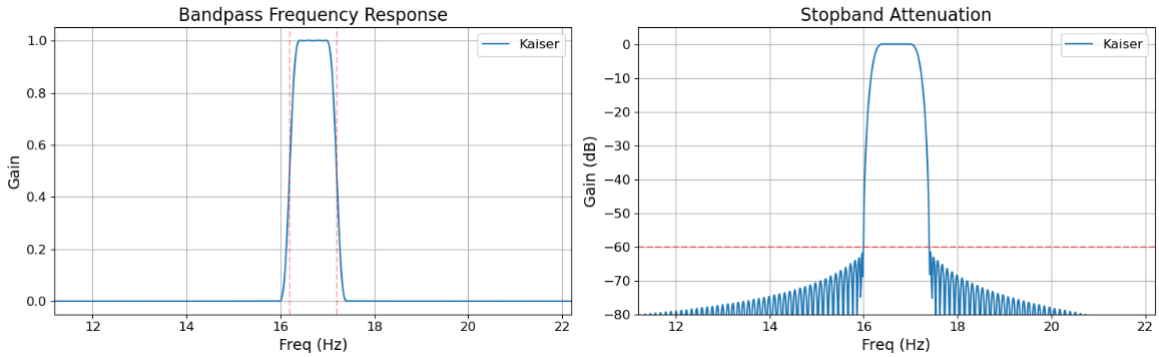

Figure 3.7: Kaiser window function used in bandpass filtering. Left: The intended bandpass response of the allowed frequencies. Right: The intended stopband attenuation portrays the suppression of frequencies outside of the window limits. Frequency 16.7 Hz was chosen as an example.

3.5.3 Transient frequency bands analysis

Spectrogram analysis was performed on the surveys with varying velocities (Surveys 8-12) to identify fluctuating frequency bands within the 30-60 Hz range. A window size of 4096 samples was used to increase frequency resolution, as preliminary analysis indicated potential overlap between bands at certain times.

Comparison of surveys 5 and 6 was performed to determine whether these bands were associated with propeller noise. Every velocity dataset underwent a bandpass filtering procedure with a passband of $[30, 60]$ Hz with a fifth-order Butterworth filter. This was done to isolate the frequency bands of interest.

Synchrosqueezing with no window overlap, [54], was applied to the spectrogram of the isolated signal of every velocity dataset. Lastly, a ridge extraction was performed targeting four ridges, with a binwidth of 5 and a penalty score of 40. The first and last seconds of the results were removed to avoid edge effects. Results from all velocity datasets were concatenated. The frequency as well as the magnitude of the spectrogram were plotted against velocity for each ridge and sensor.

3.6 Signal modelling

This brief section addresses Research Question 2: The goal is to acknowledge the possibility of extracting specific parameters from the data to create a model that essentially replicates the original noise signal.

An Autoregressive Moving Average (ARMA) model was trained on the envelope, extracted with the Hilbert-transform, from the dataset of Survey 6, using only two auto-regressive coefficients to capture the temporal dependency of the analysed frequency band. This model was then used in the multiplicative model for the specific frequency 16.7 Hz, Equation 2.13.

It predicted the noise signal from Survey 5, which was chosen because it contained only data originating from the drone itself, without soil or spatial GMF variations. As such, the survey is suitable for the evaluation of the consistency of the persistent frequency bands across multiple datasets. Acknowledging every noise source, however small, can aid in differentiation between soil variations or a deeply buried anomaly.

3.7 Feature extraction

This section addresses Research Question 3: The extraction of features from data and corresponding anomaly detection and signal modelling of the whole field.

3.7.1 Signal-To-Noise Ratio

The coordinates of the planted anomaly objects were used to determine which sensor was closest to the object at a particular time. Based on this timestamp, the corresponding data from all five sensors was extracted. For the analysis of the Signal-to-Noise Ratio (SNR), two cases were considered. In the first case, unfiltered data was analyzed to observe the sensor's response to the anomaly. The second case utilizes a 2 Hz low-pass filter on the dataset. A window size of 5 seconds before and after the timestamp was used to extract the data.

Visual analysis was used to estimate the anomaly's duration, which in this case study spanned approximately 700 samples or 3.5 seconds. The anomaly was excluded from this extracted signal. The variance of the remaining signal was calculated, which relates to the background noise variance σ_{bg}. The anomaly peak-to-peak difference was calculated and divided by the σ_{bg}, for both cases, Equation 3.3. This allowed evaluation of the effects of the noise on masking the signal.

$$SNR = \frac{Anomaly_{max} - Anomaly_{min}}{\sigma_{bg}} \tag{3.3}$$

Equation 3.3: Signal-to-Noise Ratio.

Each sensor's signal was also normalized to its respective maximum and minimum value. Regardless of the sensor's spatial offset to the anomaly, by normalizing all signals, they fall in the range from 0 to 1. This can be used to assess whether deviations in amplitude or time alignment could show inherent sensor noise. Significant deviation of one sensor from the others could indicate a defective sensor.

3.7.2 Wavelet detection

Wavelets are short localized functions whose shape and scale can adapt to the signal. Wavelets can be modulated, meaning they can dilate and shift in the time-frequency, thus enabling multiresolution analysis (MRA). MRA entails the approximation of a signal by the superposition of multiple wavelets at different scales.

High-frequency, small-scale wavelets capture fine details, while low-frequency, large-scale wavelets capture large details. Summing the finer and coarser components reconstructs the original signal [16].

For this application, the level 3 and 7 sym7 wavelet from the package pywavelets was chosen, as shown in Figure 3.8. The sym7 is a symmetric wavelet which resembles the magnetic anomaly shape: a negative peak followed by a positive peak, with a zero crossing in between, which is typical for anomalies.

A simple matched-filter procedure was used: sliding the wavelet along each timestep of the timeseries, computing the product by multiplying the wavelet with the data, and flagging locations where the response peaks. A large product response indicates a match and vice versa. This section serves as a proof of concept for automatic detection of anomalies.

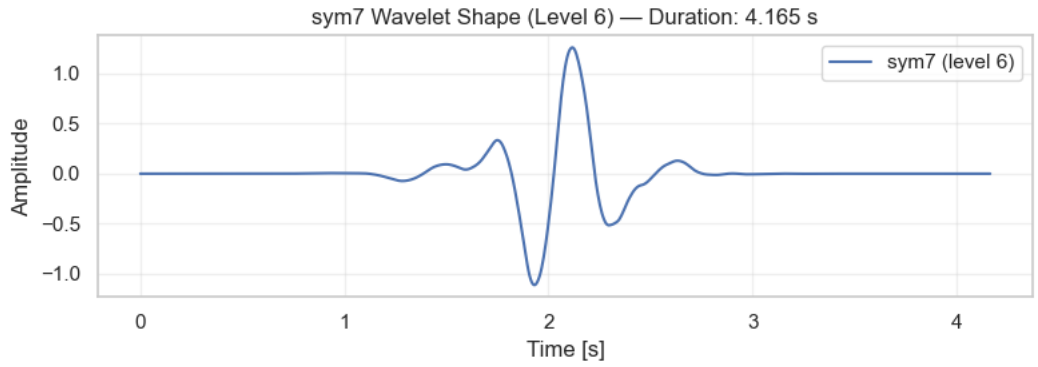

Figure 3.8: Example of a sym7 mother wavelet used for anomaly detection. It is symmetric and resembles the dipole-like shape of a magnetic anomaly.

3.8 Tieline correction

Tieline corrections applied to magnetic data serve the purpose of removing diurnal and heading effects [13]. The tieline correction was performed using overlapping datapoints between Survey 4 (tieline) and Survey 2 (constant heading, 0.87 m) as illustrated in Appendix A.2. The overlapping datapoints for each sensor are shown in Appendix A.3, A.4, A.5, A.6, A.7. A nearest neighbour approach was used to match the overlapping data points between the two surveys.

The signals from both surveys were filtered with a fifth-order 2 Hz low-pass filter. This step removes direction-dependent propeller noise, so that only the GMF and its variations are included. Residuals are created by subtracting the overlapping tieline datapoints from the matched Survey 2. The residuals are then linearly interpolated across the entire extent of Survey 2. The interpolated residual surface is removed from the original signal. The corrected data were then plotted for each sensor.

During data visualisation, each track exhibited a particular slope. To correct for this, a linear regression was fitted. A 2 Hz low-pass filter was applied before slope correction, after which the residuals were calculated and subtracted. This procedure serves as a pseudo-bandpass filter, essentially removing the low-frequency from the signal (slope) as well as the high-frequency components (UAV-induced noise). The corrected data was visualised for each sensor individually, as well as for a fused dataset combining all sensors.

4 Results

This section presents the characterization of noise sources across surveys, including persistent frequency bands and propeller-induced interference (surveys 7-12). Next, the section about signal modelling is presented, followed by anomaly detection, and impact assessment of flight heading and altitude on the signal (surveys 1-3). Lastly, tieline correction is introduced. Sensor is shortened to S, e.g., Sensor 1 is S1.

4.1 Noise characterization

Noise characterization is based on two surveys: survey 6 (no propeller activity) and survey 5 (propeller activity). The former establishes baseline noise levels, while the latter provides insight into propeller-induced interference.

4.1.1 Persistent frequency bands

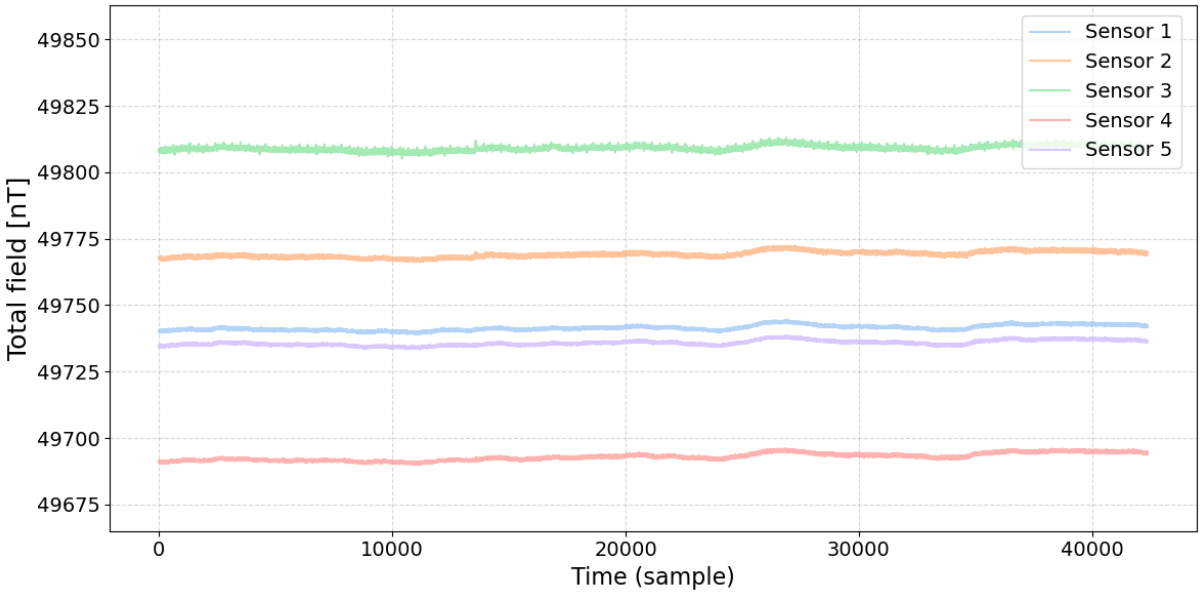

Figure 4.1: Unfiltered total field [nT] measurement of survey 6 (no propeller activity) for all sensors.

Survey 6 establishes baseline noise levels as shown by the unfiltered signal in Figure 4.1. High-frequency fluctuations can be observed in all sensors, with the largest amplitudes in S2-3. Visual observation of the fluctuations is challenging, the fourth difference in the following figure complements this analysis.

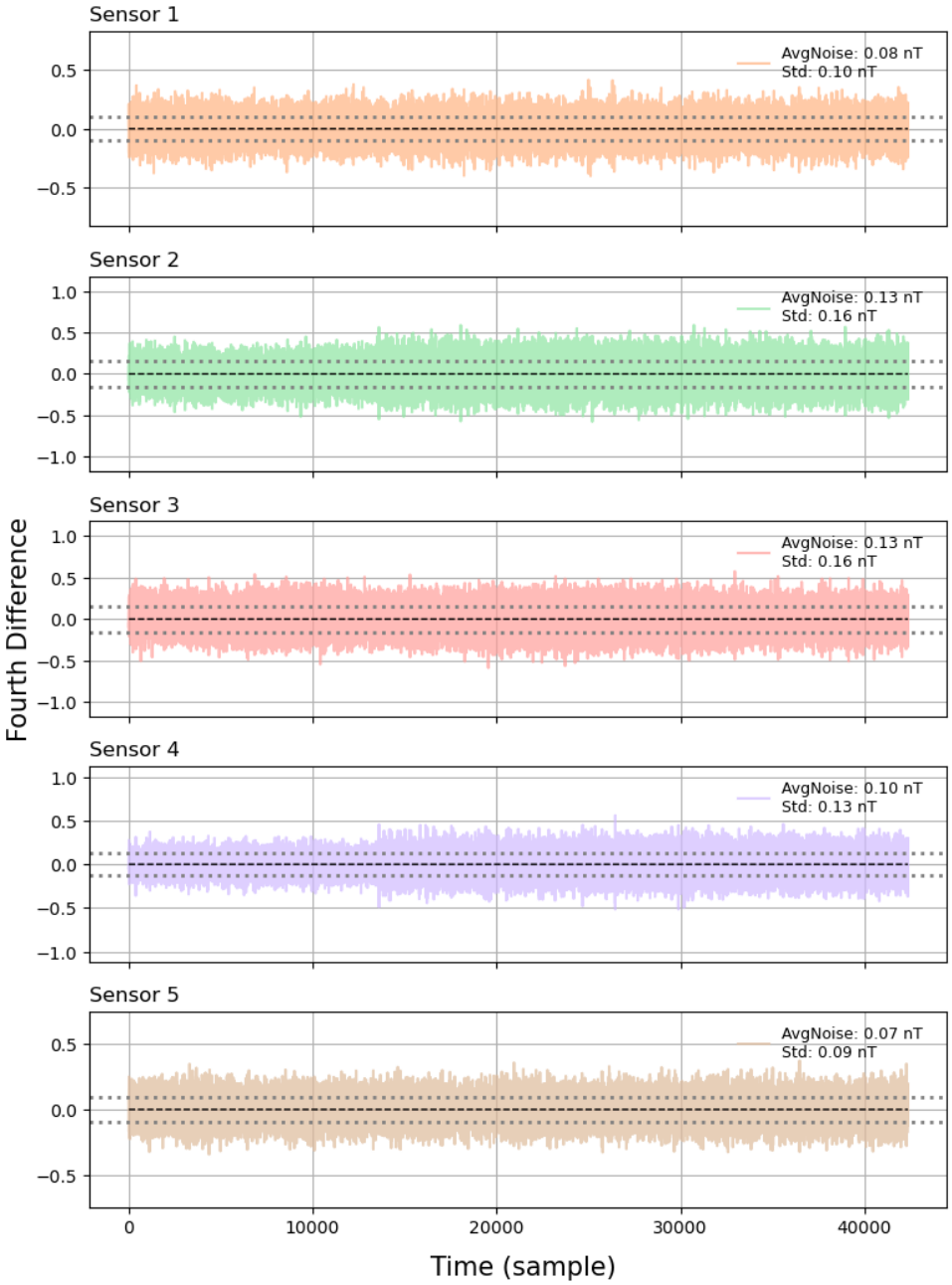

Figure 4.2: Fourth difference of the signal for survey 6 (no propeller activity) for all five sensors. Each sensor colour represents the signal, the dotted grey lines represent one standard deviation from the mean of the fourth difference. The black line represents the trend line.

High frequency variations of S1 to S5 stay within ± 0.5 nT with a majority of data points remaining within one standard deviation (SD) of the mean. S1 and S5 record the lowest SD (σ=0.09-0.1 nT), while S2-3 show the highest (σ=0.16 nT), S4 is inbetween (σ = 0.13 nT). All sensors exceed the industry limit of ± 0.1 nT for the fourth difference envelope, [12].

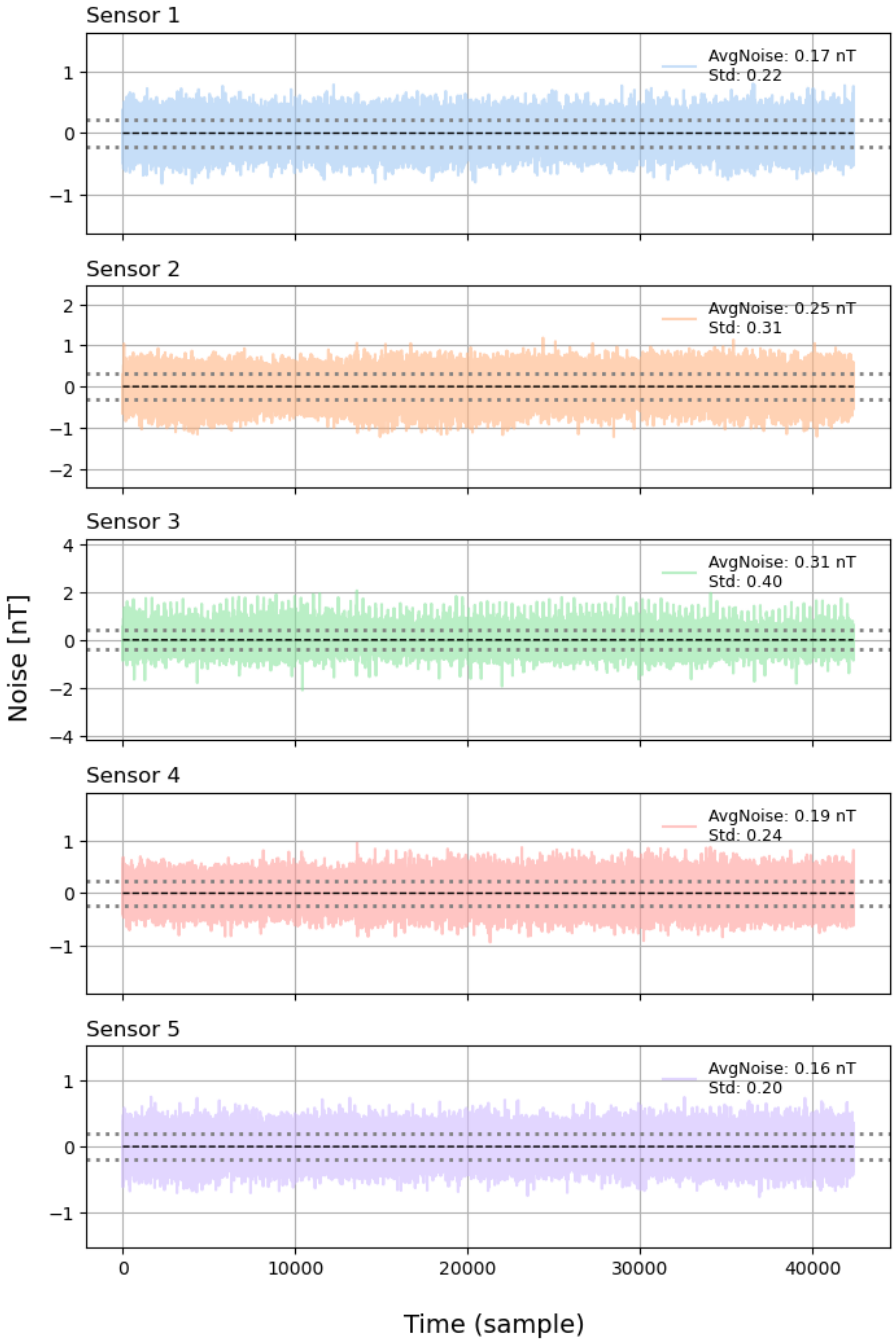

Figure 4.3: Remaining total field signal [nT] after applying 2Hz high-pass filter on survey 6 (no propeller activity).

Figure 4.3 illustrates the residual noise [nT] after applying a 2 Hz high-pass filter on the data from survey 6. S1 and S5 exhibit the lowest average noise equal to 0.17 nT (σ = 0.22 nT) and 0.16 nT (σ = 0.2 nT). S4 records higher noise 0.19 nT(σ = 0.24 nT). S2 and S3 portray the most significant noise with values equal to 0.25 nT (σ = 0.31) and 0.31 nT (σ = 0.4 nT). The noise amplitudes are bounded by 1 nT for S1, S4, and S5, but reach 1.2 nT for S2 and 2 nT for S3.

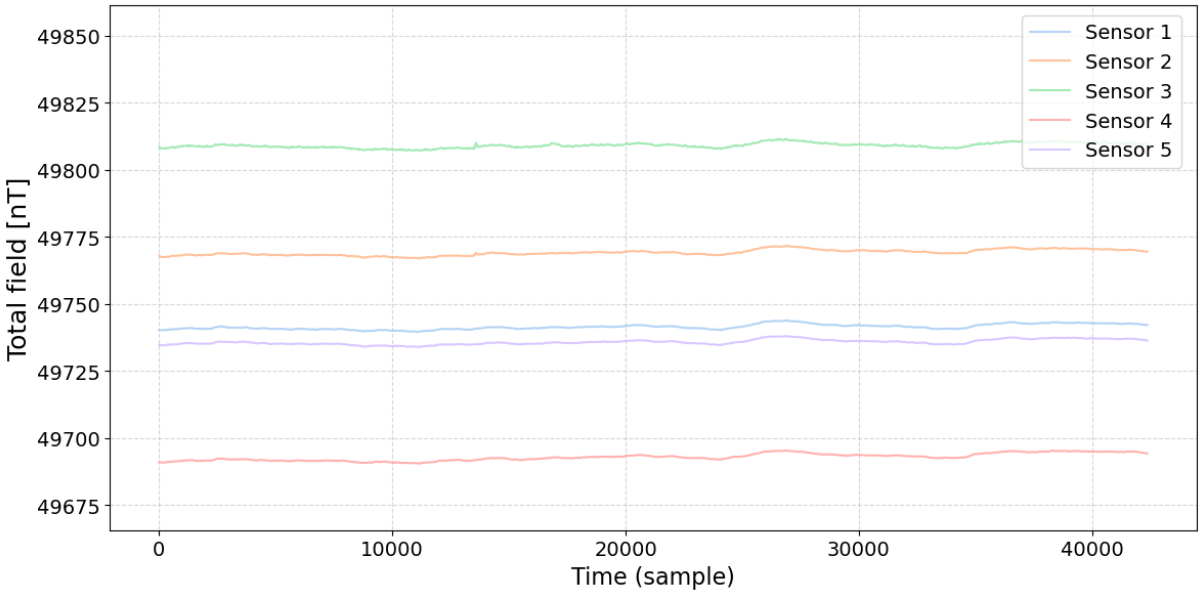

Figure 4.4: Total field [nT] measurement of survey 6 (no propeller activity) for all sensors after applying a 2 Hz lowpass filter.

Figure 4.4 displays the 2Hz lowpass filtered signal for each sensor during survey 6, revealing offsets between sensors. S1 and S5 record similar values with an absolute offset of 5.7 nT. S2 and S3 measure higher discrepancies (27.73 and 67.55 nT) relative to S1. S4 records lower values than S1 with an offset of 48.44 nT. Other inter-sensor offsets are available in Table 4.1.

Table 4.1: Average inter-sensor offsets [nT] for survey 6 (no propeller activity) after applying a 2Hz lowpass filter.

Reference Sensor	**Sensor 1**	**Sensor 2**	**Sensor 3**	**Sensor 4**
Sensor 2	27.73			
Sensor 3	67.55	39.88		
Sensor 4	-48.44	-76.26	-116.05	
Sensor 5	-5.70	-33.43	-73.23	42.76

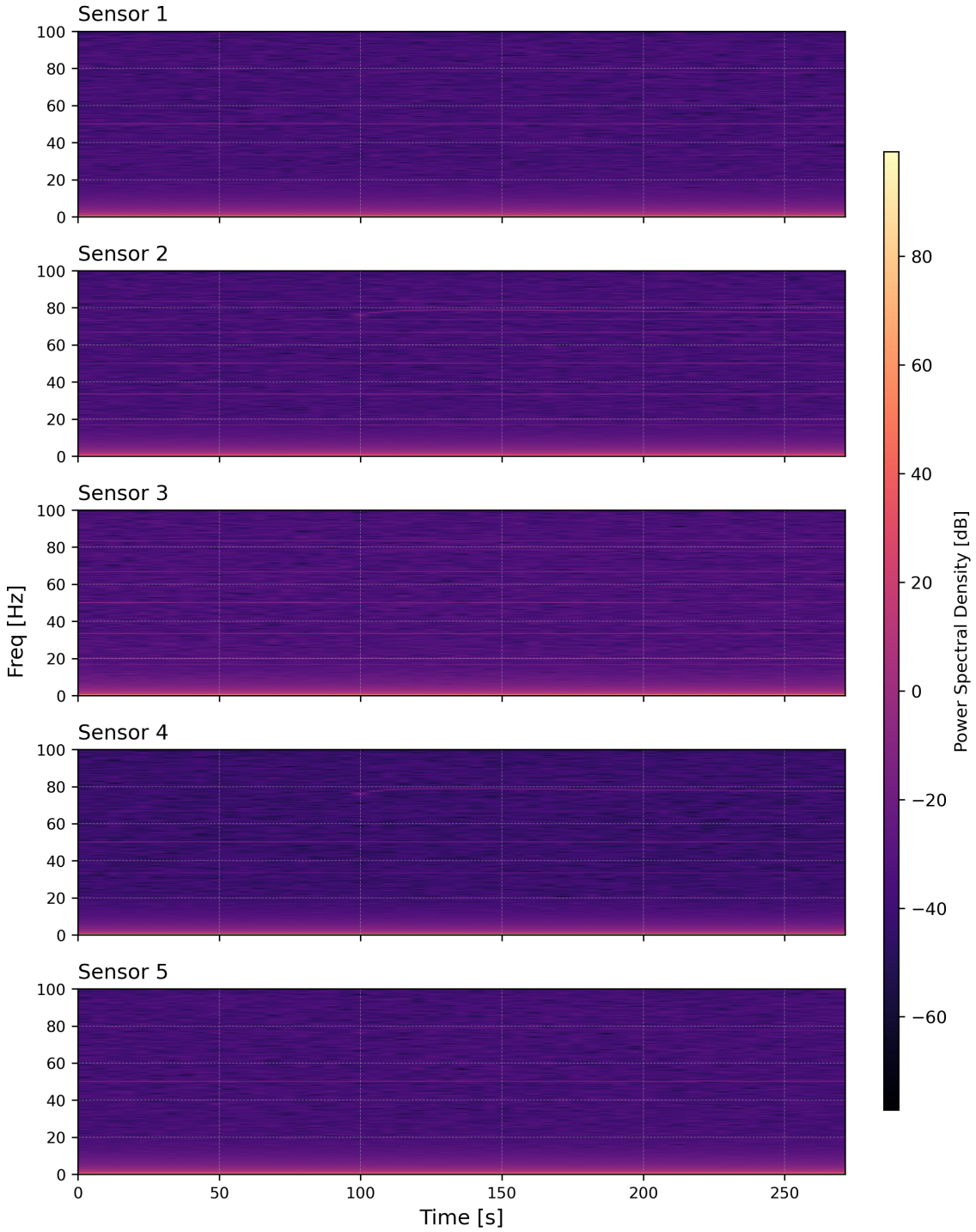

Figure 4.5: Spectrogram of magnetic field data recorded during Survey 6 for all five sensors. The data shown is unfiltered, with a Nyquist frequency of 100 Hz and a window size of 1024 samples. The spectral density of the frequencies is shown in decibels (dB).

The spectrogram presents the unfiltered time-frequency content of survey 6. Low-frequency components (the Geomagnetic field (GMF)) dominate the spectral density, thus obscuring the visualization of high-frequency noise components. However, S3 reveals weak persistent frequency lines at 16.7, 20, 33.4, 50, 66.7, and 83.5 Hz. All bands, except the 20 Hz, are harmonics of 16.7 Hz. Additionally, S2 and S4 also portray a transient frequency starting around 100 s, which fluctuates between 76-78 Hz.

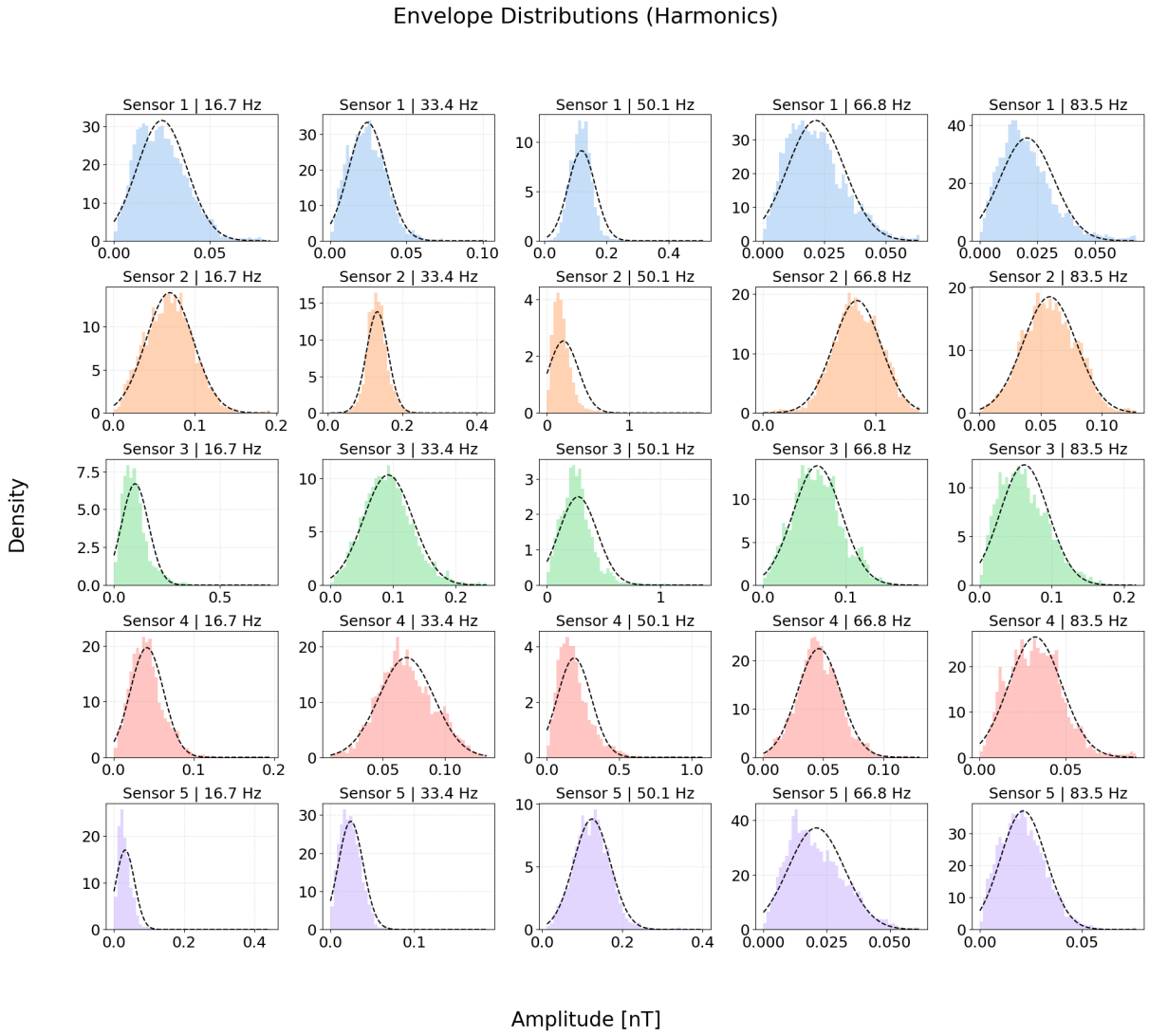

Figure 4.6: Amplitude distributions of 16.7 Hz harmonic envelopes (survey 6). Columns: harmonic frequencies; rows: sensors.

The Hilbert envelope of the harmonics approximates truncated Gaussians, except for the 50 Hz in S2 and S3, which is a multimodal distribution. Harmonics in S1, S4, and S5 have minimal amplitude between 0 and 0.08 nT. The 50 Hz harmonic exhibits larger amplitudes between 0.025 and 0.225 nT. S2 and S3 experience the most decisive influence. All harmonics present values between 0 and 0.23 nT except the 50 Hz band, which fluctuates between 0-0.3 nT for S2 and 0-0.5 nT for S3.

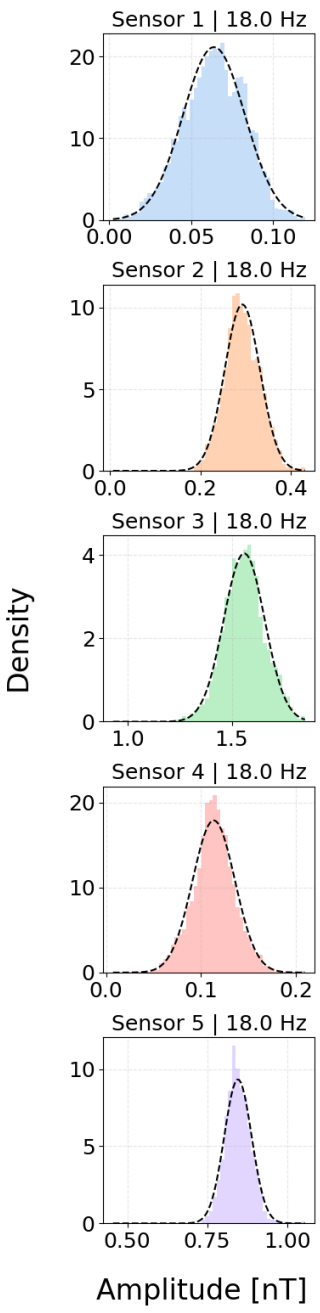

Figure 4.7: Distributions of the 18 Hz frequency line with a fitted Gaussian distribution.

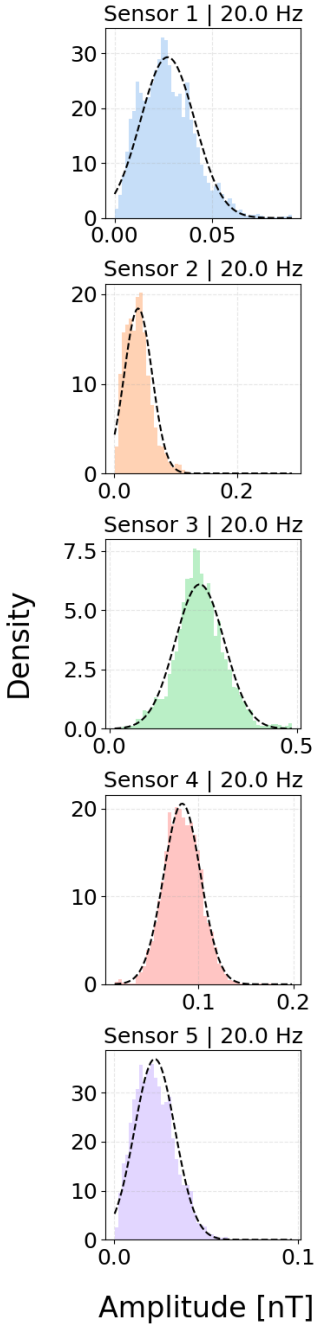

Figure 4.8: Distributions of the 20 Hz frequency line with a fitted Gaussian distribution.

The 20 Hz band follows a similar trend, but now Sensors 3-4 exhibit the highest amplitudes. The 18 Hz distributions will be covered in the next section.
A recapitulation of this subsection: Under stationary conditions, high-pass residuals are sub 0.5 nT on average (best on S1/S5), inter-sensor offsets are large yet stable, and persistent narrowband lines are part of the 16.7 Hz harmonic family and are strongest in S2-3.

4.1.2 Transient frequency bands analysis

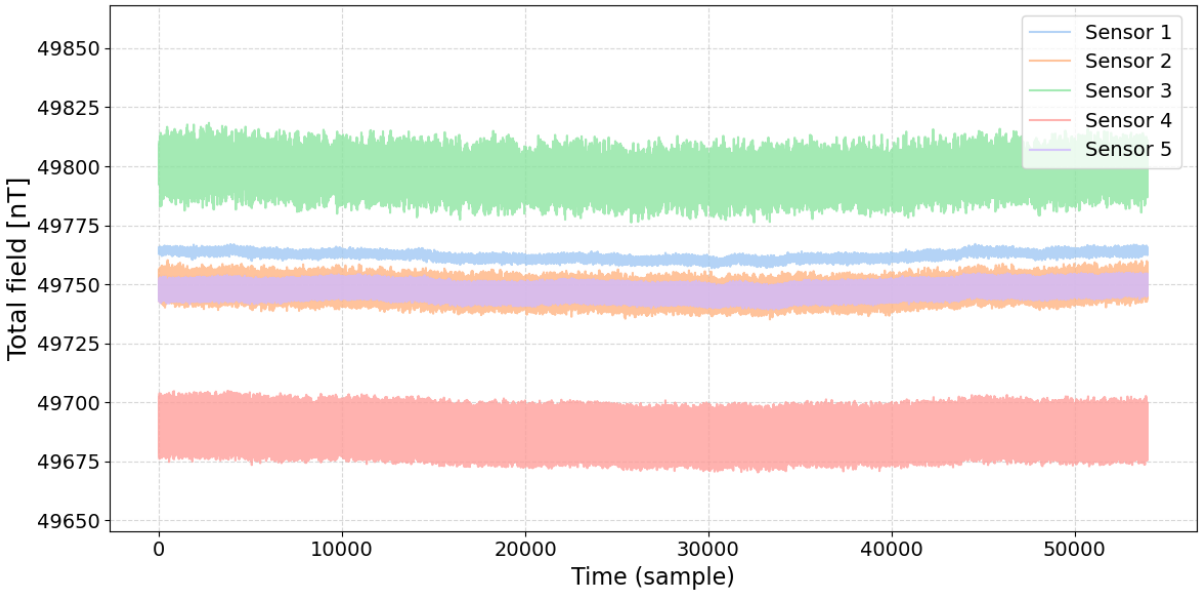

Figure 4.9: Total field [nT] measurement of Survey 5 (propeller activity) for all sensors.

Figure 4.9 portrays the unfiltered data from survey 5 (propeller activity). Compared to survey 6, the high-frequency fluctuations increased. S3 and S4 exhibit the most pronounced changes with broader amplitude spread and sharper outliers. It indicates significant propeller-induced interference.

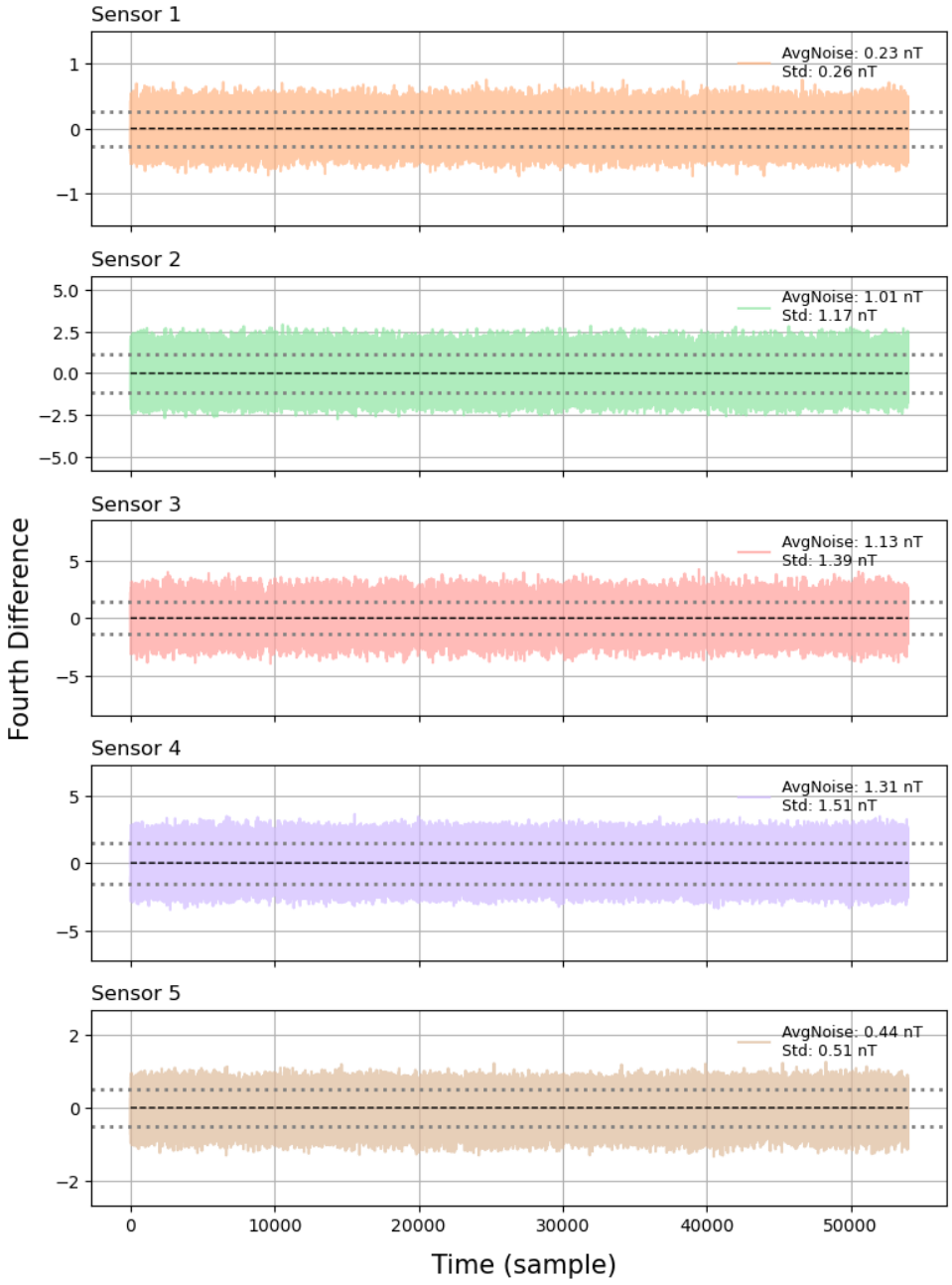

Figure 4.10: Fourth difference of the signal for Survey 5 (propeller activity) for all five sensors. Each sensor colour represents the signal. The dotted grey lines represent one standard deviation from the mean of the fourth difference. The black line represents the trend line.

The fourth difference confirms this increase. The σ of the high-frequency noise increases significantly across all sensors, for example, ranging from 0.26 to 1.51 nT, which strongly supports the presence of propeller-induced interference. All sensors exceed the industry limit of $\pm$ 0.1 nT for the fourth difference envelope, [12], with noise levels 3 (S1) to 13 (S4) times higher than survey 6. Similar noise ranges are present in survey 7 at 30 m AGL, Appendix A.18.

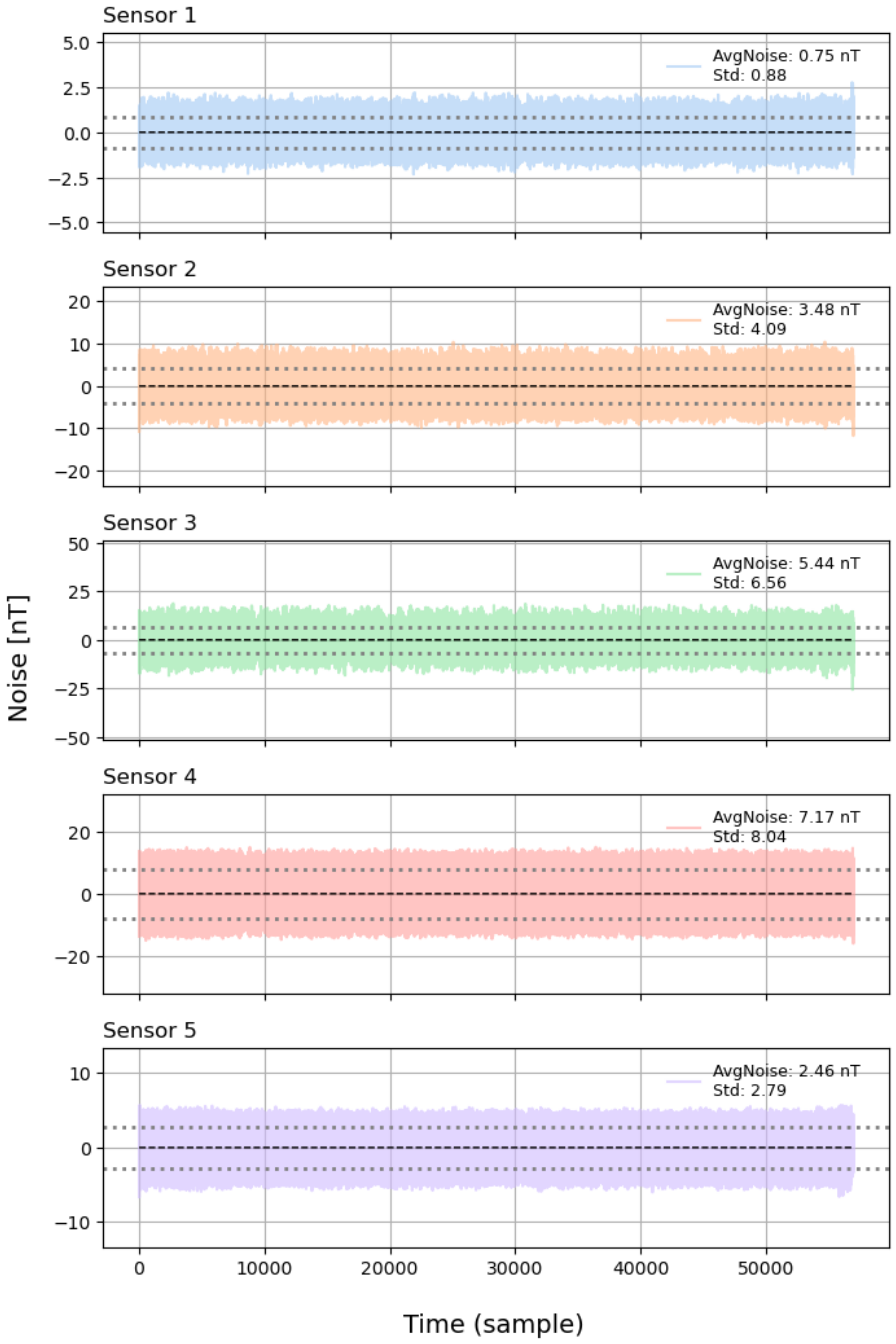

Figure 4.11: Remaining total field signal [nT] after applying 2 Hz high-pass filter on survey 5 (propeller activity).

Figure 4.11 reveals the amount of noise [nT] from survey 5 after applying a 2 Hz high-pass filter. Noise levels are substantially elevated compared to survey 6. Values range from 0.75 nT in S1 (σ= 0.88 nT), 3.48 nT in S2 (σ= 4.09 nT), 5.44 nT in S3 (σ=6.56 nT), 7.17 nT in S4 (σ=8.04 nT), and 2.46 nT in S5 (σ=2.79 nT).

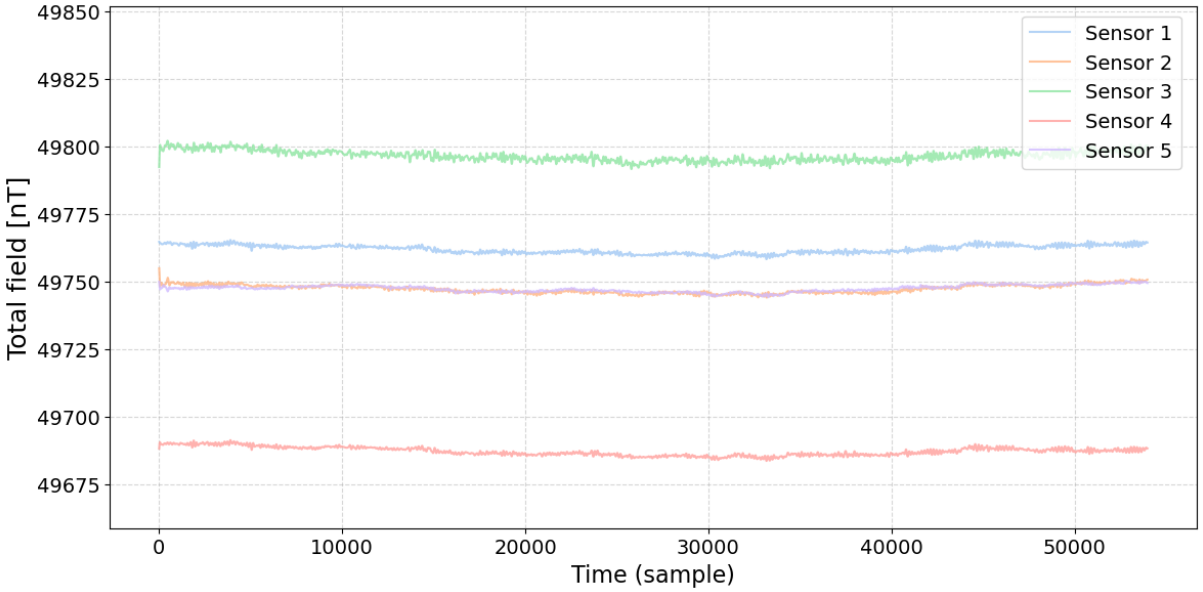

Figure 4.12: Total field [nT] measurement of survey 5 (propeller activity) for all sensors after applying a 2 Hz lowpass filter.

The 2 Hz lowpass-filtered signals, Figure 4.12, deviate from the measured values in the stationary survey 6. The following values represent the intra-sensor offsets between survey 5 and survey 6. Compared to their previous values, S2 records 21.77 nT and S3 12.46 nT less on average. S1 and S5 record around 20.65 nT and 11.67 nT more on average, respectively. S4 remains nearly unchanged.

Inter-sensor offsets become more pronounced, Table 4.2. The offset between S1-2 has decreased to 14.67 nT, between S1-3 to 34.48 nT. Lastly, the offsets between S1-4 and S1-5 have increased to values of 74.63 nT and 14.44 nT. The discrepancy between S1 and S5 has increased, while S2 and S5 show noticeable overlap.

Table 4.2: Average inter-sensor offsets for survey 5 (propeller activity) after applying a 2Hz lowpass filter.

Reference Sensor	**Sensor 1**	**Sensor 2**	**Sensor 3**	**Sensor 4**
Sensor 2	-14.67			
Sensor 3	34.48	49.15		
Sensor 4	-74.63	-59.94	-109.18	
Sensor 5	-14.44	0.17	-48.90	60.28

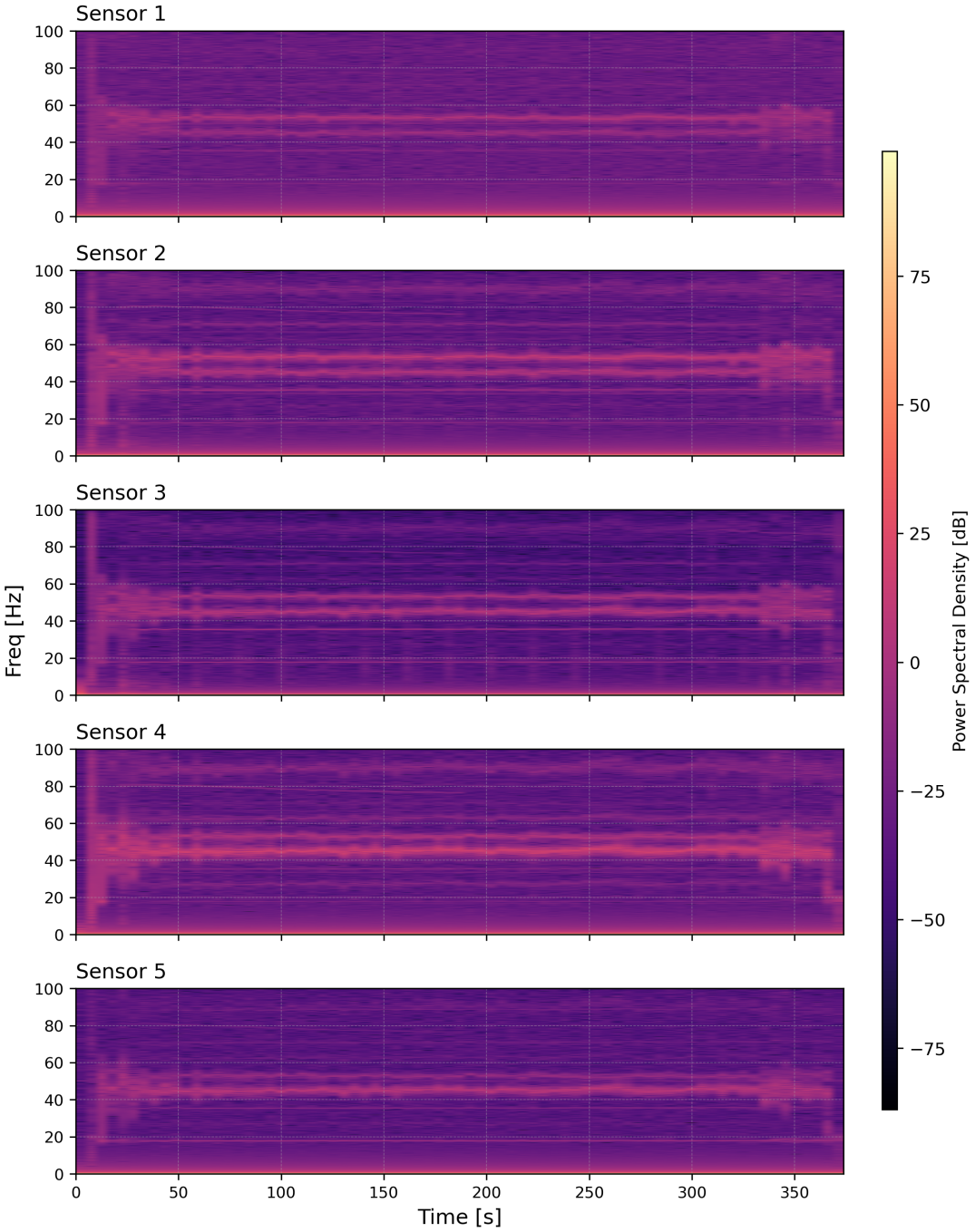

Figure 4.13: Spectrogram of magnetic field data recorded during Survey 5 for all five sensors. The data shown is unfiltered, with a Nyquist frequency of 100 Hz and a window size of 1024 samples. The spectral density of the frequencies is shown in decibels (dB).

Spectrograms of survey 5, Figure 4.13, reveal additional propeller-induced components. Two primary ridges emerge: one at 42-47 Hz (with visible harmonics at 94-98 Hz) and another at 50-57 Hz (harmonics not visible due to Nyquist density). Additional bands appear at 18 Hz and its harmonics at 36, 54, and 72 Hz. The 54 Hz harmonic is likely masked by the dominant propeller noise. Spectral density is strongest in S5, followed by S3. Figure 4.7 portrays the presence of the 18 Hz band in survey 5. S3 and S5 show peaks of 2 nT and 1.4 nT, S2 reaches 0.43 nT. All distributions are Gaussian distributions.

Surveys 8-12 reveal there are four distinct frequency ridges instead of two, Appendix A.13 through A.17. Significant overlap makes initial visualisation challenging, but SSQ-STFT isolates these components successfully.

Synchrosqueezed analysis of survey 7 (30 M AGL) at 1.5 $\frac{m}{s}$ isolates these ridges, Figures A.23, A.24, A.25, A.26, and A.27. Ridge one spans 40-44 Hz, ridge two 45-47 Hz (occasional overlap with ridge 1), ridge three 49- 51 Hz, and ridge four 51-60 Hz. The frequencies can be converted to RPM values by multiplying with a factor 60 $\frac{s}{min}$. As a result, ridge one varies roughly between 2300-2700 RPM, ridge one 2700-2900 RPM, ridge three 2900-3100 RPM and ridge four 3100-3400 RPM.

S4 and S5 lack ridge one due to spectral overlap with ridge two. Furthermore, S5 barely records minimal energy in ridge 4. Spectral energy distributions vary significantly across sensors. The following values, Table 4.3, are visually estimated from the previous figures and represent the expected noise for each ridge based on the sensor position.

Table 4.3: Visual estimates of Spectral energy distribution (dB) across sensors and frequency for survey 7 based on Figures A.23, A.24, A.25, A.26 and A.27.

Sensor	**Ridge 1**	**Ridge 2**	**Ridge 3**	**Ridge 4**
1	30	20	15	40
2	40	30	20	45
3	35	45	25	35
4	25	50	35	20
5	40 (overlap)		25	10–15

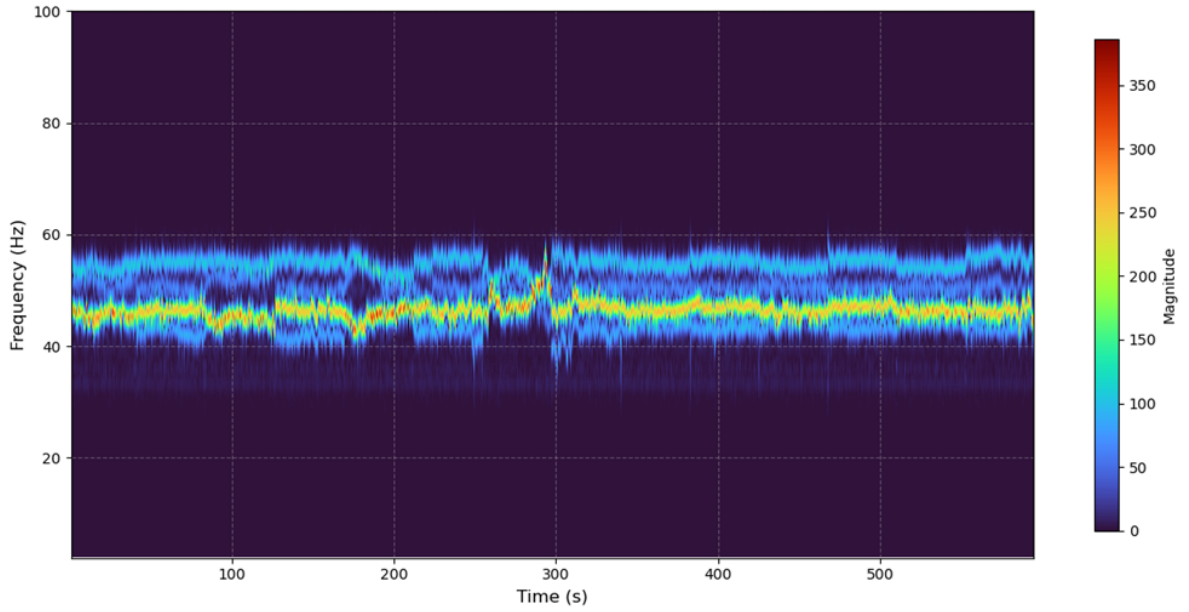

Figure 4.14: The Short Time Fourier Transform (STFT) of S3, with a window size of 1024, for Survey 7 (30 m AGL) after bandpassing the signal for frequencies between 37 and 60 Hz. The magnitude of the STFT is shown instead of spectral density in decibels (dB) for better visualization.

An interaction pattern is observed among these four ridges, the SSQ-STFT of S3 from survey 7 is used for this explanation, as shown in Figure A.25. Between 90 and 120 seconds, ridge one increases in frequency while ridge two decreases, resulting in a complete overlap. Ridge three and four do not exhibit the same dynamic, but ridge three slightly mimics the behaviour of ridge one. Between 160 and 190 seconds, ridges three and four overlap and increase in frequency. Meanwhile, ridges one and two overlap and decrease in frequency. Between 270 and 300 seconds, ridges three and four both decrease in frequency while ridges one and two increase, resulting in a complete overlap of the four ridges. These interactions can be linked to drone movement.

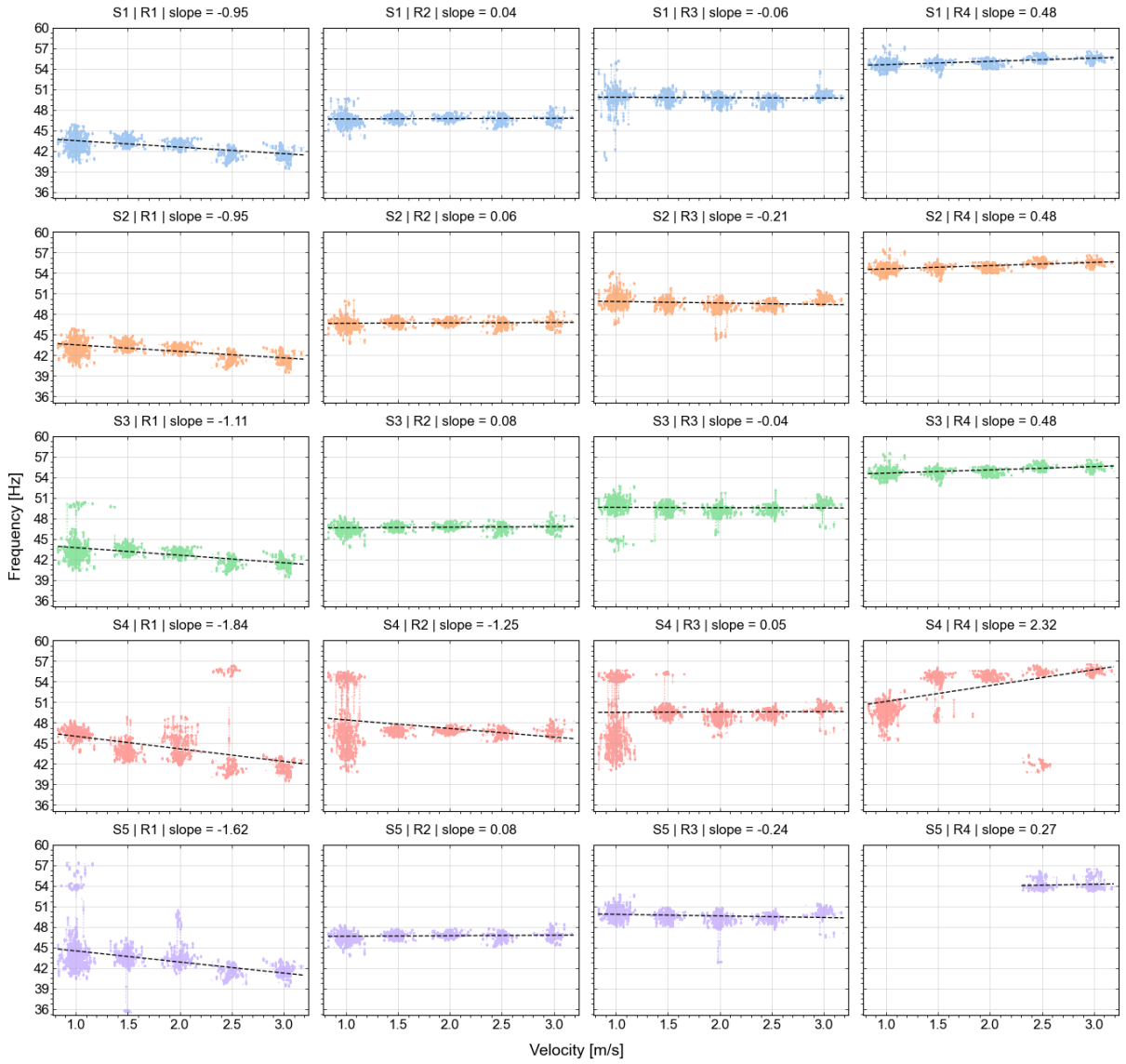

Figure 4.15: Frequency (Hz) of the extracted ridges from surveys 8-12 plotted as a function of the velocity ($\frac{m}{s}$). The columns represent the ridges, while the rows represent the 5 sensors. The slope of the five datasets is also supplied.

Figure 4.15 displays the linear evolution of each ridge frequency as a function of velocity. The columns represent the four ridges, and the rows represent the sensors. S1, S2, and S3 show the clearest patterns, while S4 and S5 have slight overlap between ridges. For instance, ridge one fluctuates initially around 45 Hz at 1 $\frac{m}{s}$ and decreases to 42 Hz at 3 $\frac{m}{s}$ across all sensors. S4 shows the most significant overlap between ridges two and three, leading to smearing over a broader range of frequencies.

1. **Ridge one:** displays a consistent negative slope of -0.95 Hz per $\frac{m}{s}$ in S1 and S2. S3, S4, and S5 have similar slopes, but overlapping datapoints influence these.
2. **Ridge two:** shows a negligible slope between 0.04 and 0.08 Hz per $\frac{m}{s}$ across all sensors. S4 is an exception with a negative slope of -1.25 Hz per $\frac{m}{s}$ due to overlap between ridges one and four. Ridge two remains stable between 45–48 Hz. This suggests that ridge two may be less involved in movement control.
3. **Ridge three:** behaves similarly to ridge two, with negligible slopes varying between -0.06 and 0.05 Hz per $\frac{m}{s}$. S2 and S5 show steeper negative slopes of -0.21 and -0.24 Hz per $\frac{m}{s}$ due to overlap. Ridge three typically fluctuates between 48–51 Hz, with a noticeable decline up to 2.5 $\frac{m}{s}$, followed by an increase at 3 $\frac{m}{s}$.

4. **Ridge four:** displays a positive slope of 0.48 Hz per $\frac{m}{s}$ for S1, S2, and S3. S4 shows heavy overlap with other ridges, while S5 records an extremely weak signal from ridge four that cannot be reliably extracted. Ridge four fluctuates between 54–57 Hz.

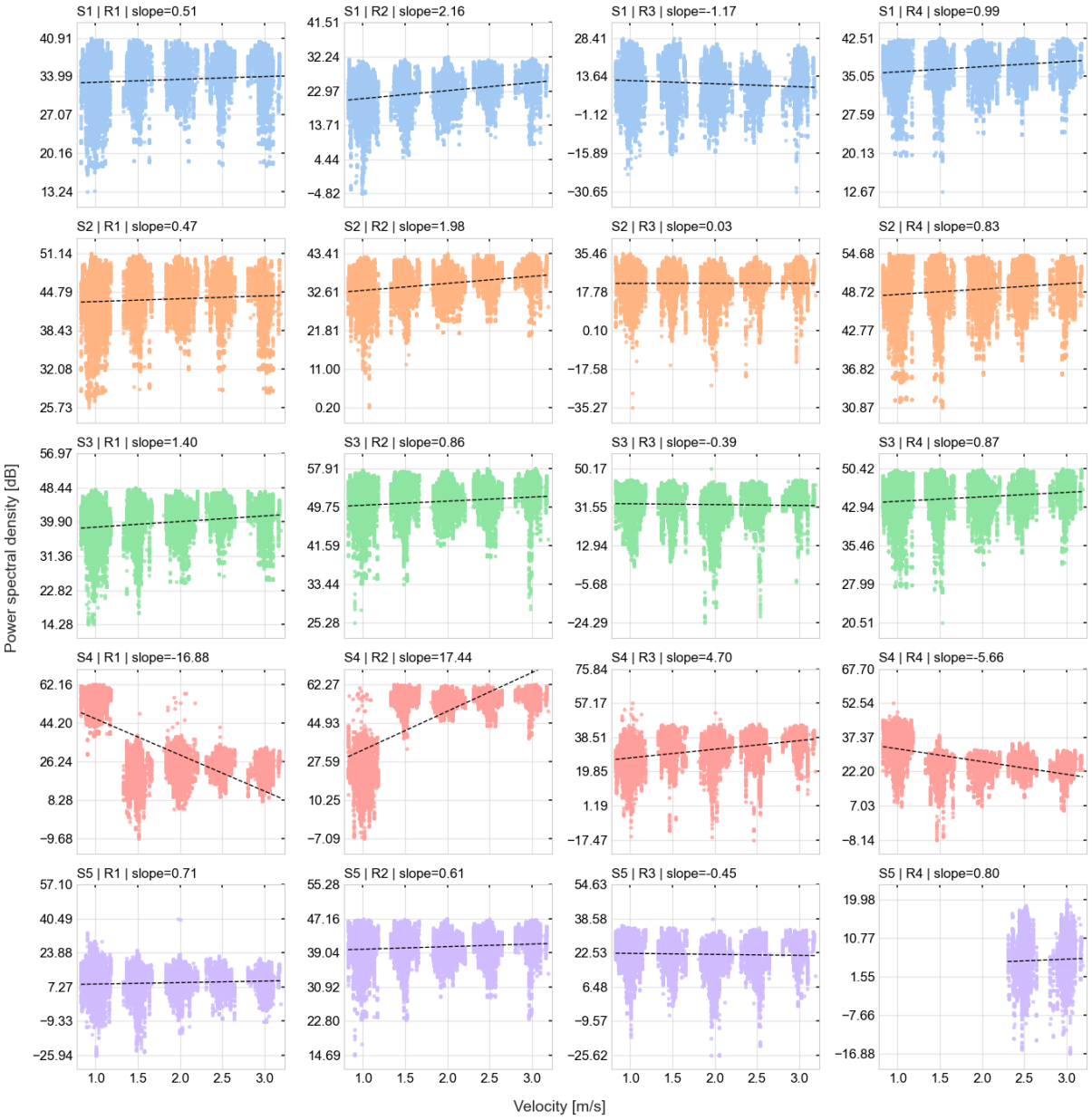

Figure 4.16: Magnitude [-] of the extracted ridges from Surveys 8-12 plotted in function of the velocity [m/s]. The columns represent the ridges, while the rows represent the 5 sensors. The slope of the five datasets is also supplied.

Figure 4.16 illustrates the linear change of the emitted noise for each ridge as a function of velocity. Due to excessive overlap, S4 will be excluded from the analysis.

1. **Sensor 1:** Ridge one, two, and two increase in spectral density with slopes of 0.51, 2.16, and 0.99 dB per $\frac{m}{s}$. Ridge three decreases with a slope of -1.17 dB per $\frac{m}{s}$. Ridge one and four are the most energy-dense (20-43 dB), while ridge two and three are weaker (max. 32.24 and 28.41 dB).

2. **Sensor 2:** Similar slopes are observed for ridge one, two, and four (0.47, 1.98, and 0.83 dB per $\frac{m}{s}$). Ridge three exhibits only a minimal slope. Ridge one and four fluctuate mostly between 32-55 dB, ridge two ranges from 11-44 dB, and ridge three remains within 0-36 dB.

3. **Sensor 3:** Ridge one shows a steeper slope of 1.4 dB per $\frac{m}{s}$, about three times larger than for the previous sensors. Ridge two is halved (0.86 dB per $\frac{m}{s}$). Ridge three decreases slightly (-0.39 dB per $\frac{m}{s}$). Ridge four has a slope of 0.87 dB per $\frac{m}{s}$. Ridge two is the most energetic (33-58 dB); other ridges range from 20 to 51 dB.

4. **Sensor 5:** Ridge one, two, and four increase with slopes of 0.71, 0.61, and 0.8 dB per $\frac{m}{s}$, respectively, while ridge three decreases (-0.45 dB per $\frac{m}{s}$). Ridge one and four have lower energy (≤ 25 dB). Ridge two is the most energetic (22-48 dB), and ridge three peaks at 39 dB.

A recapitulation of this subsection: Under propeller activity, high-frequency noise increases significantly, with S3 and S4 showing the most pronounced changes. Four ridges are identified at 40-44, 45-47, 49-51, and 51-60 Hz. These ridges exhibit a clear interactive pattern among themselves in frequency. Additionally, the ridges either increase or decrease in frequency and amplitude as a function of velocity. The spectral energy distribution varies significantly across sensors, due to sensor position.

4.2 Noise modelling

The ARMA model captured the envelope characteristic of the 16.7 Hz component, which led to the multiplicative model replicating the noise signal quite accurately. Figure 4.17 shows significant overlap between the original signal from survey 5 and the synthetic data acquired from the trained model on the data from survey 6. The first 40 seconds show high signal peaks, which are present due to drone movements leading to overlap between the 16.7 Hz band and other noise sources.

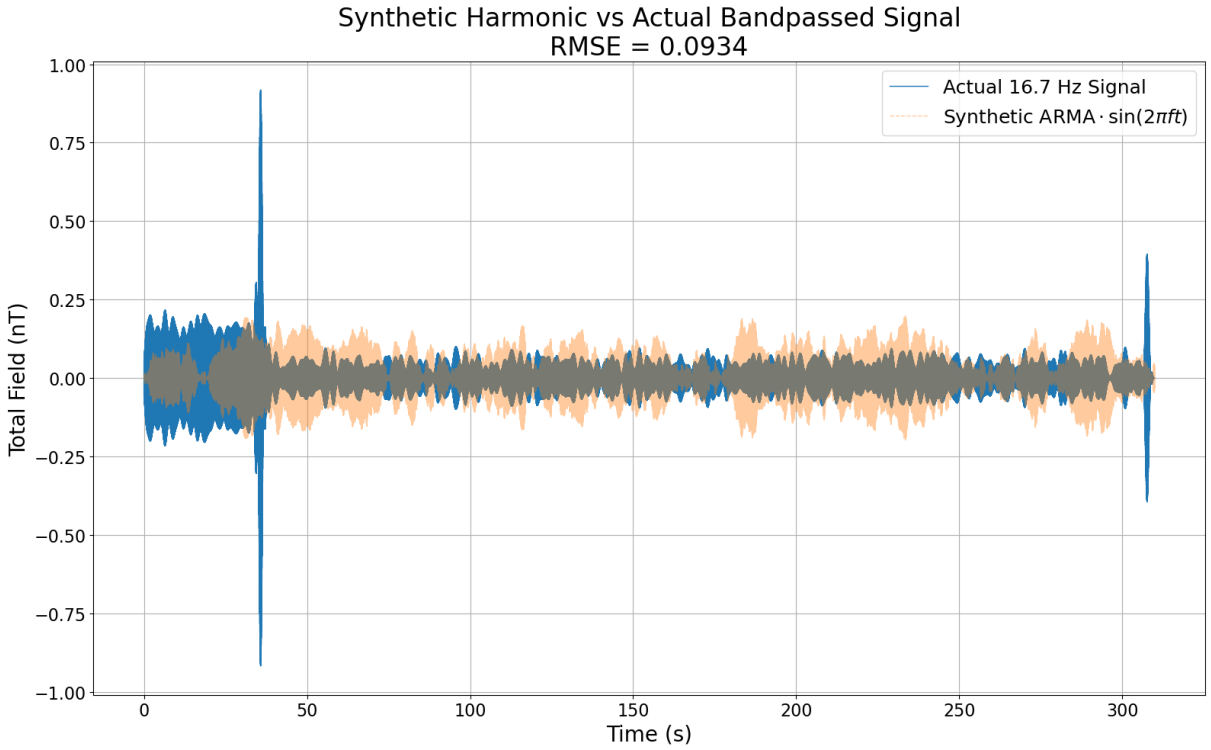

Figure 4.17: Multiplicative model of the 16.7 Hz frequency band of S3 from survey 6. The blue signal represents the original signal from survey 5. The orange signal represents the predicted signal by the ARMA model multiplied by a sinus function with a frequency equal to 16.7 Hz.

4.3 Anomaly Detection

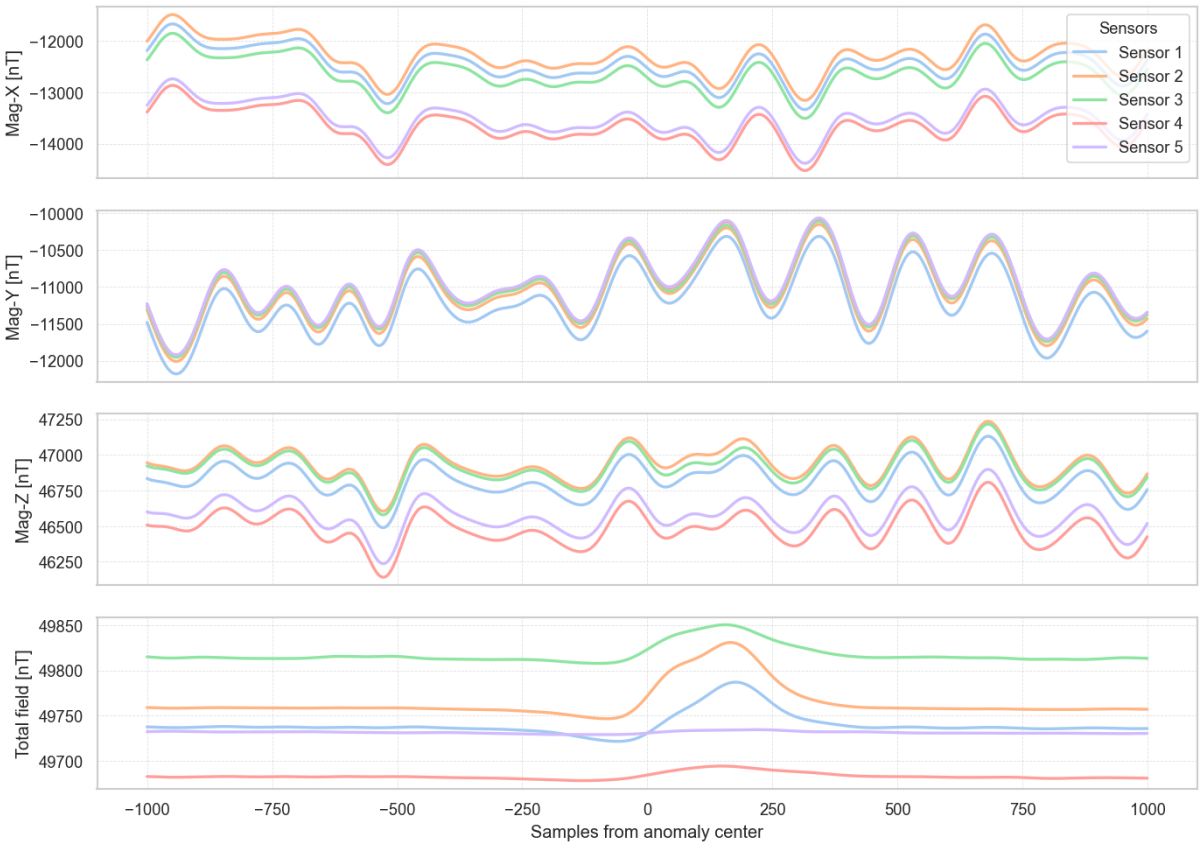

Figure 4.18: The X-, Y- and Z- component and total field [nT] of the 10 sec anomaly window after applying a 2Hz lowpass filter for Survey 2 (constant heading 0.87 m AGL) for all 5 sensors.

Figure 4.18 shows the signal of the extracted anomaly window with a 2 Hz lowpass filter for survey 2. In survey 2, S2 shows the clearest two-peak anomaly in the total field. A negative trough followed by a first and secondary peak with a small trough in between. S1 is similar with weaker peak separation. S3 barely separates the peaks, while S4 only records a hill-shaped anomaly without clear peak separation, and S5 fails to detect the anomaly.

Component-wise, the X-component experiences an initial increase around -100 samples. The signal then follows a decrease-first peak-decrease-second peak-sharp decrease pattern. The Z-component behaves identically, only with a less pronounced secondary decrease after the first peak. The Y-component displays a simple increase-decrease-increase pattern. The unfiltered data can be observed in Appendix A.30.

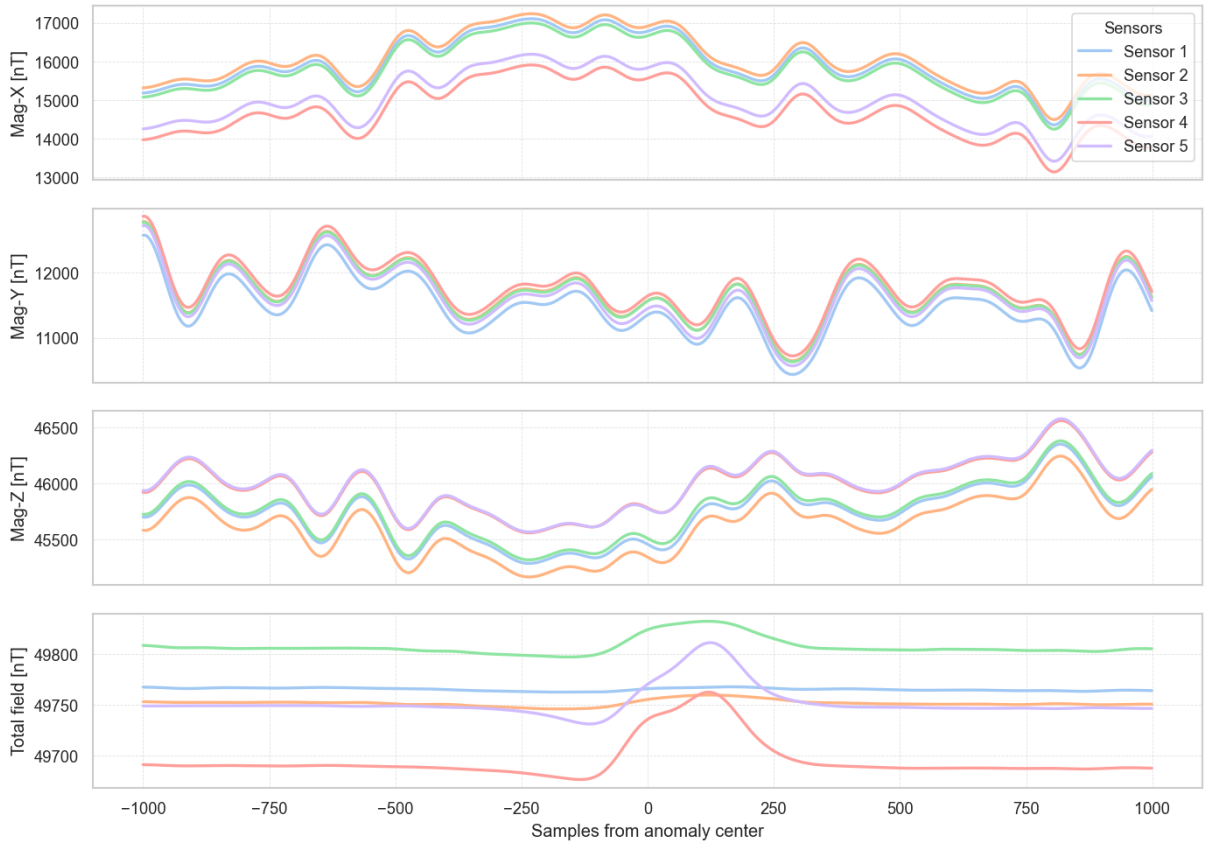

Figure 4.19: The X-, Y- and Z- component and total field [nT] of the 10 sec anomaly window after applying a 2Hz lowpass filter for Survey 3 (variational heading 0.87 m AGL) for all 5 sensors.

Figure 4.18 shows the signal of the extracted anomaly window with a 2Hz lowpass filter for survey 3. S4 and S5 detect the clearest two-peak anomalies, while S1 and S2 barely detect the anomaly. S3 records a broad hill-shaped anomaly. S2 barely records the anomaly, while S1 fails to detect the anomaly.

Component-wise, both the X- and Z-components show the first peak-decrease-second peak-sharp decrease pattern, with the Z-component's secondary peak increase more pronounced. The Y-component alternates between decreases and increases but is less tied to the anomaly. The unfiltered data can be observed in Appendix A.31.

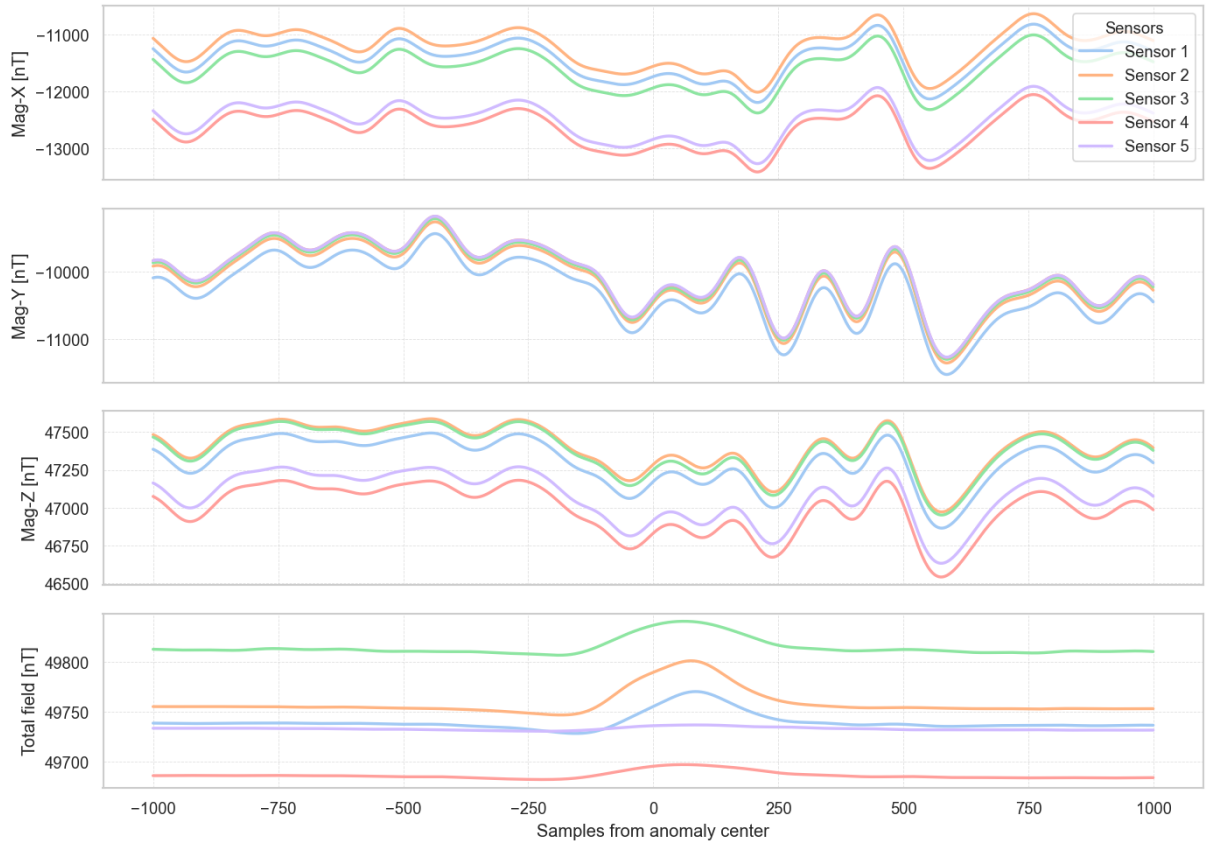

Figure 4.20: The X-, Y- and Z- component and total field [nT] of the 10 sec anomaly window after applying a 2Hz lowpass filter for Survey 1 (constant heading 1.17 m AGL) for all 5 sensors.

Figure 4.19 shows the signal of the extracted anomaly window with a 2Hz lowpass filter for survey 1. Survey 1 resembles survey 2, though with reduced two-peak resolution. The small trough between the peaks is no longer visible in any sensor. The unfiltered data can be observed in Appendix A.29.

Tables 4.4 and 4.5 show the Signal-To-Noise Ratio (SNR) and the Peak-to-Peak (p2p) amplitude difference of the extracted anomaly window for surveys 1, 2, and 3. Filtering with a lowpass 2 Hz filter substantially increased SNR (for instance, S2, from 21.01 to 75.27).

Table 4.4: Signal-to-noise ratio for all 5 sensors across survey 1, 2, and 3 before and after 2 Hz high-pass filtering.

Sensor ID	**Cte 1.17 m**		**Cte 0.87 m**		**Var 0.87 m**	
	Unfiltered	Filtered	Unfiltered	Filtered	Unfiltered	Filtered
1	28.23	39.61	43.47	67.71	5.75	4.48
2	15.01	61.44	21.01	75.27	7.12	14.82
3	8.98	30.32	10.24	33.86	9.21	27.81
4	4.72	16.57	4.94	19.08	13.49	69.68
5	6.58	8.63	6.35	7.33	30.41	76.86

Simultaneously, the p2p amplitude decreases for each sensor, indicating the removal of additive noise above the 2 Hz limit. Table A.2 in the Appendix shows the variance of the background variations.

Table 4.5: Peak-to-peak amplitudes for all 5 sensors across survey 1, 2, and 3 before and after 2 Hz high-pass filtering.

Sensor ID	**Cte 1.17 m**		**Cte 0.87 m**		**Var 0.87 m**	
	Unfiltered	Filtered	Unfiltered	Filtered	Unfiltered	Filtered
1	47.13	42.19	69.59	65.55	8.41	5.24
2	72.16	54.58	101.56	84.37	29.80	13.68
3	60.43	34.15	68.71	42.75	61.33	35.15
4	45.61	15.03	46.57	16.15	112.01	86.03
5	13.24	6.34	12.49	5.49	90.49	80.10

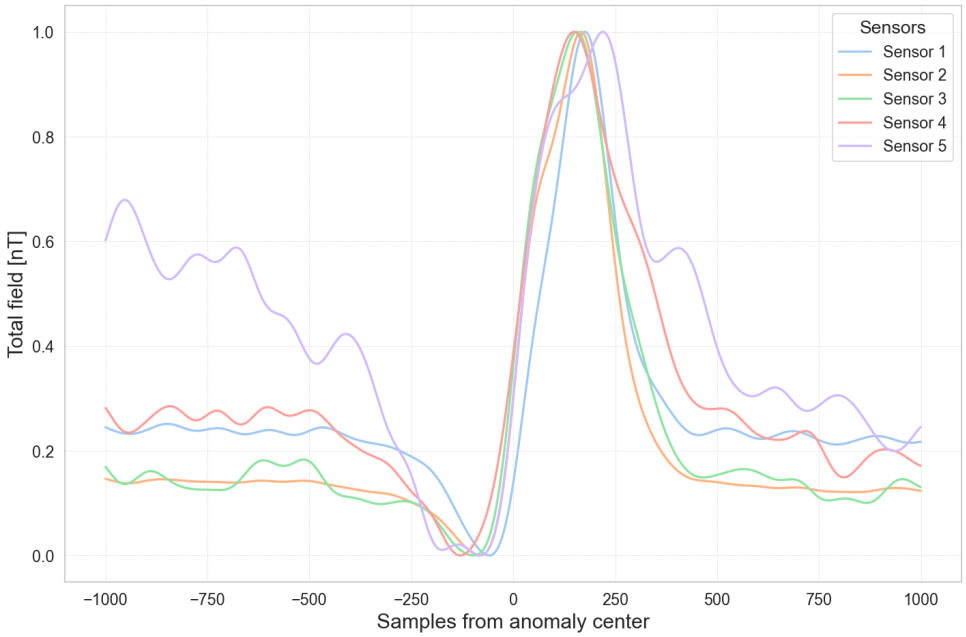

Figure 4.21: The normalized total field [-] of the 10 sec anomaly window for survey 2 (constant heading 0.87 m AGL) for all 5 sensors.

Figure 4.21 shows the normalized signal of the anomaly window for survey 2 after lowpass filtering. Outside the interval of [-250, 250] time samples, the signals converge to nearly identical normalized values, provided the SNR is sufficiently high. This can be observed in S5 which shows almost similar values inside and outside the anomaly window. A counterintuitive result: S5 records a weak signal, but it detects the two-peak pattern, unlike the other sensors.

The normalized signals of survey 1 and 3 are present in Appendix A.19 and A.20, respectively. The normalized signals of surveys 1 and 3 show similar patterns. In survey 1, all sensors seem to measure similar values outside the anomaly window, except for S5. In survey 3, S1 shows the most evident two-peak pattern, while having similar values outside and inside the anomaly window.

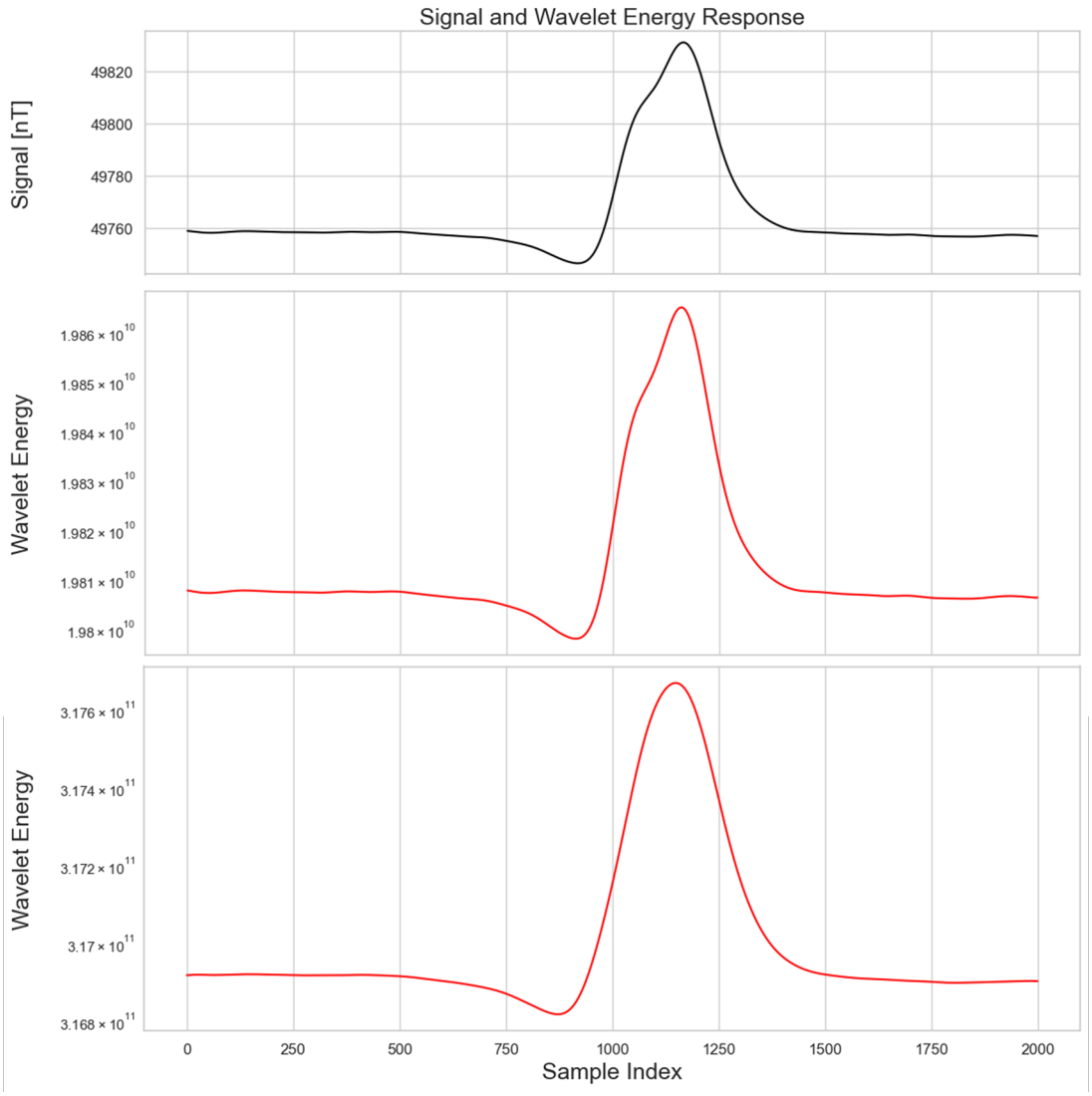

Figure 4.22: Match filtering of the anomaly signal for survey 2 with the sym7 wavelet. The first wavelet filtering is done with a level 3 sym 7 wavelet, the second with a level 7 sym 7 wavelet.

Figure 4.22 shows the match-filtered results of the wavelet with the anomaly signal. The first wavelet filtering, done on level 3, resembles a two-peak pattern. The second wavelet filtering only shows a single peak. A loss of resolution arises from increasing the wavelet level, which implies a larger time window for the wavelet.

4.3.1 Directional discrepancies

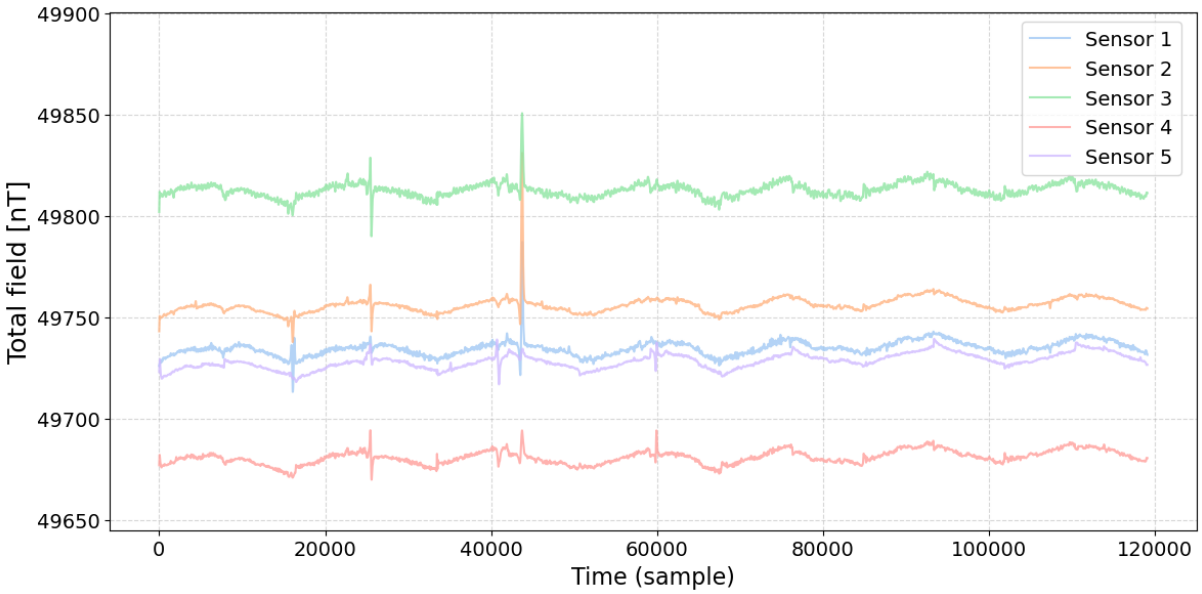

Figure 4.23: Total field [nT] measurement of survey 2 (constant heading 0.87 m AGL) for all sensors. The signal was processed with a 2Hz lowpass filter.

Comparison between surveys 2 and 3 allows for the determination of the magnetic response to flight orientation. Survey 2 displays a leveled zigzag pattern, Figure 4.23. Once again, the offsets between the sensors are visible. S4 records the lowest values, S1 and S5 are similar, while S2 and S3 measure the highest values.

Appendix A.22 shows the components for survey 2. Discontinuities arise from the removal of the off-track points during drone turns. The X-component alternates between two bands, one ranging from -11000 to -14000 nT and another from -14000 to -17000 nT. The Y-component has a consistent pattern with values between -9000 and -13000 nT. The Z-component displays the same jumps with values ranging from 45000 to 47500 nT.

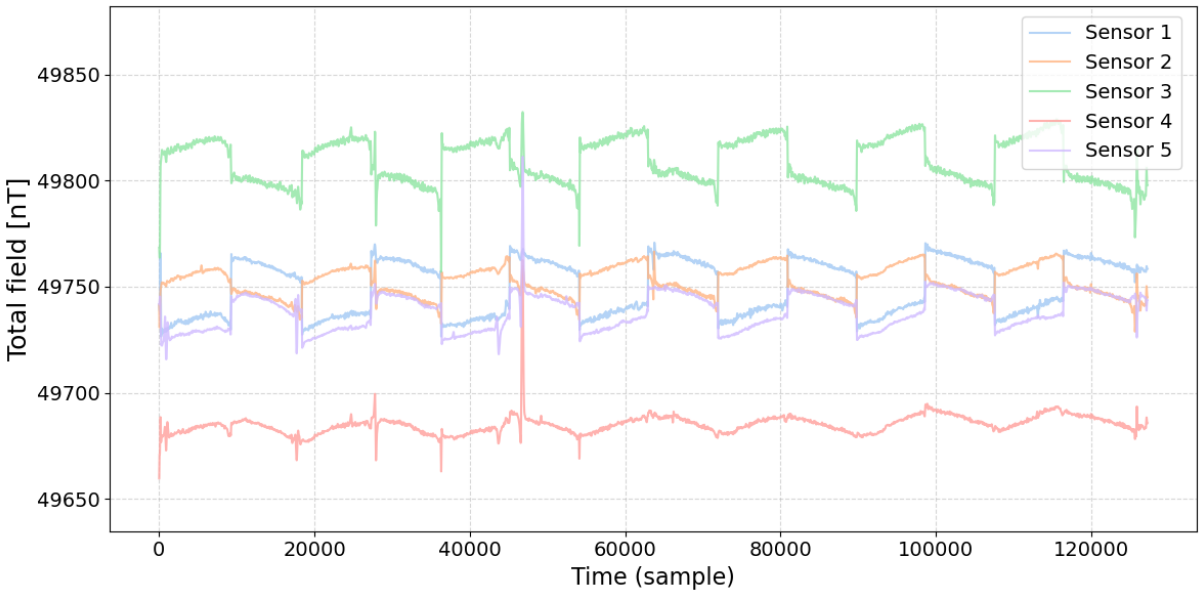

Figure 4.24: Total field [nT] measurement of survey 3 (variational heading 0.87 m AGL) for all sensors. The signal was processed with a 2Hz lowpass filter.

Survey 3 displays similar jumps across all components but with larger discrepancies. S3 records the highest values and S4 the lowest. S4 maintains a consistent pattern, while others vary. S1 and S5 behave similarly and opposite to S2.

Appendix 4.24 the X-component varies between ± 20000 nT and the Y-component between ± 15000 nT. The Z-component shows a consistent pattern with values mainly ranging from 44500 to 47500 nT. S5 exhibits a more levelled signal, while the remaining sensors alternate between jumps. This contrasts with survey 2, where component variation was more limited and consistent per flight leg.

In both surveys, each track contains a consistent slope, whether positive or negative, of similar magnitudes per track and survey. This is also visible in the components. The slope has values of ± 0.008-0.012 $\frac{nT}{sample}$. As a result, the slope- and tieline-correction is applied.

A recapitulation of this subsection: A 2 Hz lowpass filter substantially improves the SNR of a signal, thus making anomalies more visible. The individual components, as well as the total field strength of an anomaly window, show a clear two-peak pattern. This pattern becomes less pronounced the further the sensor is positioned from the anomaly. The pattern can be used for anomaly detection, as is indicated by the wavelet match filtering. The signal shows discrepancies depending on the flight direction. Component analysis shows which components cause these differences.

4.4 Tieline correction

Figure 4.25 shows the slope- and tieline-corrected, 2 Hz lowpass filtered data from survey 2. The left plot shows precise data, with slight artefacts visible in the middle of the field. For instance, the linear artefact present on the track which crosses the anomaly. NNbathy in the right plot shows less sensor positioning preference and almost accurately pinpoints the anomaly. Edge artefacts are visible from interpolation. The compensated signal for each sensor from this figure is included in Appendix A.32.

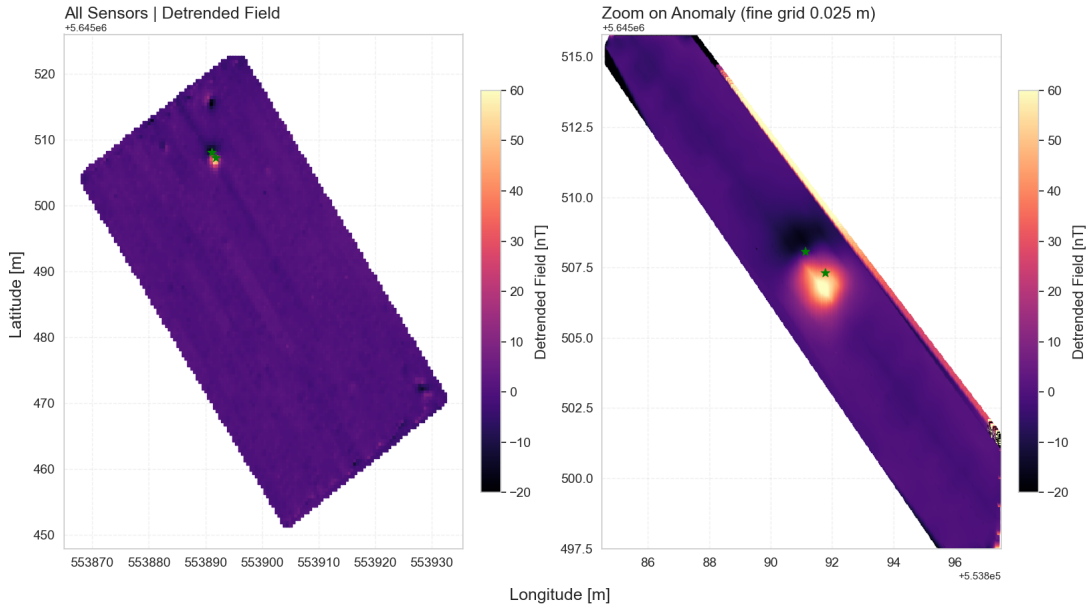

Figure 4.25: Left: Slope- and tieline-corrected, 2 Hz lowpass filtered data from survey 2 interpolated with NNbathy at 0.5 m. Right: Zoomed-in view of the anomaly area, interpolated at 0.025 m.

Table 4.6 summarizes the 2 Hz lowpass filtered total field strength of survey 2 for each sensor before and after tieline correction with survey 4, Figure 4.26. It also lists the relative anomaly position based on Natural-Neighbour bathymetry (NNbathy) and notes visual artefacts observed in the interpolated grids.

Table 4.6: Tieline correction (unfiltered vs tieline-corrected) summary across all sensors.

Sensor	Raw Range [nT]	Tieline Range [nT]	Position	Artefacts
1	49725-49780	49720-49775	Correct	Slope presence
2	49740-49820	49740-49815	Left	Slope presence
3	49790-49850	49790-49845	Left	Horizontal ridges and slope
4	49665-49700	49660-49695	Left	Slope and dead pixel
5	49718-49739	49715-49735	Right	Slope presence

Table 4.7 summarizes the 2 Hz lowpass filtered and slope correction total field strength of survey 2 for each sensor before and after tieline correction with survey 4, Figure 4.27. It also lists the relative anomaly position based on (NNbathy) and notes visual artefacts observed in the interpolated grids.

Table 4.7: Tieline correction (unfiltered vs slope- and tieline-corrected) summary across all sensors.

Sensor	Raw Range [nT]	Tieline Range [nT]	Position	Artefacts
1	-12 up to 35	-12 up to 35	Correct	Slope presence
2	-11 up to 60	-11 up to 55	Left	Slope presence
3	-20 up to 31	-20 up to 31	Left	Horizontal waves, slope
4	-10 up to 11	-10 up to 15	Left	Stark slope, dead pixel
5	-11 up to 7.5	-11 up to 7.5	Right	Slope presence

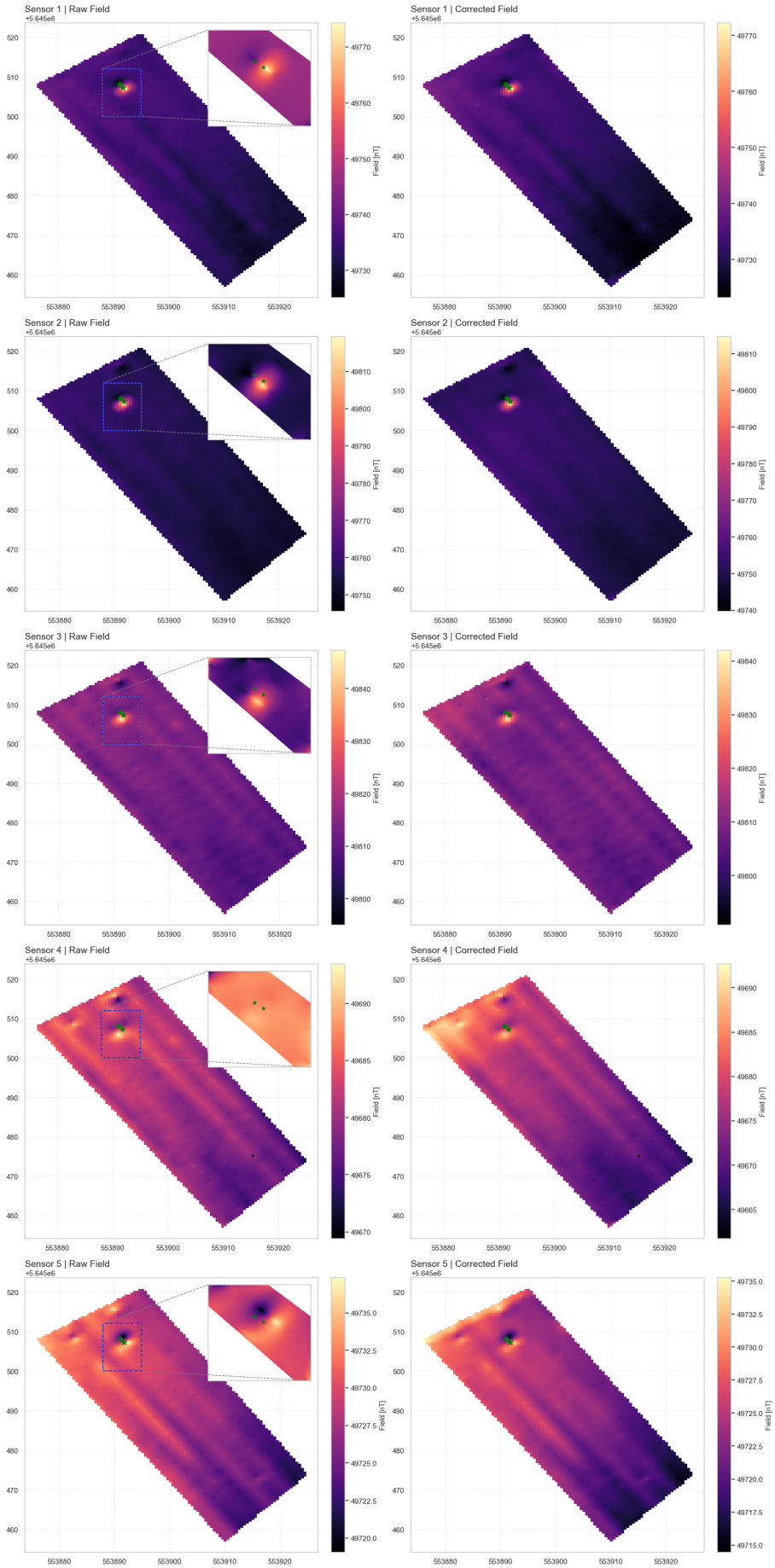

Figure 4.26: Left: The 2 Hz lowpass filtered data from survey 2 (constant heading 0.87 m AGL) interpolated at 0.5 m with NNbathy, the zoom in plot focuses on the planed anomaly and is interpolated at 0.025 m. Right: The tieline "corrected" data interpolated at 0.5 m with NNbathy.

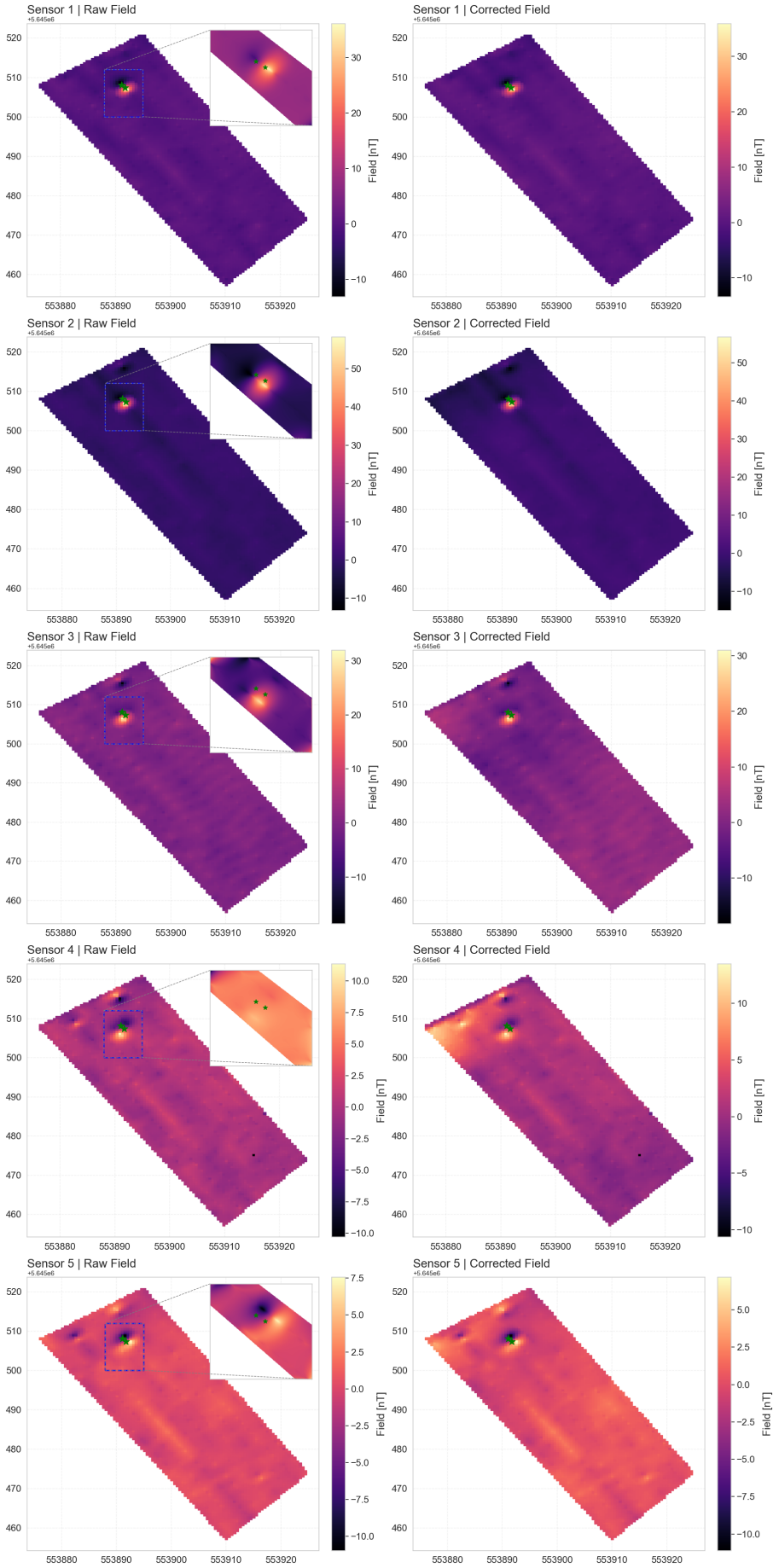

Figure 4.27: Left: The 2 Hz lowpass filtered and slope-corrected data from survey 2 (constant heading 0.87 m AGL) interpolated at 0.5 m with NNbathy, the zoom in plot focuses on the planed anomaly and is interpolated at 0.025 m. Right: The tieline "corrected" data interpolated at 0.5 m with NNbathy.

5 Discussion

The discussion chapter provides a critique of the methodology and a reflection of the results, structured according to the research questions. It also includes suggestions for improvement.

5.1 Methodology critique

5.1.1 Experiments

The experiments performed in this study enabled a baseline assessment of UAV-induced noise and anomaly detection. They provided helpful information about which noise sources are present, when they occur, and how they evolve under different flight conditions. Additionally, the data allowed for appropriate visualisation of anomaly signals at different orientations and heights. However, these experiments remain largely preliminary, and a more extensive experimental design is required for a more comprehensive assessment of noise and limitations.

First, only two identical planted anomalies were used, composed of strong ferromagnetic material. These anomalies ensured a strong and distinct signal, but it does not provide insight into the detection performance in less ideal conditions. Conditions such as, for instance, a magnetically weaker and/or deeper planted anomalies, differential drone altitude, or velocity. The spherical shapes of the anomalies allowed for easily detectable and interpretable signals. Thus, the effect of geometry remains unknown, as more complex shapes were not tested. Additionally, the flights were performed along one axis, as such, diagonal or perpendicular flights were omitted.

Second, the lack of benchmarking makes the data analysis less scientifically grounded. Confirming the results with previous literature can compensate for this issue, however, on-site magnetic surveys would significantly contribute to the development of noise thresholds, sensor sensitivity, and anomaly comparison [68].

To address these limitations, a more comprehensive experimental design should be developed. The design is inspired by [42, 56] and should include the following:

1. Shape variation: Testing multiple anomaly shapes, such as rectangular, cylindrical, and triangular, in both horizontal and vertical orientations.
2. Depth variation: Testing anomalies at different depths, ranging from 0 to 1 meter, with increments of 10 cm and an additional depth of 2 m, to assess the sensor's lower detection limit.
3. Above Ground Level (AGL) variation: Flying at multiple heights, ranging from 0.5 m to 1.5 m above the anomaly, to assess the influence of the mineralogy's magnetic properties on the background signal.
4. Velocity variation: Testing the drone at different velocities, while flying at fixed AGL and anomaly depth. It determines the pitch's influence on the noise.
5. Directional variation: Conducting diagonal and perpendicular flights, at different heights above the anomaly. Ideally, one anomaly should be selected, and a wind rose pattern should be executed across it.

6. Benchmarking: Performing a ground-based survey ideally with the same sensors to compare with the drone-based survey [68]. This would allow for a quality assessment of the drone-based survey and noise threshold determination.

7. Calibration flight: This should be applied to every survey campaign. It accounts for heading errors and diurnal variations.

The recommended values are based on insights and intuition gained from the results of this study and are not validated by existing literature. Further procedural details can be executed on-site. While these recommendations were envisioned before the survey campaign, time limitations prevented their implementation.

For future survey campaigns, it might be worthwhile to consider comparison with satellite imagery and infrastructure maps to correlate organized anomalies to infrastructure [56]. Additionally, a multi-sensor approach could allow for better detection of anomalies, as mentioned in [42] with the integration of GPR and magnetometers. Lastly, the use of a stationary base station could allow for better compensation of diurnal variations.

5.1.2 Track detection and Geolocalisation

An open-source track detection allows the user to carefully analyse tracks without requiring multiple exports of the same PMXLF format file from a patented application. For example, it allows for the opportunity to analyse oversampled sections of the surveys when the drone decelerates or accelerates, before or after a track switch.

These sections could be locally downsampled without having to downsample the entire survey. However, these features, such as oversampling analysis and a functioning track labeling and selection function, have not yet been implemented. Track identification could be incorporated by clustering the "on_{track}" datapoints and then assigning a track number to each cluster [14].

Although the geolocalisation used in this study is not perfect, it is sufficient for the research questions posed. However, the errors occurring from the lack of telemetric and the initial offset of the GPS receiver to the sensor can substantially impact the data [40]. For future research, it is recommended to update the code and to integrate telemetric data, such as roll, yaw, and pitch, to ensure coordinate accuracy. At the time of writing, new developments make it possible to extract this data.

5.2 Noise characterization

5.2.1 Persistent frequency bands

The high-frequency noise characterized by the fourth difference within Survey 6 exceeds the industry threshold of $\pm$ 0.1 nT. In this case, drone electronics noise without propeller activity heavily impacts the signals' credibility according to this benchmark. The impact increases even more as the propellers are turned on in Survey 5, essentially rendering the signal unusable.

This kind of metric is beneficial as it provides a quick means to assess signal credibility. However, its applicability must be evaluated with the study's context. The rationale for how this industry standard was determined is unclear. It is possible that it was developed for specific survey types or strict regulations within a particular country, where tolerance for noise is low. The aircraft and setup used in this study differ substantially from the original research [13].

Future research should either determine whether this threshold is applicable or instead establish a value tailored to the specific DJI-Sensys setup. This can be achieved by comparing ground-based surveys to the drone-based surveys with the aid of pairwise correlation matrices [68].

Most of the noise that remains after high-pass filtering the signal from Survey 6 lies between 0 and 0.5 nT with occasional outliers at 1 or 2 nT, depending on the sensor. The 16.7 Hz harmonics family contributes to the majority of the electronic noise. The position of the sensor to the electronics matters, particularly to the Skyhub, which is the reason why S2 and S3 experience the highest noise from the 16.7 Hz harmonics family, and why S1 and S5 record Gaussian noise.

This finding is consistent with Stele et al. [68], who also observed a 16.7 Hz signal in a magnetically quiet area of the test site. However, this does not imply that the frequency band purely originates from the Skyhub. In Stele et al. [68] the noise is additive and a combination of the Skyhub and the railway system, whereas in this study it can be exclusively linked ot the Skyhub.

The 83.5 Hz harmonic of 16.7 Hz, also observed in [68], is present in this study as well, but the distribution differs. In this study, it is the strongest in S2 and S3, while their data shows the highest amplitudes in S3 and S4. Either way, the influence is minimal. The 20 Hz frequency is also observed and appears to be electronics noise, though it is not clear where it originates from.

The 50 Hz noise appears to be a superposition of the noise from the Skyhub and a separate electronic component, most likely the power grid, as in [68]. It's probably also the reason its distribution follows a Laplace distribution. Ultimately, the electronics noise is additive, indicating that a low-pass filter can remove it.

Overall, electronics noise contributes for at most 10 % of the total interference and can be reliably removed with low-pass filtering due to its additive nature.

5.2.2 Propeller noise

In survey 5, propeller activity produced high-pass noise outliers of up to 20 nT at certain times, indicating that propeller motors account for the majority of the noise. The contribution is estimated at 80 % of the total noise. The ridges in Figure 4.15 and 4.16 likely correspond to the labelling shown in Figure A.1. Propeller 1 through 4 will be referred to as P1-P4.

P1 and P4 generate the strongest noise at S1, likely due to P1's closer positioning and P4's higher rotation frequency, which emits more energy. A similar pattern occurs at S2, where the spectral densities of P1 and P4 are alike due to the sensor positioning. S3, located at the center of the setup, should record similar noise from pairs P1 & P2, and P3 & P4, given their equal distances.

In practice, P2 is substantially stronger than P1, and P4 is stronger than P3. In the case of S4 and S5, P2 is the most decisive influence. S5 is positioned far enough that it barely registers P4. Hence, the detection limit of a propeller to a sensor is estimated to be at 1.5 m.

Overall, the propeller ranges suggest that P4 is the most powerful propeller, followed by P3, P2, and P1. This disproportionate thrust could destabilize the drone. It is hypothesized that the P1-P2 pair generates the same forward thrust, which also holds for the P3-P4 pair.

However, the Skyhub's positioning shifts the drone's center of gravity, thus requiring compensation by accelerating P4 more than P3 and decelerating P1 more than P2. If true, this would mean the Skyhub's position increases noise emission for nearby sensors. It is a hypothesis that can be tested in the future.

Stele et al. [68] observed increased interference frequencies at higher flight speeds, attributed to faster motor rotation. This claim is confirmed by the analysis in Figures 4.15 and 4.16.

Frequency analysis provides a useful proxy for unavailable propeller data. It delivers the propellers' interplay and approximates individual propeller noise contributions. Figure A.25 revealed some of the interplay between the propellers. It can be hypothesized that P1 and P2 have a coupled interaction system. When P1 changes, P2 also changes, though this coupling weakens at higher velocities. P2 maintains a constant frequency slope as a function of velocity.

At lower velocities, the propellers mimic each other's behaviour, while at higher velocities, P2 assumes a stabilizing role compared to P1. A similar dynamic occurs for P3 and P4, where P3 assumes the stabilizing role. The interplay between P1 and P4 directs the drone velocity. P1 reduces RPM while P4 increases it, to generate forward thrust. Both P2 and P3 maintain stable frequency across velocities, though P3 rotates slightly faster than P2 to maintain that forward thrust.

It is important to note that these inferred couplings between the propellers do not represent the actual RPM values. The methodology only provides an educated guess of the macro-scale propeller behaviour and interplay. Micro-scale behaviour, such as instantaneous RPM, cannot be extracted reliably. This limitation arises from the time-frequency conversion, which depends on the window size and the SSQ-STFT compression. Multiple errors accumulate without a reliable way to quantify them.

5.2.3 Vibrational noise

This section interprets the 18 Hz frequency band observed in Survey 5 as vibrational noise. Detailed analysis of vibration propagation within the drone frame is beyond the scope of this thesis. The following claims are hypothesis-driven, based primarily on the gathered data. Although direct literature confirming the origin of the 18 Hz band for the DJI Matrice is lacking, vibrations within drones are well known to affect sensor performance and data quality [29]. However, a study observed vibrations ranging from 160-175 Hz within a DJI mini 2 combo, suggesting the 18 Hz band may represent a subharmonic of 162 Hz [1]. Even if the 18 Hz band does not originate from vibrations, the concepts discussed here remain relevant as potential sources of error if untested or unaccounted for.

The 18 Hz band is consistent with the visible vibrations at the end of the frame where S5 was mounted during the survey campaign. These vibrations originate from propeller RPM, which is transmitted through the drone's legs into the frame and subsequently the sensors. Data suggests that P2 and P4 contribute the most to the vibrational noise, inferred from Figure 4.7. The left side of the drone records intense vibrational noise while the right side is comparatively unaffected. The noise is evident in S1, which measures no vibrational noise and only white Gaussian noise.

S3 experiences the most intense noise due to its central position in the setup. The precise mechanism of the vibration transmission remains unclear. One hypothesis is that internal cooling fans, central within the drone, are the source of the noise. Another hypothesis is that P2 creates constructive interference together with waves from P1, even though their RPMs differ, as shown in Figure 5.1. Both may explain the horizontal ridges observed for S3 in Figure 4.26.

Continuing with the latter hypothesis, energy recorded in S3 may result from mechanical bending of the frame, with the average 1.5 nT signal reflecting kinetic energy. This hypothesis suggests the noise has both additive and multiplicative components. The kinetic energy is additive, visible in spectrograms, and removable by filtering. However, the sensor repositioning due to bending is multiplicative.

The repositioning is less problematic when flying over areas without any points of interest, as the Geomagnetic Field is relatively stable at the centimeter scale. However, near anomalies, these vibrations could substantially impact signal accuracy. For instance, if an anomaly is deeply buried and the drone flies at the edge of its induced field, upward frame bending could cause the anomaly signal to be missed. This risk applies to S1 and S5, given their extended freedom of movement compared to S3, anchored between two drone legs, Figure 5.1.

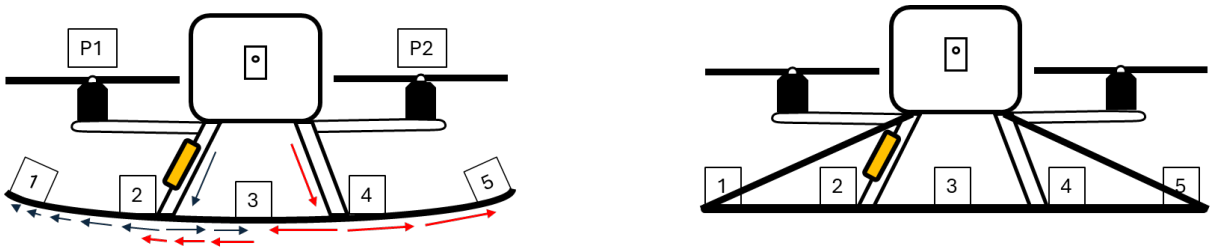

Figure 5.1: Left: Vibrational noise representation. The smaller blue arrows indicate the vibration emitted by P1, the bigger red arrows indicate the vibration by P2. Right: Additional arms compensating for noise on the setup.

S2 and S4 show similar ranges of noise, likely because their anchoring to the drone's legs dampens the vibration. It is hypothesized that S4's anchoring dissipates the kinetic energy horizontally to the rest of the frame. Vibrations originating from P2 and P3 appear to dominate over those originating from P1 and P4 at the anchorage of S2, which is the reason why S2 experiences slightly higher values than S4 in the 18 Hz frequency band. However, most of the noise is transmitted to S5, which has a Gaussian distribution around 0.9 nT.

For future research, it is recommended to perform a finite element method (FEM) analysis to model the vibrations within the drone frame [26].

5.2.4 Offset

The offsets from Figure4.12 make it hard to determine which sensors record the signal accurately. These offsets are not constant. Manual correction with the offset from Table 4.1 and 4.2 proved to be insufficient. The offsets appear to change in complete stationarity (Survey 6), in stationary flight (Survey 5), and during flight (Survey 1-3, 7). It is pretty unclear how these offsets originated.

One possibility is a perming effect from onboard electronics influencing recorded values. However, this idea does not explain precisely why the offset from Survey 6 between S2 and S5 essentially becomes non-existent in Survey 5. Another possibility is track-based bias induced by the pitch of the drone (slopes across tracks). A multifactor origin is likely, involving vibration, in- or off-flight conditions, heading, and electronics.

Stele et al. [68] observed diffuse stripes in their magnetograms, attributed to drone-borne interference frequencies propagating below 2 Hz. To minimize this effect, they recommend lower flight speeds in constant heading mode and applying a constant (CM) or moving median (MM) window filter to remove these stripes.

This is the same phenomenon described here: offsets, driven by vibration, in- or off-flight, heading and electronics. This explains why constant heading yielded the best results. As Figures 4.23 and 4.24 illustrate that these offsets worsen when the drone reorients itself 180 °during track switching. Ideally, compensation algorithms using roll, yaw, and pitch telemetry, [40], would correct these offsets, making heading irrelevant.

In the absence of this data, CM and MM filters can be applied to remove these offsets. In this study, a linear regression was fitted to each track and subtracted from the signal to test whether the slopes were consistent. This suggests that the slope relates to the pitch of the drone. However, MM filters may be preferred for time-adjusted trend removal. These filters should be applied with care according to Stele et al. [68]. Small windows risk blurring the data, while large windows can completely remove anomalies. Such filters are also applied to compensate for the GMF [61].

5.3 Noise modelling

A multiplicative model using an ARMA model was able to replicate the characteristics of the 16.7 Hz electronics noise. While an ARMA model is excessive for such a minor source, it opens up the opportunity of simulating noise sources. Most electronic and environmental noise can be approximated as Gaussian noise, thus requiring only the mean and variance. A combined model with parameters for each noise source could be developed Figure 5.2. This model can then be calibrated to internal drone telemetry to predict the actual noise emission in terms of current. This model would be theory-driven and not black-box, unlike when using machine learning methods.

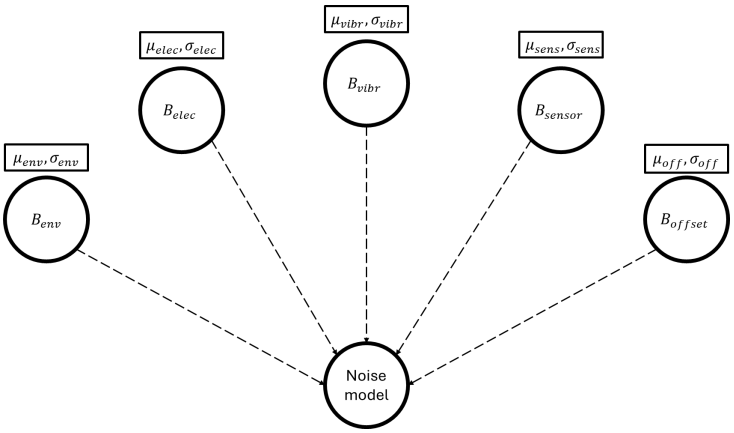

Figure 5.2: Conceptual noise model illustrating different noise sources: environmental (B_{env}), electronic (B_{elec}), vibrational (B_{vibr}), sensor (B_{sensor}), and offset (B_{offset}). Ideally, each component is described by its mean (μ) and variance (σ). These all contribute to a combined model that can predict overall noise emission.

5.4 Anomaly modelling

5.4.1 Sensor sensitivity

Normalising the signals from all sensors in survey 2 shows that each sensor is capable of detecting the anomaly, as shown in Figure 4.21. From this analysis, the lower detection limit (LDL) is estimated at 2 m, Figure 5.3, based on the minimal signal recorded by S5. The limit may be extended with further corrections. Experiments with weaker and deeper anomalies are required. Interestingly, the sensors with the lowest anomaly signals reveal the most evident two-peak pattern once normalized, highlighting the R4's sensitivity [69].

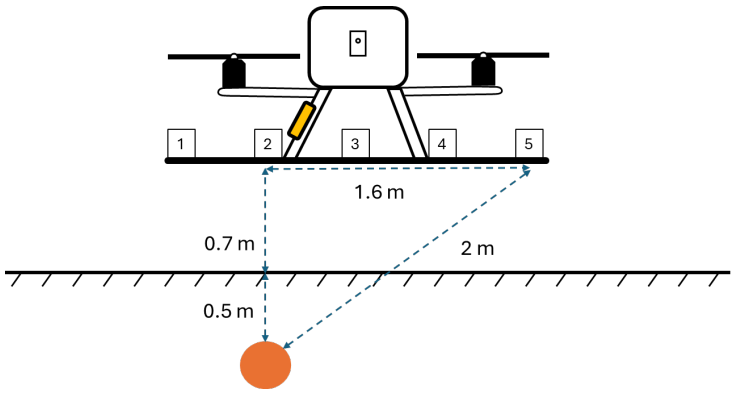

Figure 5.3: Estimated lower detection limit (LDL) of the R4 sensors for a 0.3 m ferromagnetic anomaly. Based on the signals recorded by S5 during Survey 2, the LDL is approximated at 2 m.

5.4.2 Wavelet detection

The wavelet detection algorithm serves as a proof of concept for anomaly detection. Two wavelet levels were tested to illustrate the loss of resolution with larger wavelets, Figure 4.22. For shallow, large anomalies, the signal remains strong, while deeper anomalies appear weaker and narrower. Based on expertise and preliminary anomaly signal response, a suitable wavelet can be selected. Suitable wavelets could effectively detect anomalies of different shapes, depths, and materials. This demonstrates the potential of multiresolution analysis for anomaly detection [48, 82].

5.4.3 NNbathy and artefacts

The original NNbathy interpolation produced an inaccurate field representation with plenty of artefacts. For instance, in Survey 2, S2 passed approximately 10 centimeters to the left of the anomaly, S1 (40 cm to the right of the anomaly), S3 (60 cm left), S4 (110 cm left), and S5 (160 cm left). However, single-sensor interpolation in Figures 4.26 and 4.27, biased towards the highest-value data points. Combining all sensors reduces the bias substantially, yielding a truer representation of the anomaly. Yet, a bias towards S2 is still present. A line passing through the anomaly at the position of S2 is visible in Figure 4.25.

In Figure 4.26, S3 is the only sensor that possesses horizontal ridges, likely caused by previously mentioned vibrations. The ridges form a consistent pattern across the survey. Horizontally, the signal alternates between troughs and peaks, followed by a trough and a peak once again. Vertically, the same pattern occurs at a higher frequency. Additionally, all sensors have a track-based slope referring to offset-related effects described earlier.

5.4.4 Tieline correction and compensation algorithms

A rigid frame induces larger magnetic perturbations from the aircraft but also allows correction through compensation algorithms that use scalar magnetometers and IMUs [40]. Common approaches include the Tolles-Lawson algorithm and non-linear inverse methods. For this study, given the lack of telemetry data, a simpler solution was applied. The initial tieline correction for only the lowpassed 2 Hz filtered signal proves to be quite useful as it compensates 0-5 nT from the signal and also seemingly smoothens out the survey values. However, it does not removed artefacts which have to be manually removed.

Most of the noise and artefacts were removed applying a compensation-like methodology. Figure A.32, namely the removal of a linear fit to each track. In essence this represents a pseudo-bandpass filter as it removes all frequencies above 2 Hz and the 0 Hz component from the signal. This narrows the signal down to the anomaly range. It proved to be mostly succesful as it removed a majority of the linear features but not all.

6

Conclusion

This thesis aimed to optimize high-resolution drone-based magnetometry by identifying, characterizing, and mitigating UAV-induced noise. While drone-based magnetometry offers flexibility, survey speed, and extensive area coverage, it is limited by structured interference originating from the drone. This study demonstrates that such interference can be decomposed, modelled, and filtered systematically, thereby improving anomaly detection and survey reliability.

The experimental survey campaign, using the Sensys R4 setup on a DJI Matrice 350 RTK, revealed that UAV-induced interference is not random but structural. It contains identifiable frequency bands that can be linked to propeller harmonics and electronic components. The bands are composed of transient fluctuations, persistent, and vibrational frequencies.

The first research question addressed the identification of UAV-induced interference. Persistent harmonic frequencies such as the 16.7 Hz harmonics family and 50 Hz were discovered. Additionally, transient frequencies within the 30-60 Hz frequency range. Vibrational noise was identified by analyzing the 18 Hz band. Lastly, offset discrepancies were observed, which were caused by heading changes and flight dynamics.

These features were linked to specific drone components, such as propeller motors and onboard electronics. The consistency of these interferences across different surveys confirms that these noises are inherent to the system. Frequency-domain techniques, such as Short-Time Fourier Transform and synchrosqueezing, provided a detailed characterization of these signals.

The second research question related to noise modelling. ARMA modelling enabled the prediction of recurring harmonic noise and demonstrated the potential to develop an integral system model capable of predicting noise in relation to internal telemetric data.

The third research question concerned anomaly identification. Sensor sensitivity after filtering proved consistent across all sensors. Wavelet matching showed potential in isolating anomalies from background variations or noise. Tieline correction greatly improved survey consistency, thus reducing directional discrepancies and interpolation artefacts. Comparisons with existing literature confirmed that the results are consistent with known characteristics of UAV-based magnetometry.

The scope of this thesis was limited by time and infrastructure constraints. The survey campaign was conducted over only a few hours, thus limiting the volume of obtained data. Parameters such as anomaly shape, depth, altitude, and velocities could not be exhaustively tested. Additionally, no ground-based reference survey was conducted, which would allow for qualitative comparison of the drone-based results.

Environmental limitations must also be considered. All experiments were performed under controlled conditions, limiting the insight into performance under complex background signals. In real survey environments, additional factors such as soil variability, anthropogenic noise, and uneven terrain could substantially affect sensor performance. The robustness of the proposed methodology needs to be assessed in more complex environments. Future work involves incorporating these aspects and further extending the methodology, such as improving sensor geolocalisation for signal integrity.

This thesis demonstrates that UAV-based magnetometry, despite its liability to interference, can be optimized. UAV-induced noise displayed structure and was therefore predictable. The resulting improvements in signal-to-noise ratio enabled more reliable detection of ferromagnetic anomalies at low altitudes, bringing UAV surveys closer to the performance of ground-based campaigns. The results confirm that UAV magnetometry holds significant potential for magnetometry surveys. Further hardware and processing developments will lead it to become a standard tool in archaeology, environmental monitoring, and geophysical prospecting.

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A

Appendix

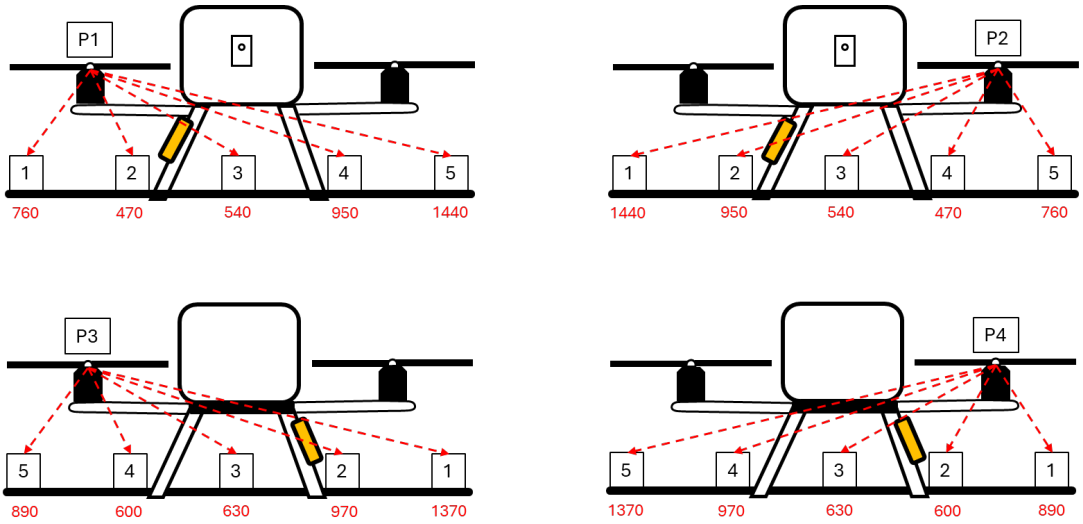

Figure A.1: Distances (mm) from each propeller to each sensor. Propeller P1 and P2 are placed in the front, P3 and P4 are placed in the back. The orange box represents the Skyhub.

Table A.1: Average offset between each sensor and reference sensors for survey 7 (30 m flight).

Reference Sensor	**Sensor 1**	**Sensor 2**	**Sensor 3**	**Sensor 4**
Sensor 2	22.20			
Sensor 3	81.69	59.03		
Sensor 4	-56.15	-78.46	-137.89	
Sensor 5	-7.51	-29.91	-89.22	48.54

Table A.2: Background noise standard deviation, after excluding the anomaly, for all 5 sensors across Survey 1, 2, and 3 before and after 2 Hz high-pass filtering.

Sensor ID	**Cte 1.17 m**		**Cte 0.87 m**		**Var 0.87 m**	
	Unfiltered	Filtered	Unfiltered	Filtered	Unfiltered	Filtered
1	1.67	1.07	1.60	0.97	1.46	1.17
2	4.81	0.89	4.83	1.12	4.18	0.92
3	6.73	1.13	6.71	1.26	6.66	1.26
4	9.66	0.91	9.43	0.85	8.31	1.23
5	2.01	0.73	1.97	0.75	2.98	1.04

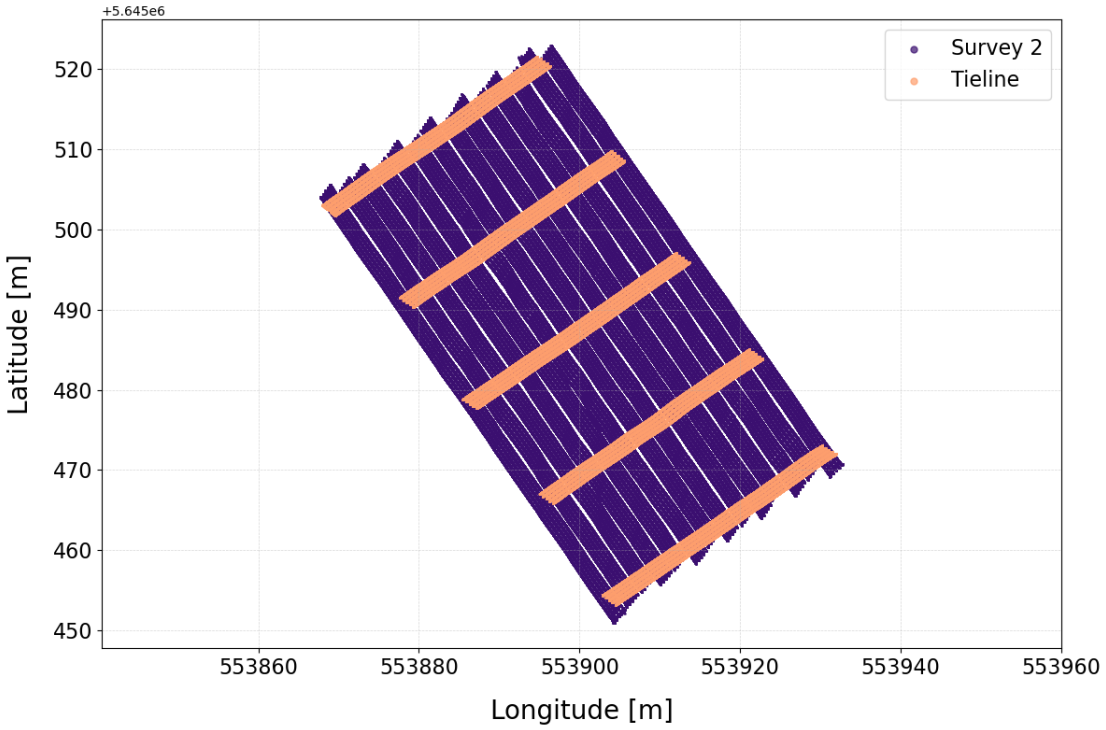

Figure A.2: Survey 2 (constant heading at 0.87 m AGL) flight lines shown in purple with the tielines from survey 4 (pink) plotted over them.

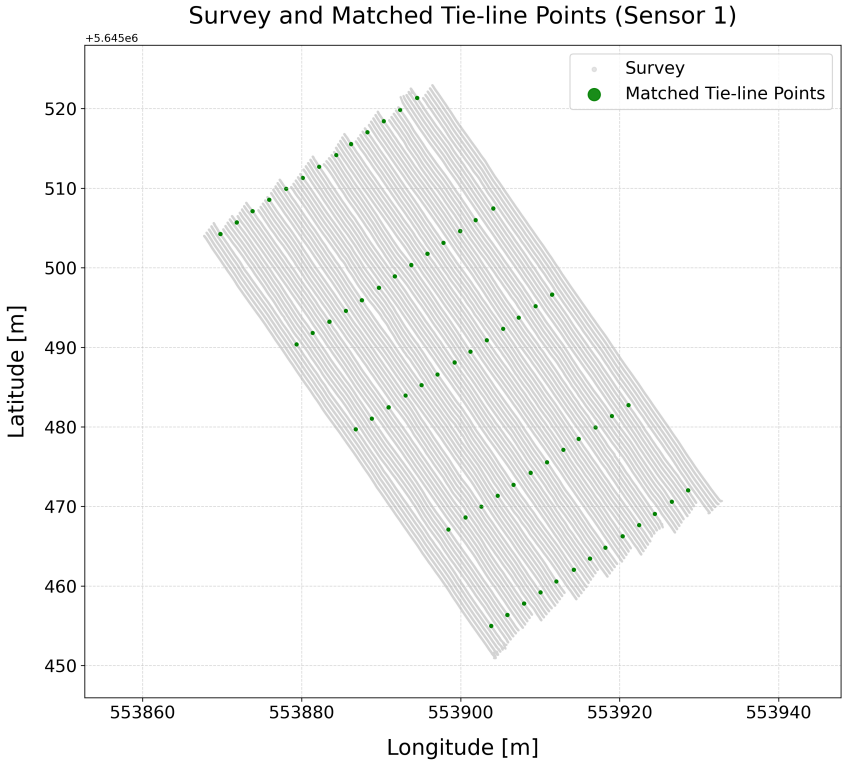

Figure A.3: Overlapping datapoints (green) between tieline (SW-NE direction) and Survey 2 (SE-NW direction with constant heading 0.87 m AGL) for sensor 1.

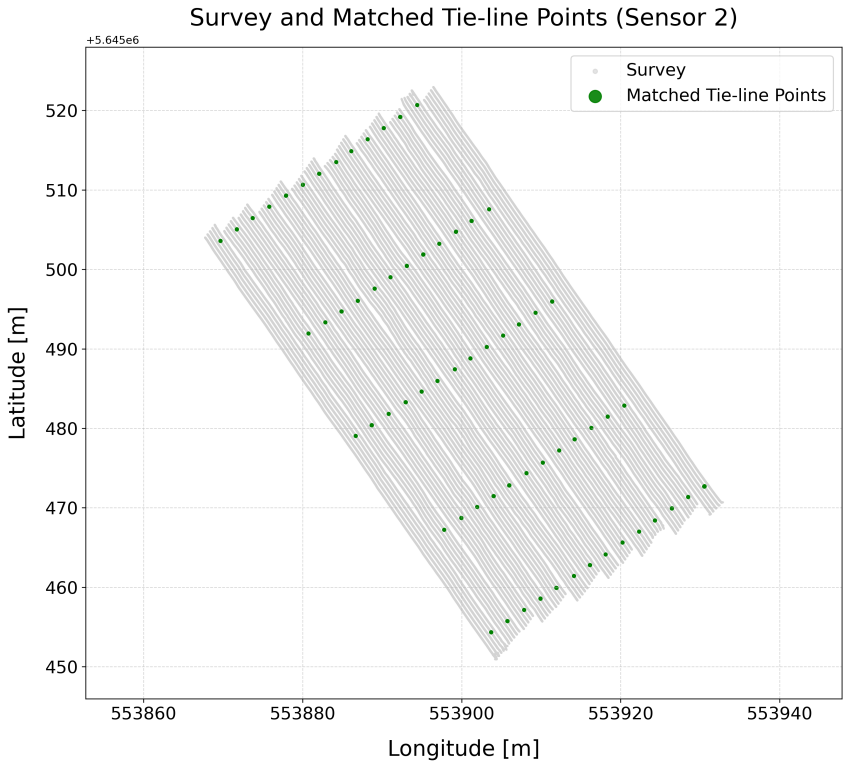

Figure A.4: Overlapping datapoints (green) between tieline (SW-NE direction) and Survey 2 (SE-NW direction with constant heading 0.87 m AGL) for sensor 2.

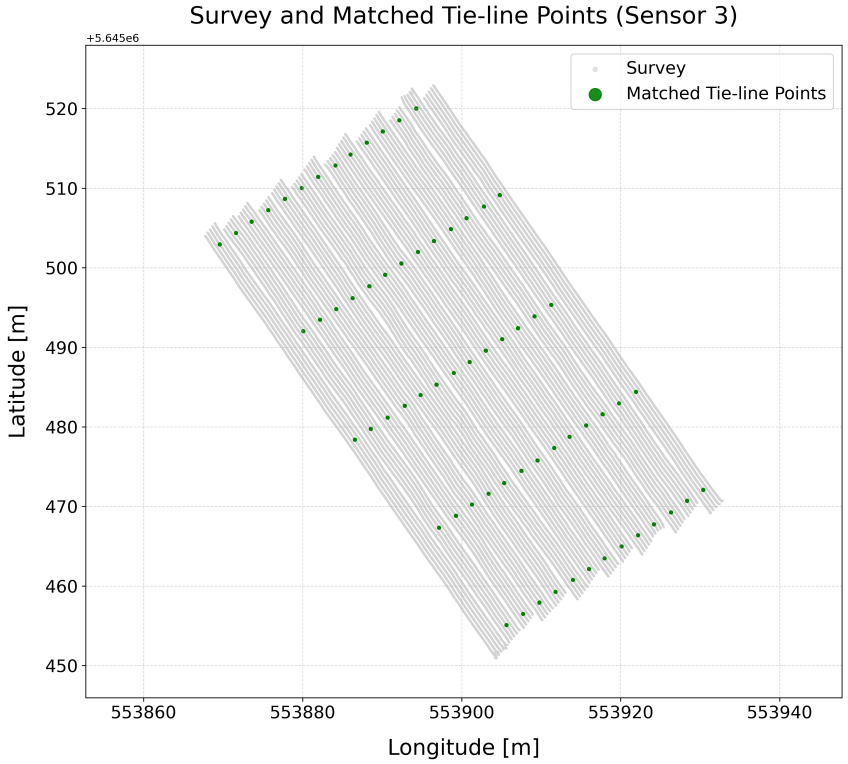

Figure A.5: Overlapping datapoints (green) between tieline (SW-NE direction) and Survey 2 (SE-NW direction with constant heading 0.87 m AGL) for sensor 3.

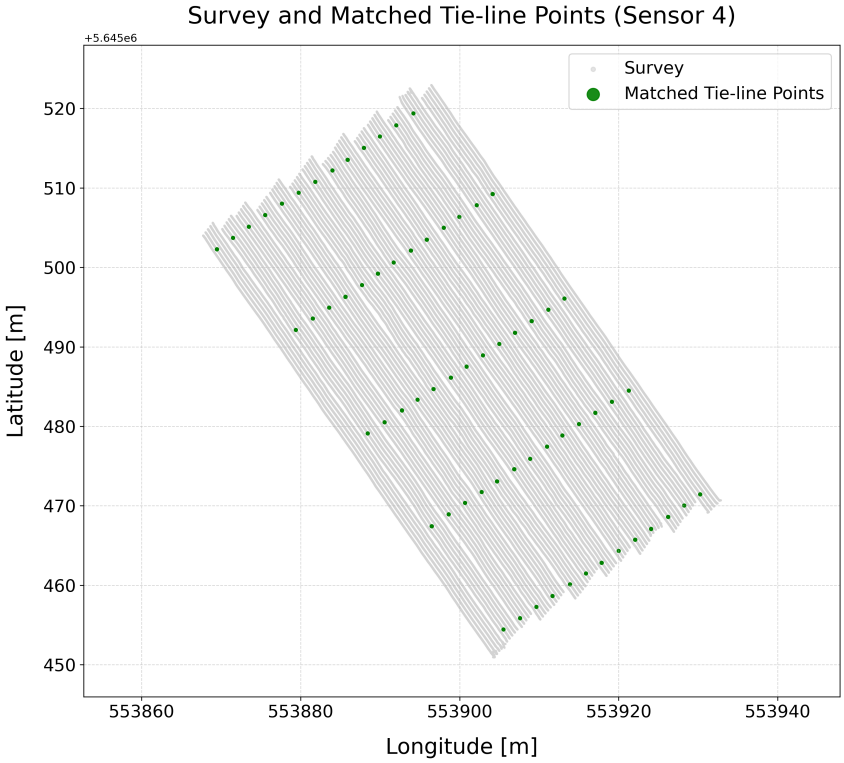

Figure A.6: Overlapping datapoints (green) between tieline (SW-NE direction) and Survey 2 (SE-NW direction with constant heading 0.87 m AGL) for sensor 4.

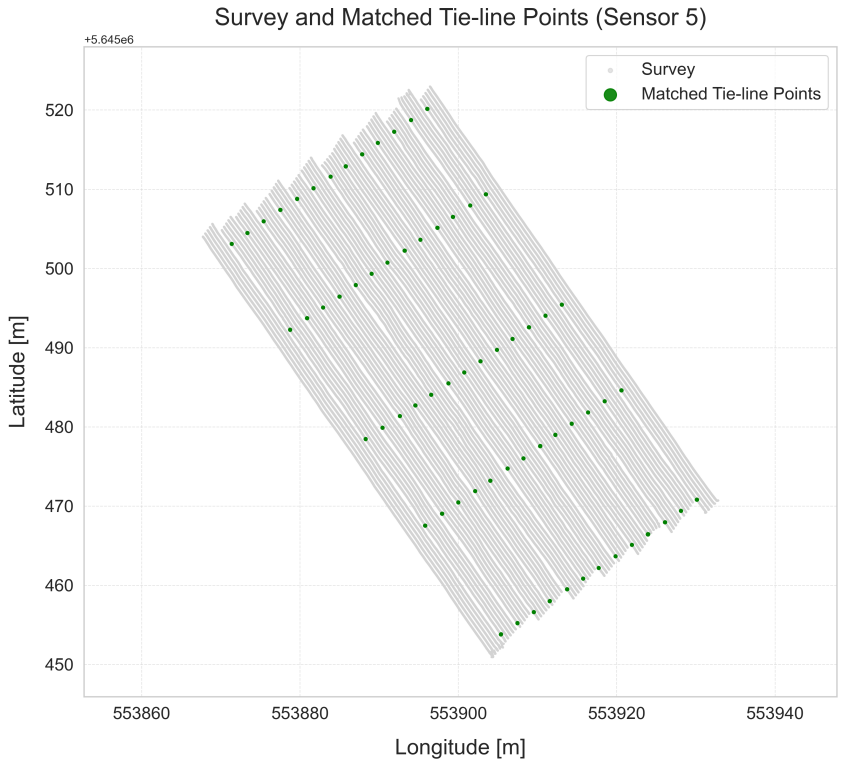

Figure A.7: Overlapping datapoints (green) between tieline (SW-NE direction) and Survey 2 (SE-NW direction with constant heading 0.87 m AGL) for sensor 5.

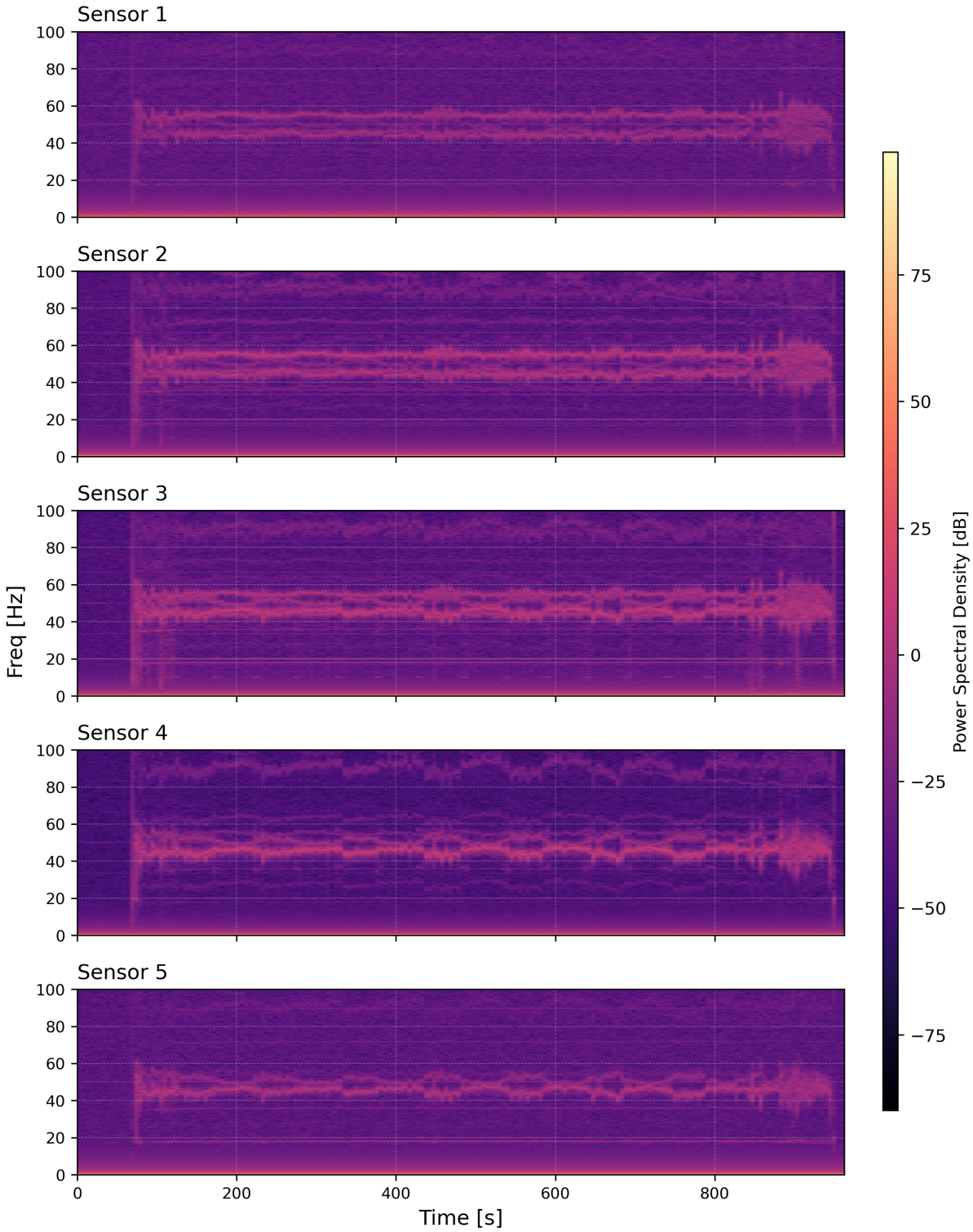

Figure A.8: Spectrogram of magnetic field data recorded during Survey 1 for all five sensors. The data shown is unfiltered, with a Nyquist frequency of 100 Hz and a window size of 1024 samples. The spectral density of the frequencies is shown in decibels (dB).

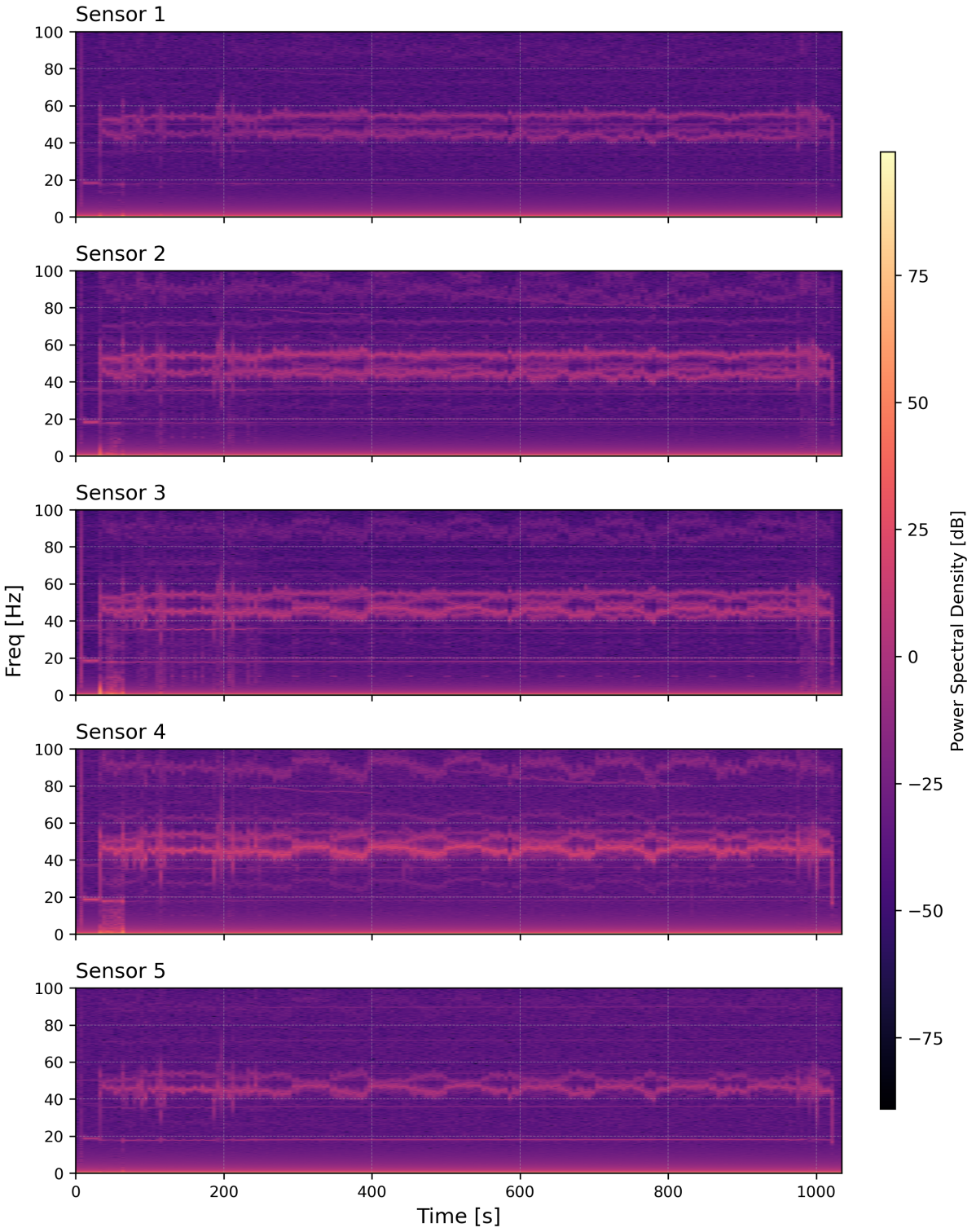

Figure A.9: Spectrogram of magnetic field data recorded during Survey 2 for all five sensors. The data shown is unfiltered, with a Nyquist frequency of 100 Hz and a window size of 1024 samples. The spectral density of the frequencies is shown in decibels (dB).

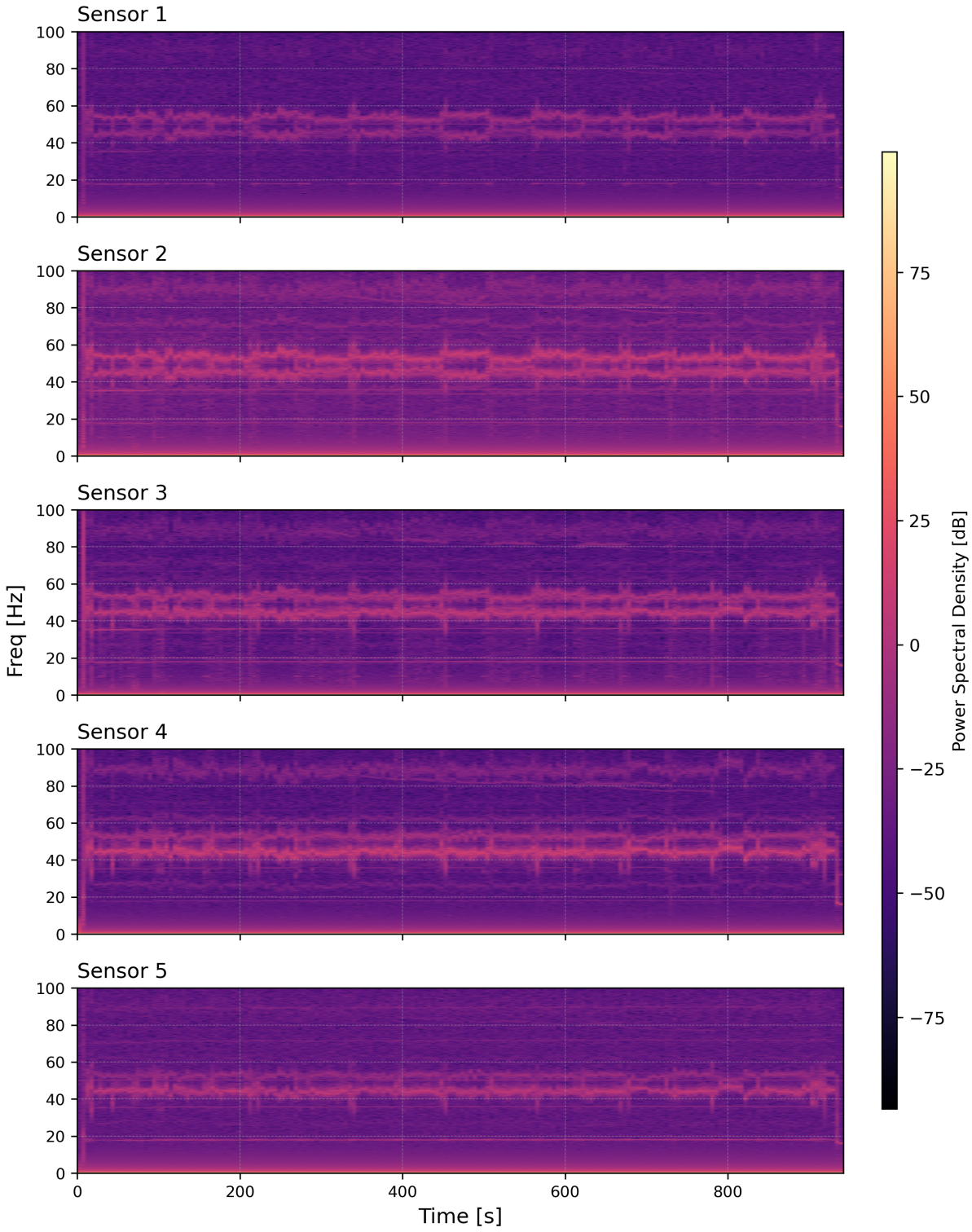

Figure A.10: Spectrogram of magnetic field data recorded during Survey 3 for all five sensors. The data shown is unfiltered, with a Nyquist frequency of 100 Hz and a window size of 1024 samples. The spectral density of the frequencies is shown in decibels (dB).

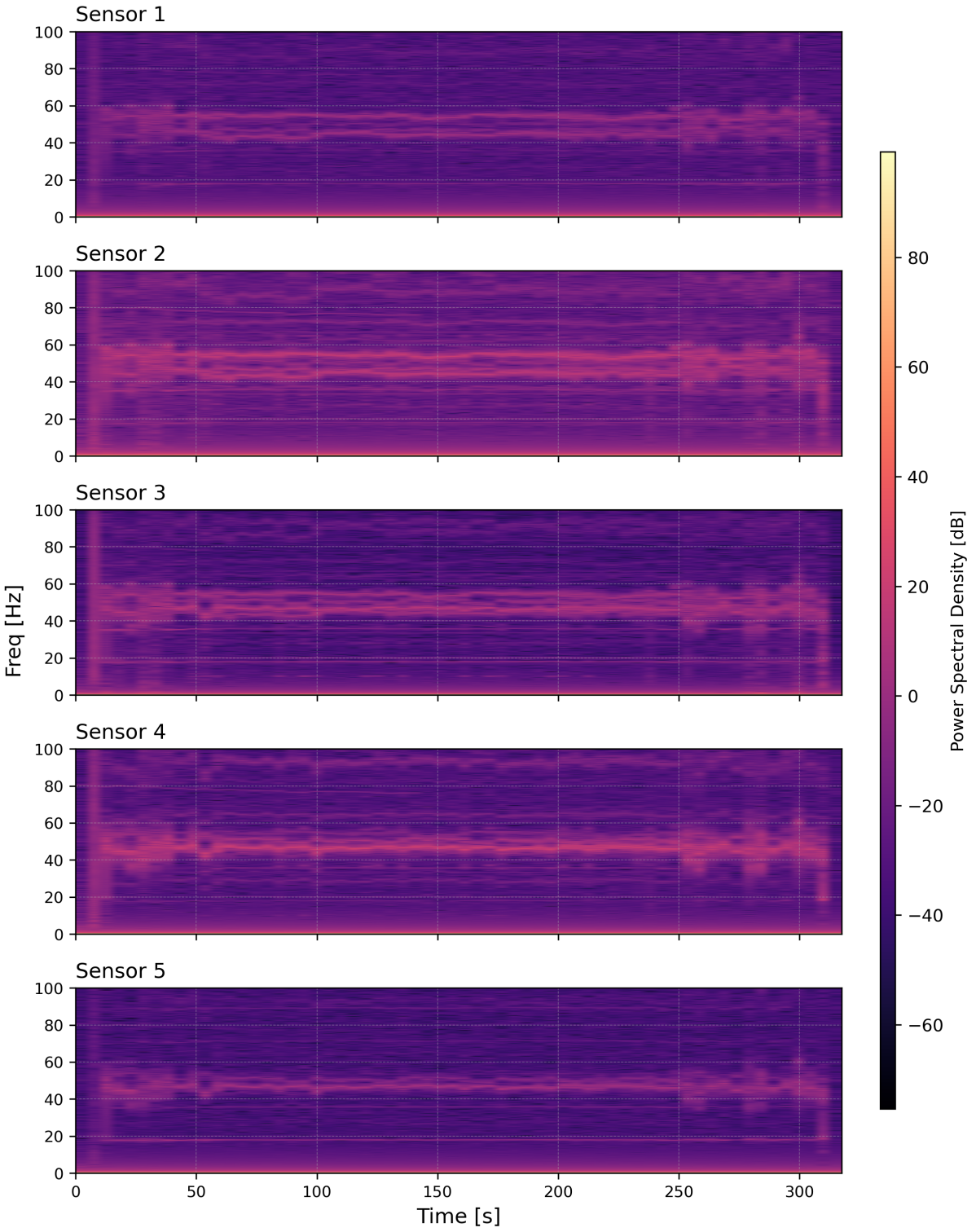

Figure A.11: Spectrogram of magnetic field data recorded during Survey 4 for all five sensors. The data shown is unfiltered, with a Nyquist frequency of 100 Hz and a window size of 1024 samples. The spectral density of the frequencies is shown in decibels (dB).

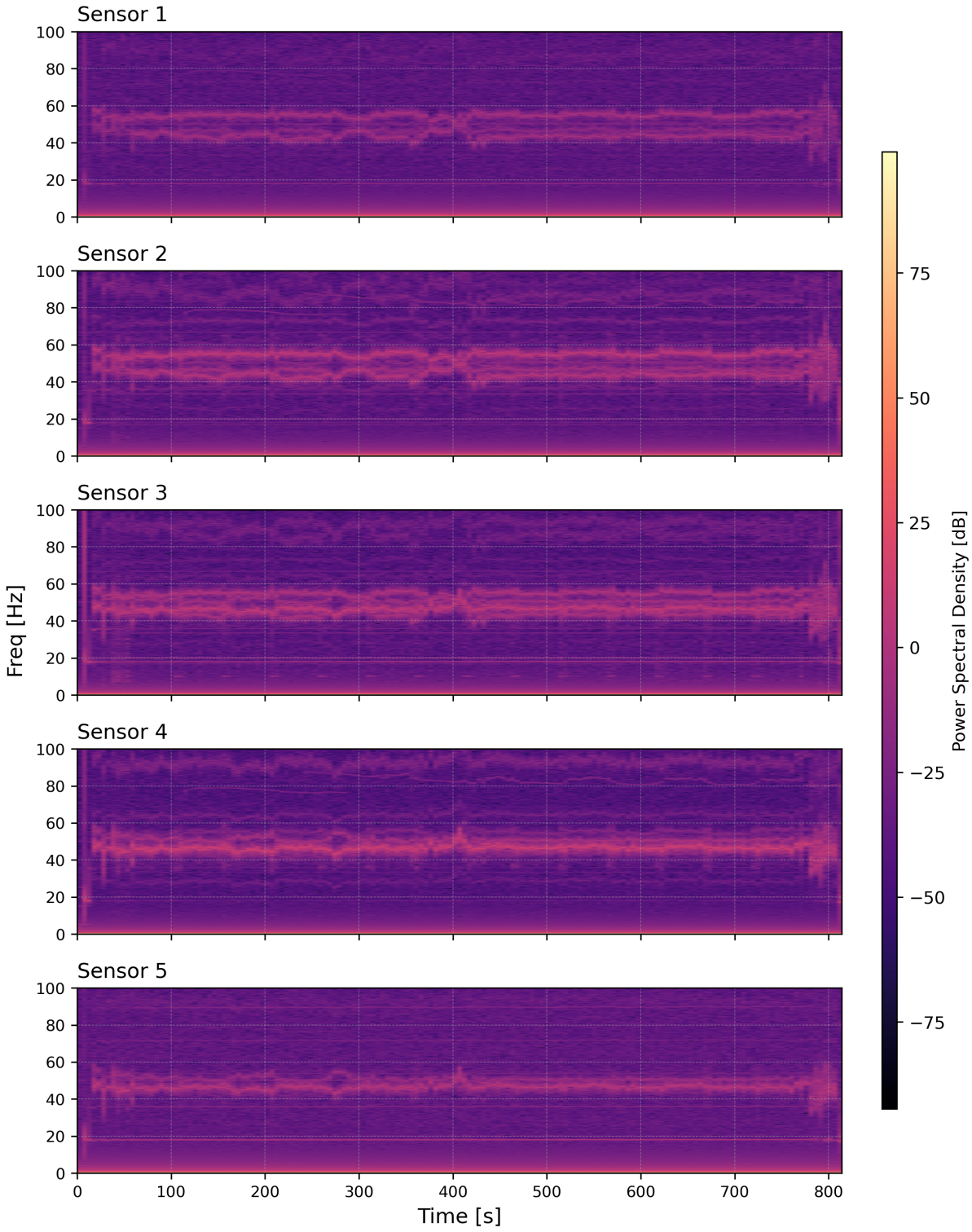

Figure A.12: Spectrogram of magnetic field data recorded during Survey 7 for all five sensors. The data shown is unfiltered, with a Nyquist frequency of 100 Hz and a window size of 1024 samples. The spectral density of the frequencies is shown in decibels (dB).

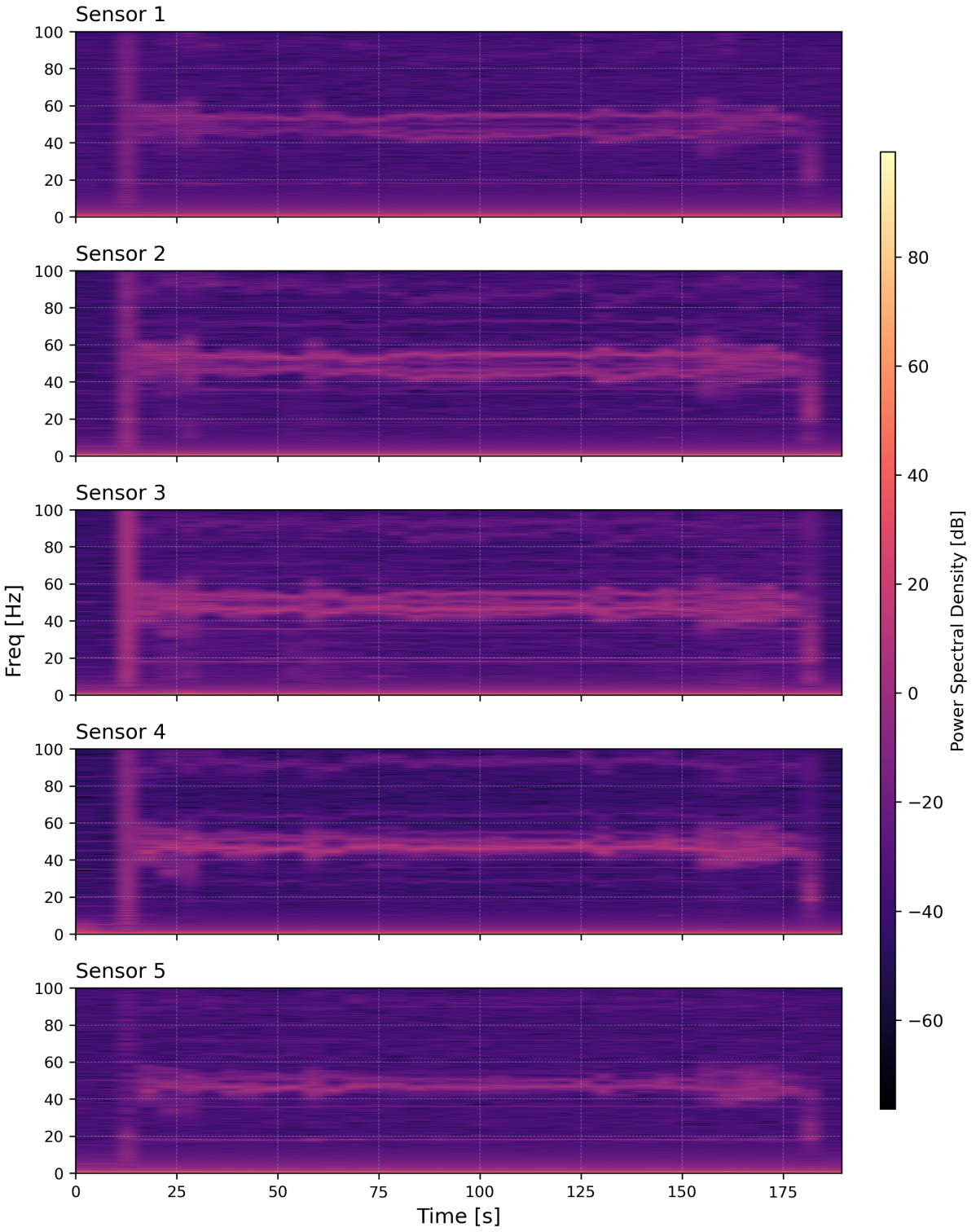

Figure A.13: Spectrogram of magnetic field data recorded during Survey 8 for all five sensors. The data shown is unfiltered, with a Nyquist frequency of 100 Hz and a window size of 1024 samples. The spectral density of the frequencies is shown in decibels (dB).

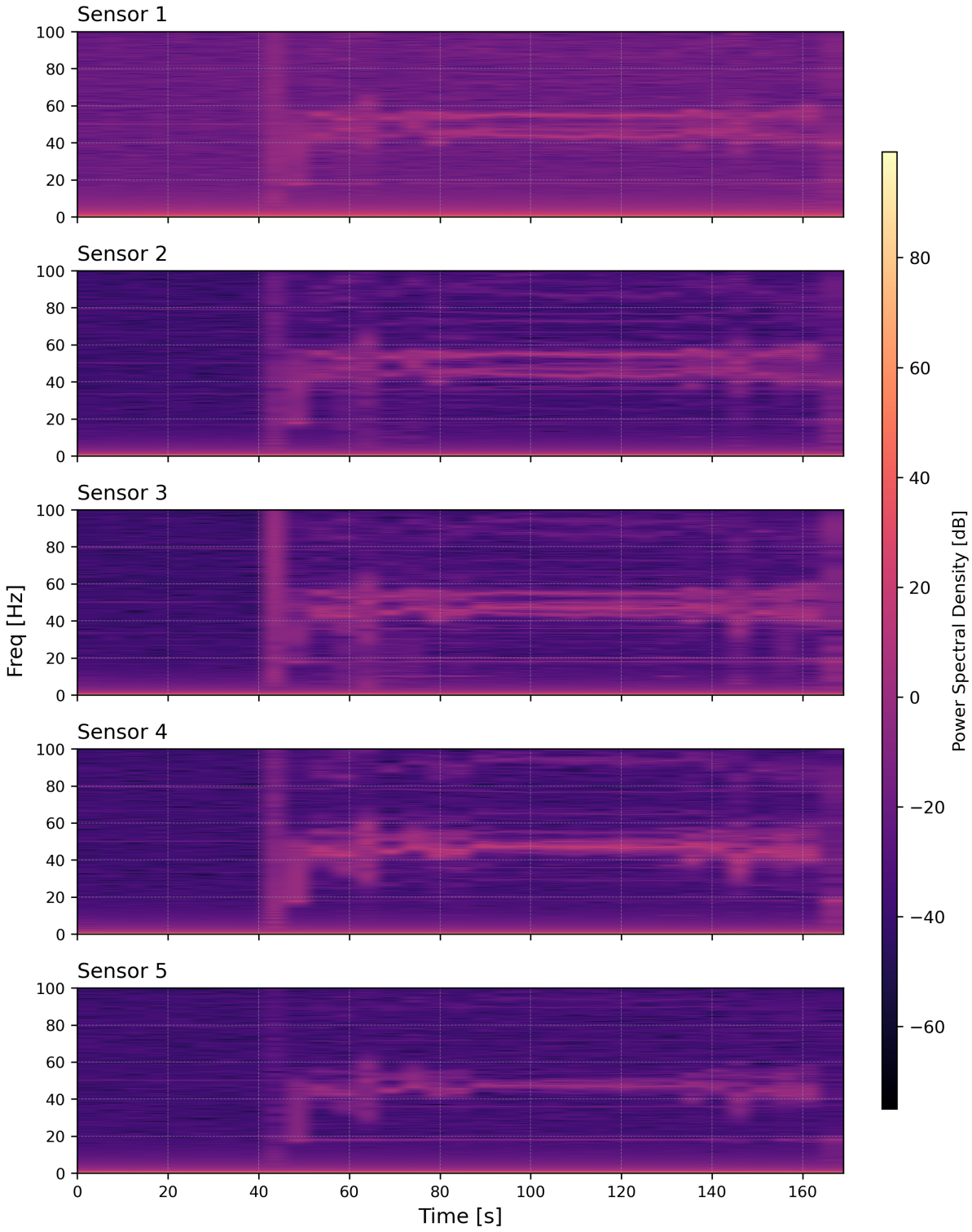

Figure A.14: Spectrogram of magnetic field data recorded during Survey 9 for all five sensors. The data shown is unfiltered, with a Nyquist frequency of 100 Hz and a window size of 1024 samples. The spectral density of the frequencies is shown in decibels (dB).

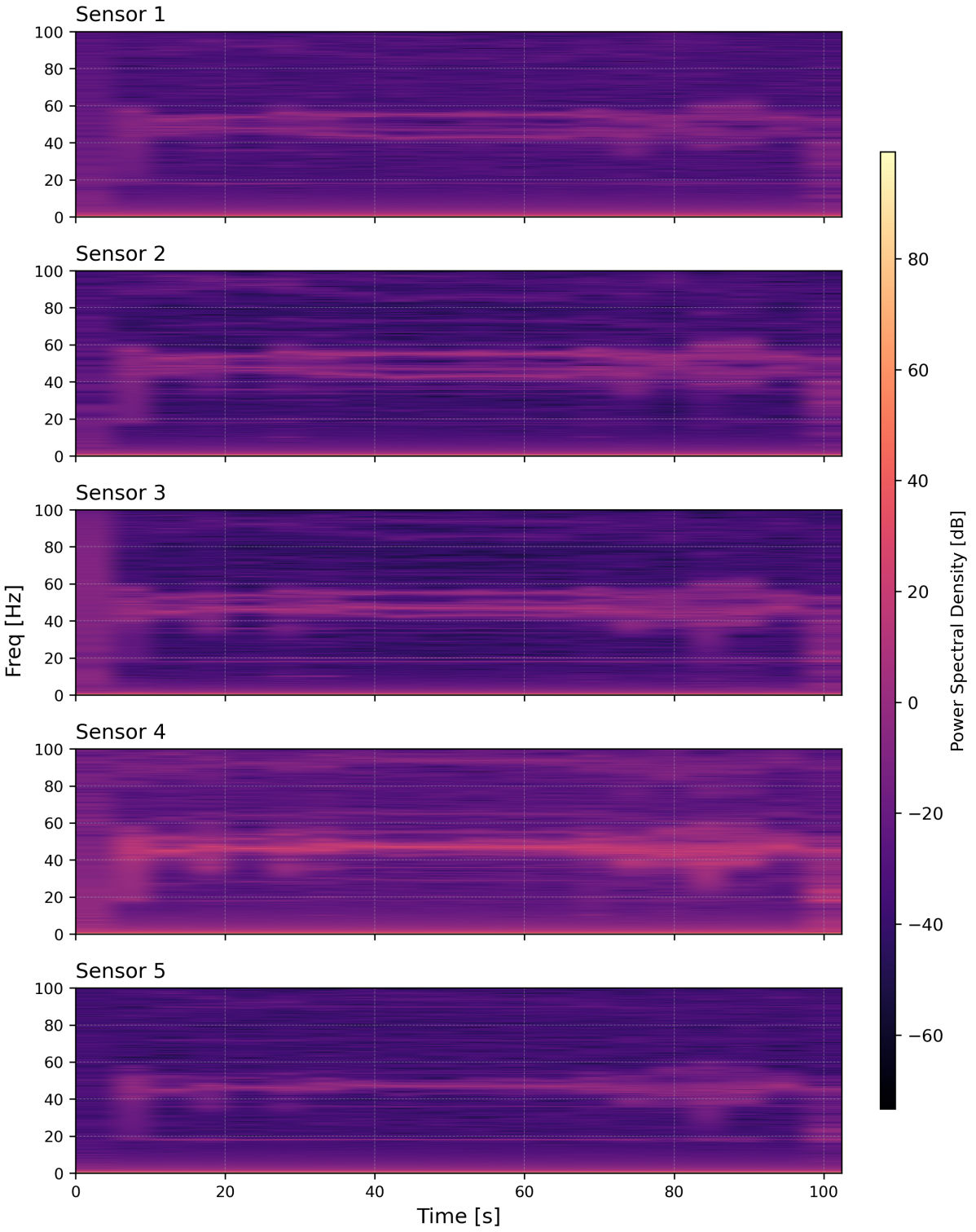

Figure A.15: Spectrogram of magnetic field data recorded during Survey 10 for all five sensors. The data shown is unfiltered, with a Nyquist frequency of 100 Hz and a window size of 1024 samples. The spectral density of the frequencies is shown in decibels (dB).

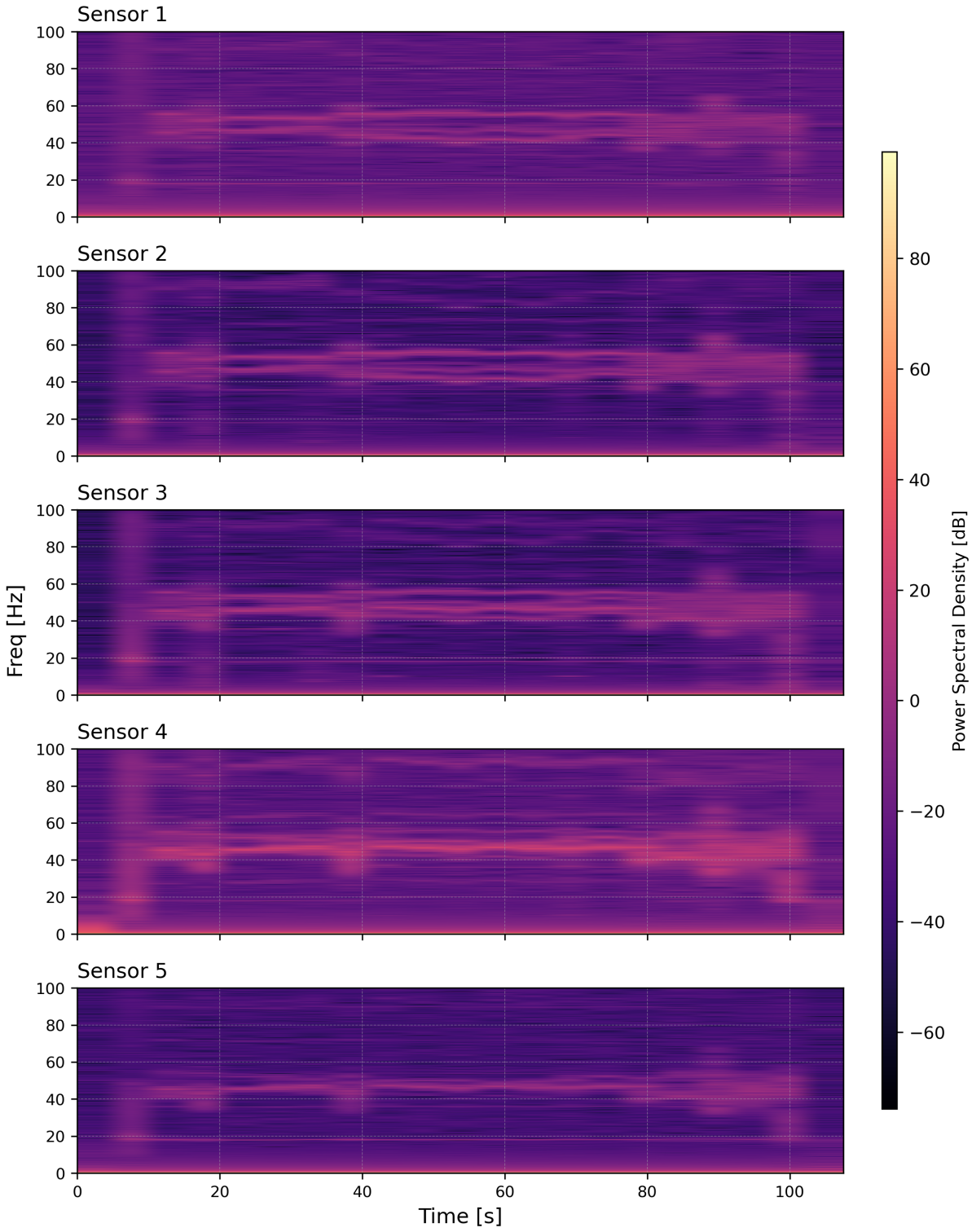

Figure A.16: Spectrogram of magnetic field data recorded during Survey 11 for all five sensors. The data shown is unfiltered, with a Nyquist frequency of 100 Hz and a window size of 1024 samples. The spectral density of the frequencies is shown in decibels (dB).

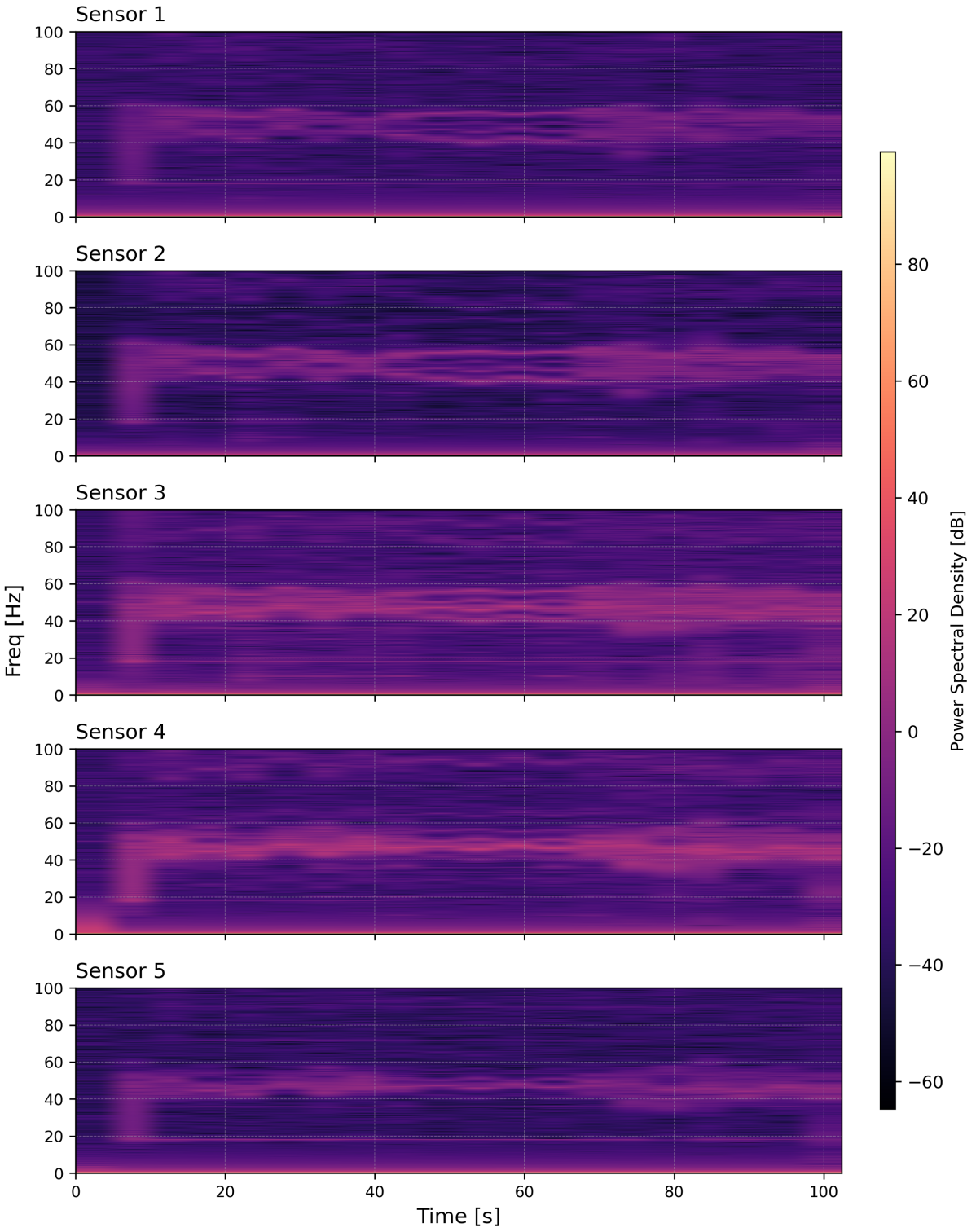

Figure A.17: Spectrogram of magnetic field data recorded during Survey 11 for all five sensors. The data shown is unfiltered, with a Nyquist frequency of 100 Hz and a window size of 1024 samples. The spectral density of the frequencies is shown in decibels (dB).

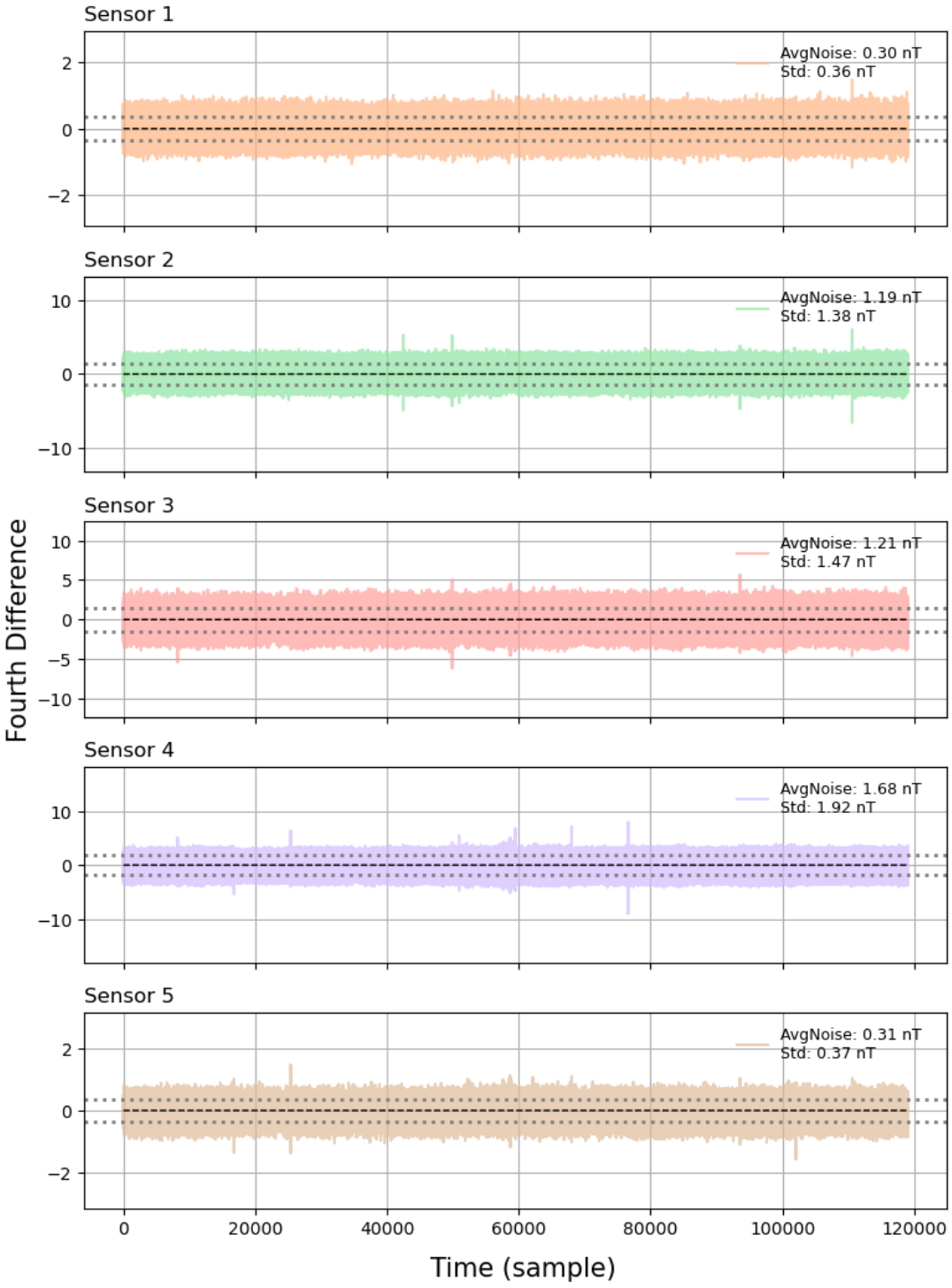

Figure A.18: Fourth difference of the signal for Survey 7 (30 m AGL) for all five sensors. Each sensor colour represents the signal, the dotted grey lines represent one standard deviation from the mean of the fourth difference. The black line represents the trend line line.

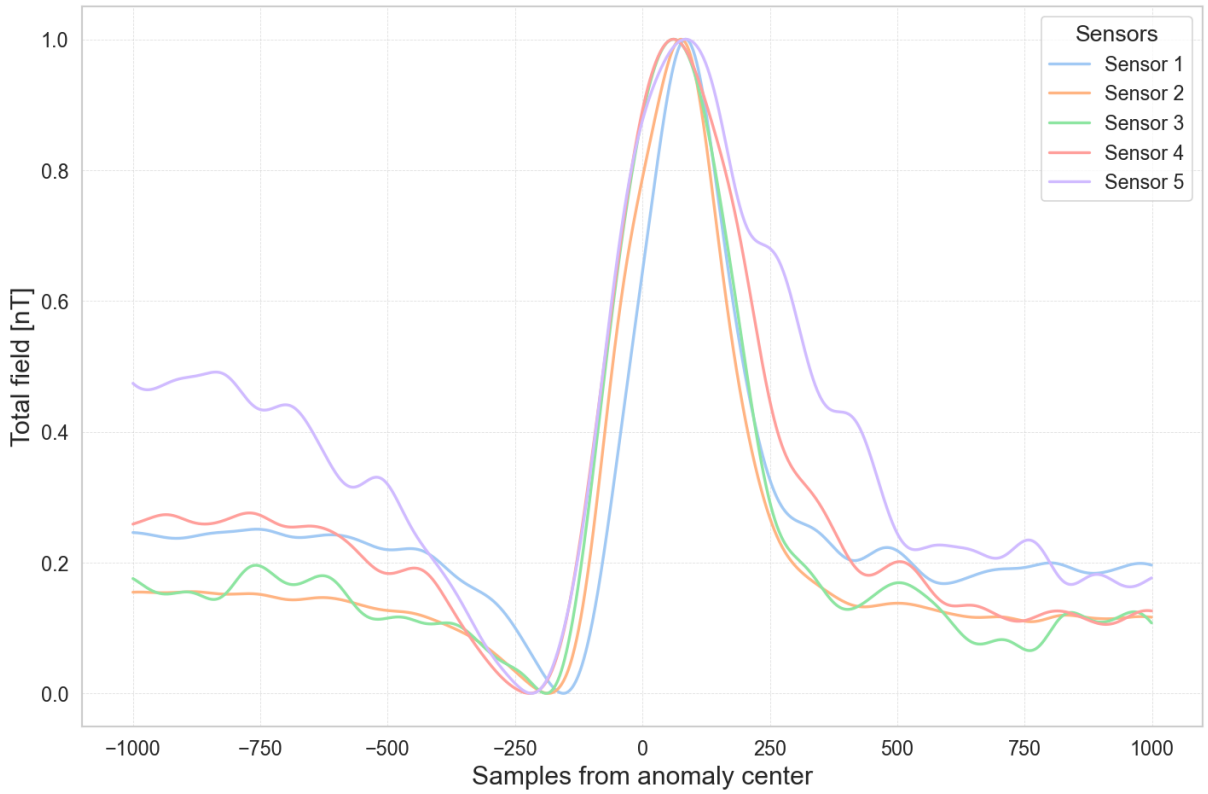

Figure A.19: The normalized total field [-] of the 10 sec anomaly window for Survey 1 (constant heading 1.17 m AGL) for all 5 sensors.

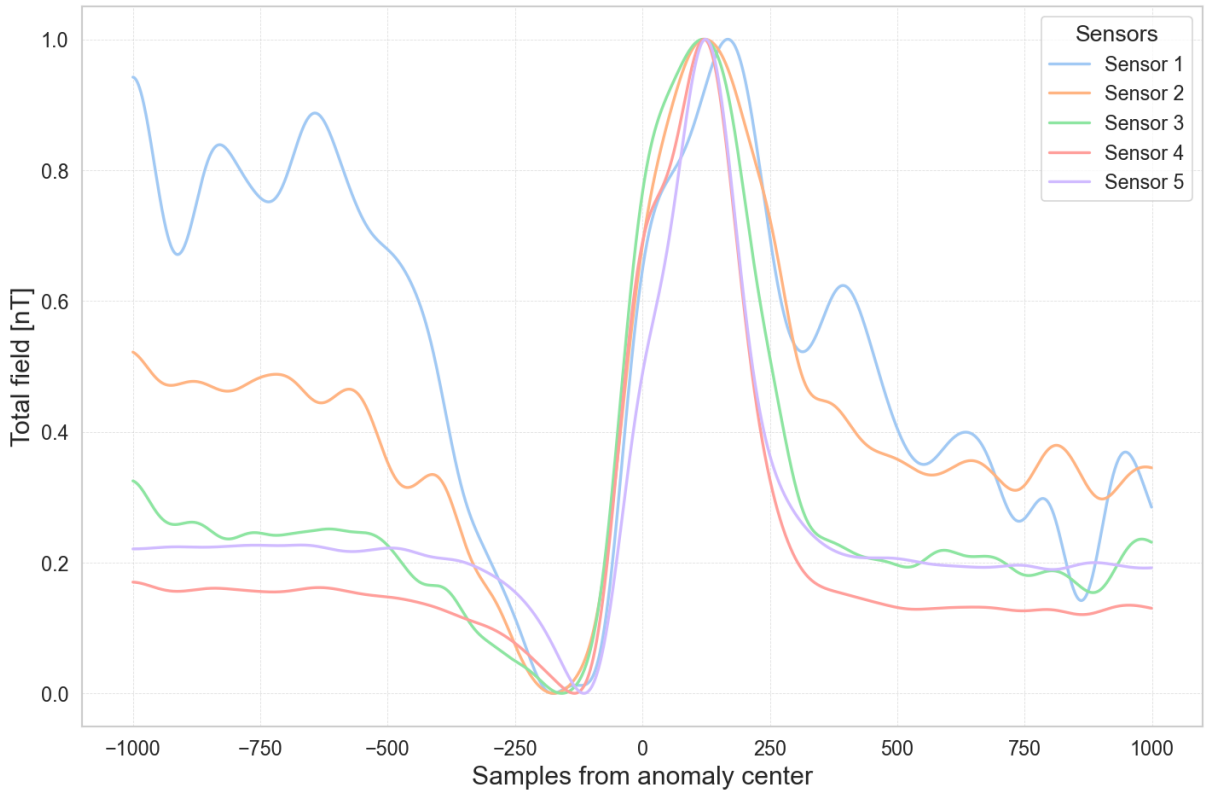

Figure A.20: The normalized total field [-] of the 10 sec anomaly window for Survey 3 (variational heading 0.87 m AGL) for all 5 sensors.

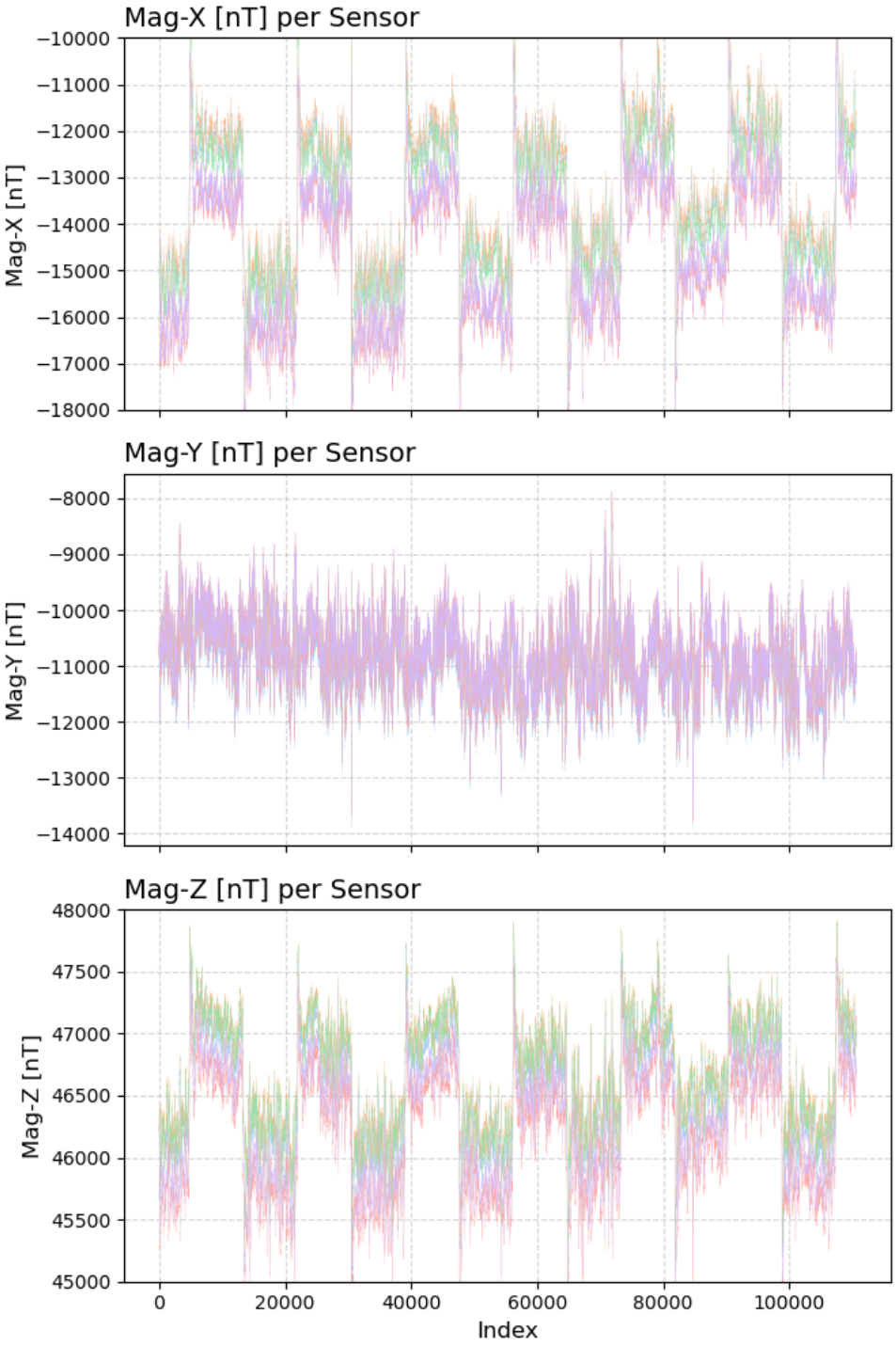

Figure A.21: The X-, Y-, Z- components [nT] of Survey 2 (constant heading 0.87 m AGL). Blue represent sensor 1, orange sensor 2, green sensor 3, red sensor 4 and purple sensor 5.

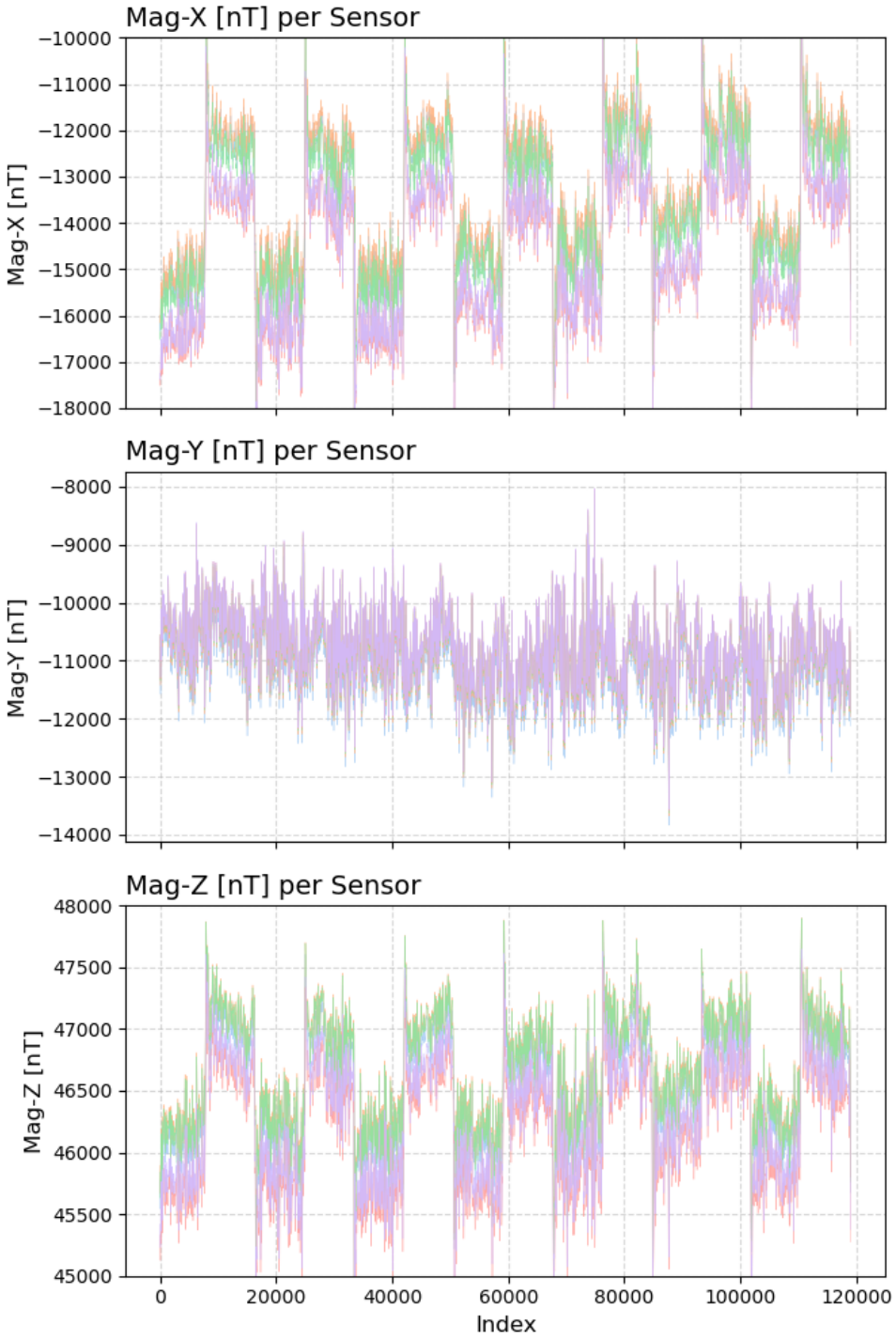

Figure A.22: The X-, Y-, Z- components [nT] after applying a 2 Hz lowpass filter of Survey 2 (constant heading 0.87 m AGL). Blue represent sensor 1, orange sensor 2, green sensor 3, red sensor 4 and purple sensor 5.

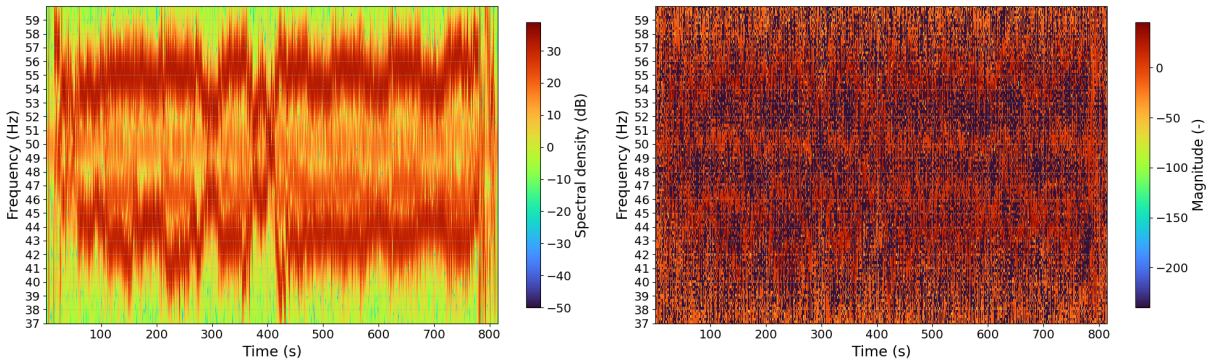

Figure A.23: Left: Short Time Fourier Transform (STFT) of sensor 1, with a window size of 1024, for Survey 7 (30 m AGL) after bandpassing the signal for frequencies between 37 and 60 Hz. Right: Synchrosqueezed STFT, the magnitude of the SSQ-STFT is shown instead of spectral density in decibels (dB) for better visualization.

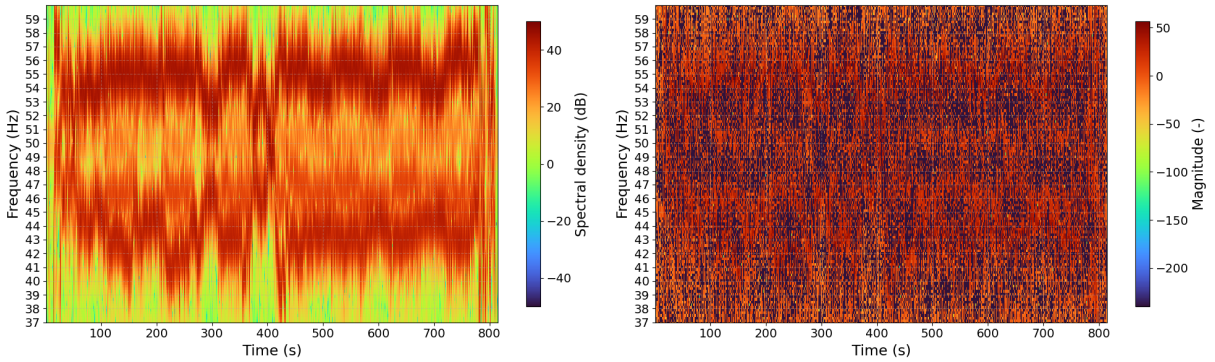

Figure A.24: Left: Short Time Fourier Transform (STFT) of sensor 2, with a window size of 1024, for Survey 7 (30 m AGL) after bandpassing the signal for frequencies between 37 and 60 Hz. Right: Synchrosqueezed STFT.

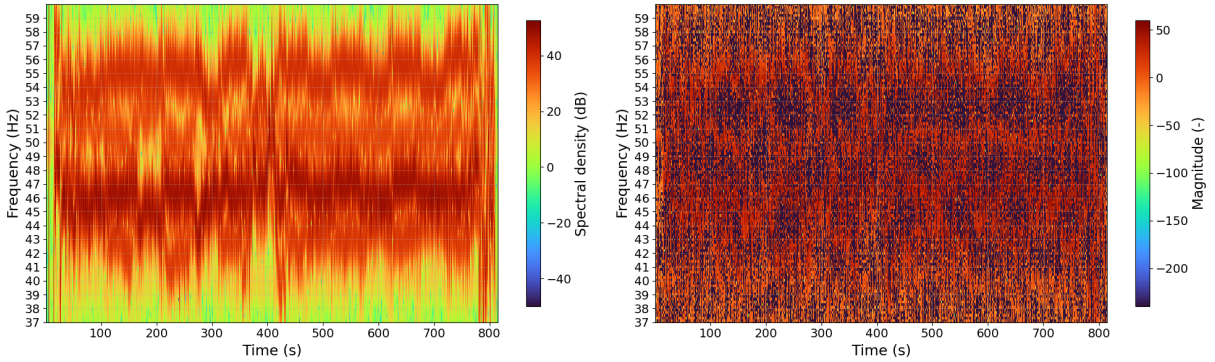

Figure A.25: Left: Short Time Fourier Transform (STFT) of sensor 3, with a window size of 1024, for Survey 7 (30 m AGL) after bandpassing the signal for frequencies between 37 and 60 Hz. Right: Synchrosqueezed STFT.

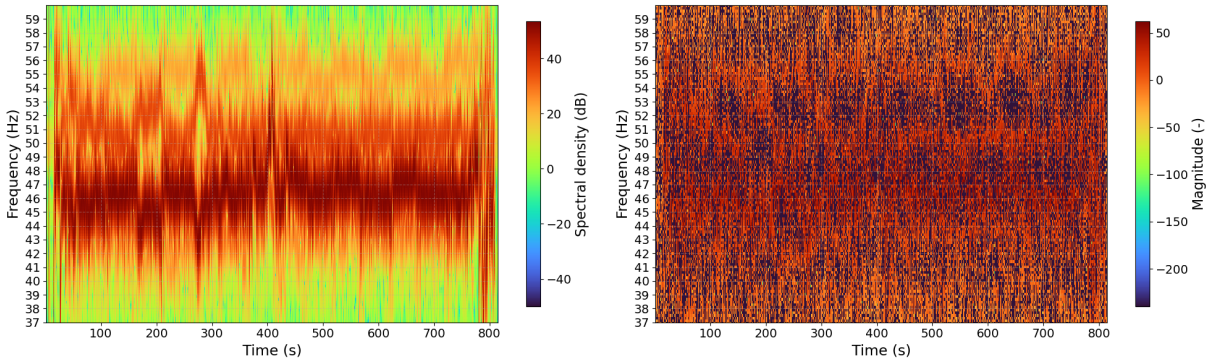

Figure A.26: Left: Short Time Fourier Transform (STFT) of sensor 4, with a window size of 1024, for Survey 7 (30 m AGL) after bandpassing the signal for frequencies between 37 and 60 Hz. Right: Synchrosqueezed STFT.

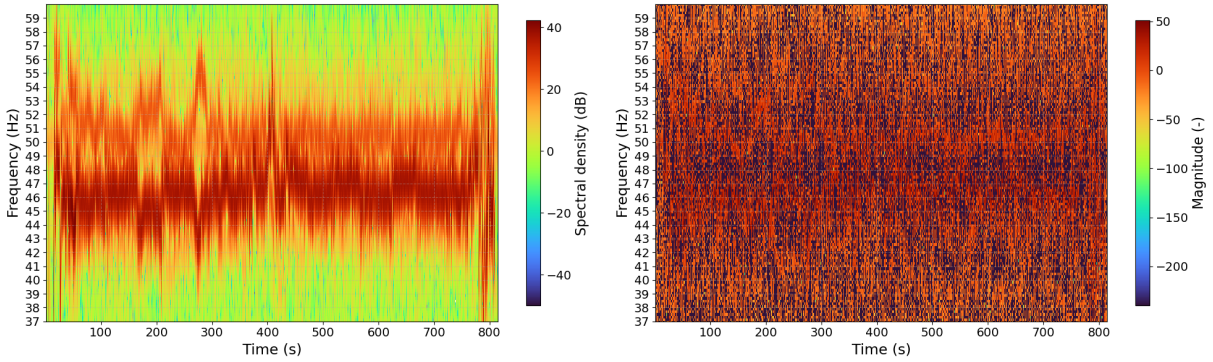

Figure A.27: Left: Short Time Fourier Transform (STFT) of sensor 5, with a window size of 1024, for Survey 7 (30 m AGL) after bandpassing the signal for frequencies between 37 and 60 Hz. Right: Synchrosqueezed STFT.

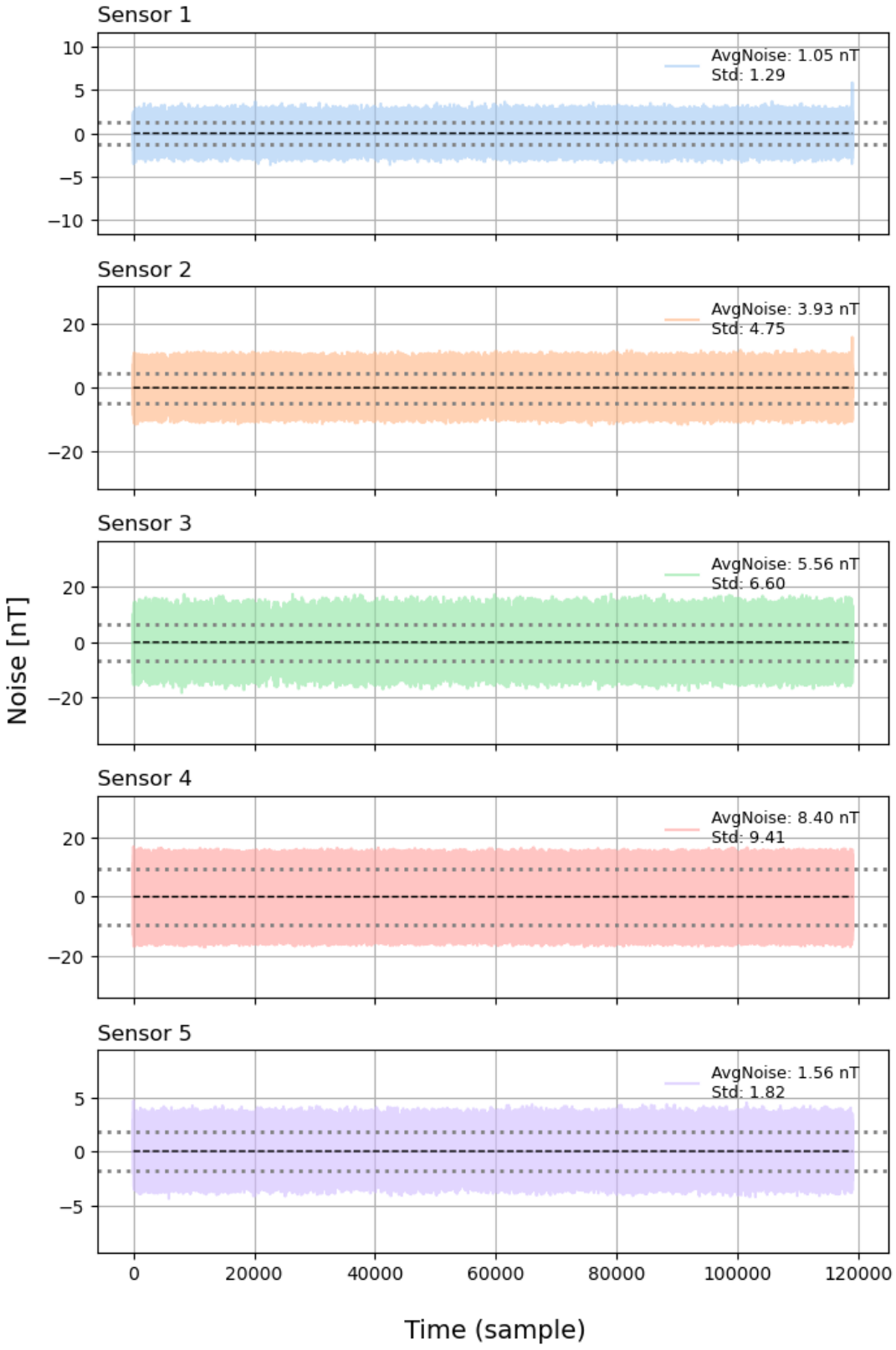

Figure A.28: Noise characterization [nT] of the signal after a 2Hz highpass filter was applied on Survey 7 (30 M).

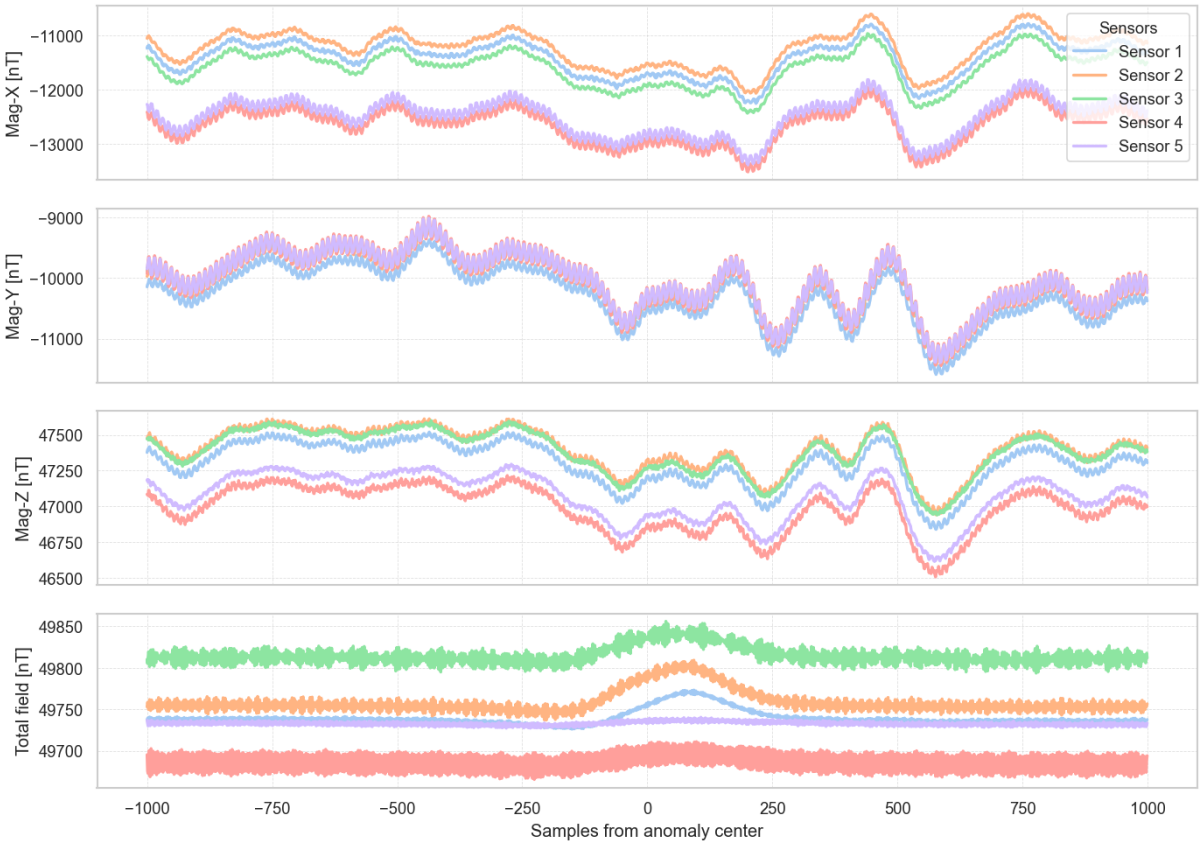

Figure A.29: The X-, Y- and Z- component and total field [nT] of the 10 sec anomaly window for Survey 1 (constant heading 1.17 m AGL) for all 5 sensors.

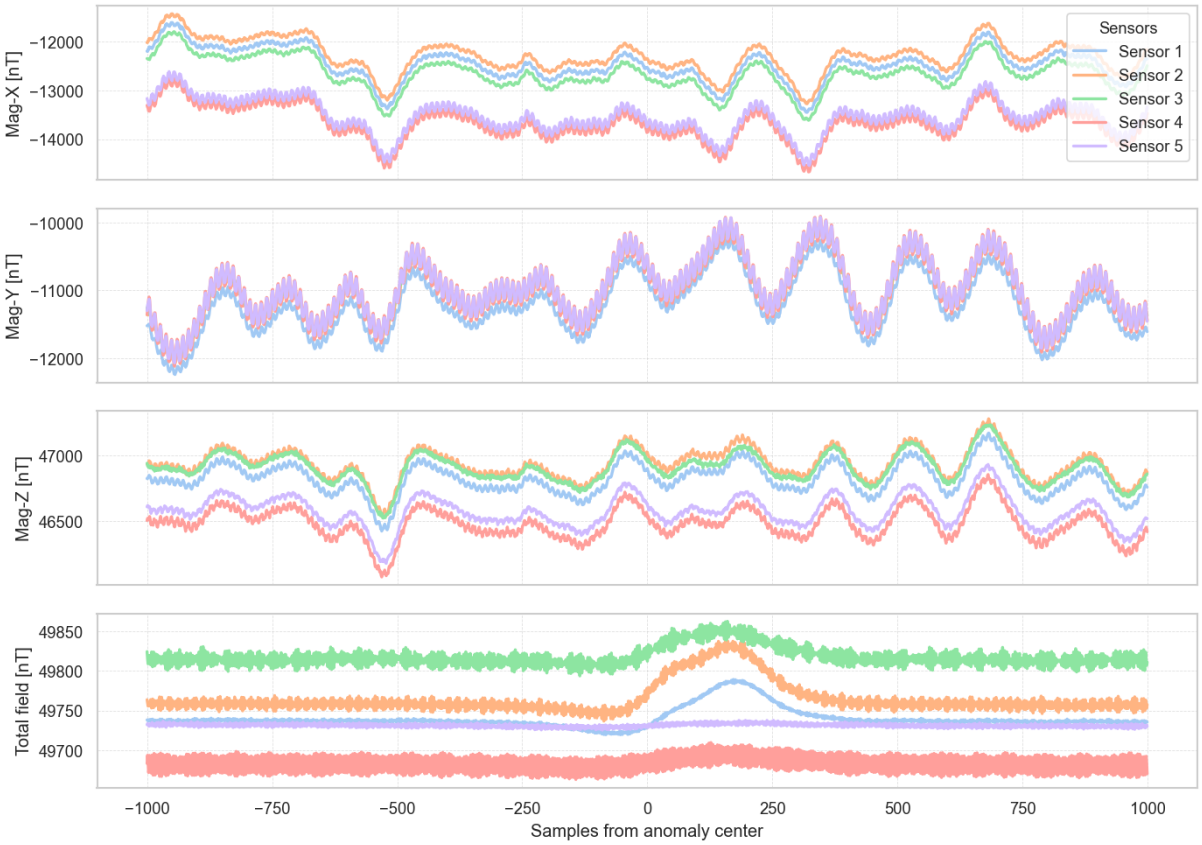

Figure A.30: The X-, Y- and Z- component and total field [nT] of the 10 sec anomaly window for Survey 2 (constant heading 0.87 m AGL) for all 5 sensors.

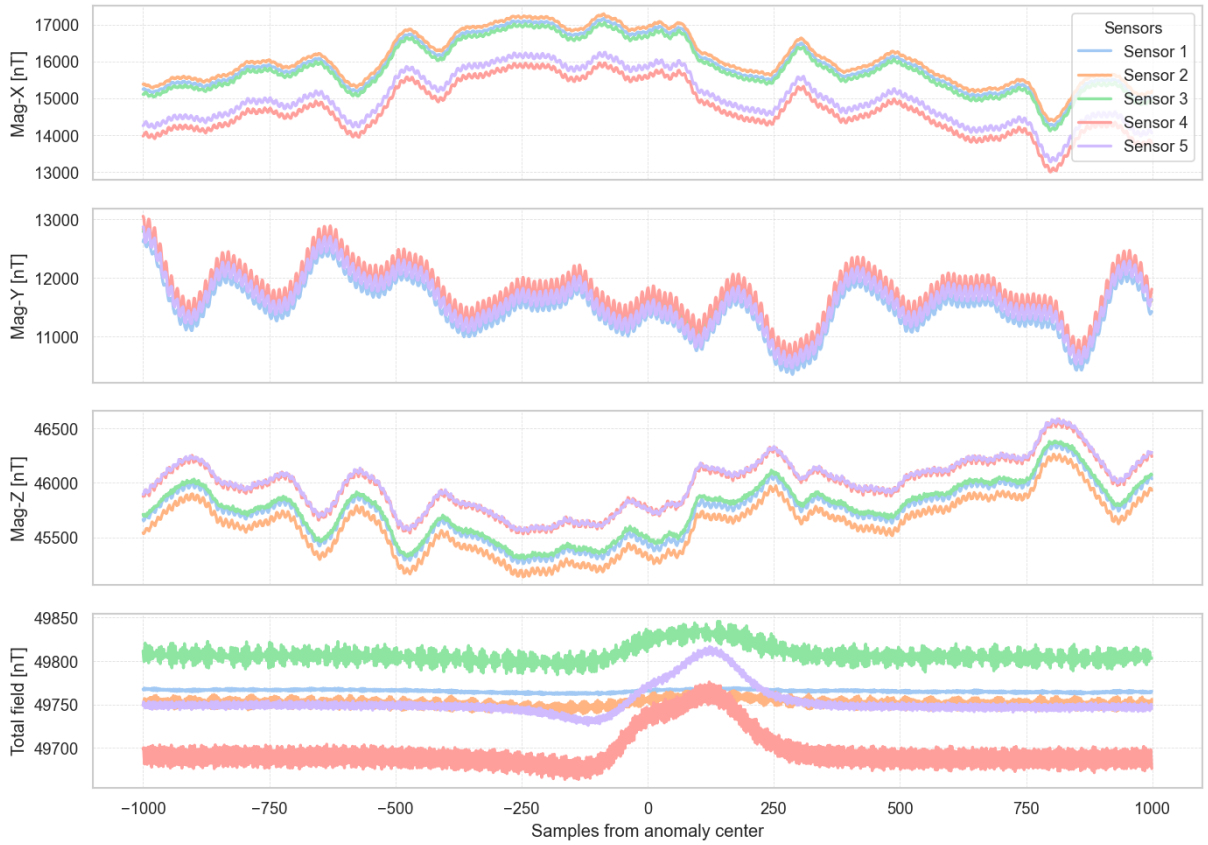

Figure A.31: The X-, Y- and Z- component and total field [nT] of the 10 sec anomaly window for Survey 3 (variational heading 0.87 m AGL) for all 5 sensors.

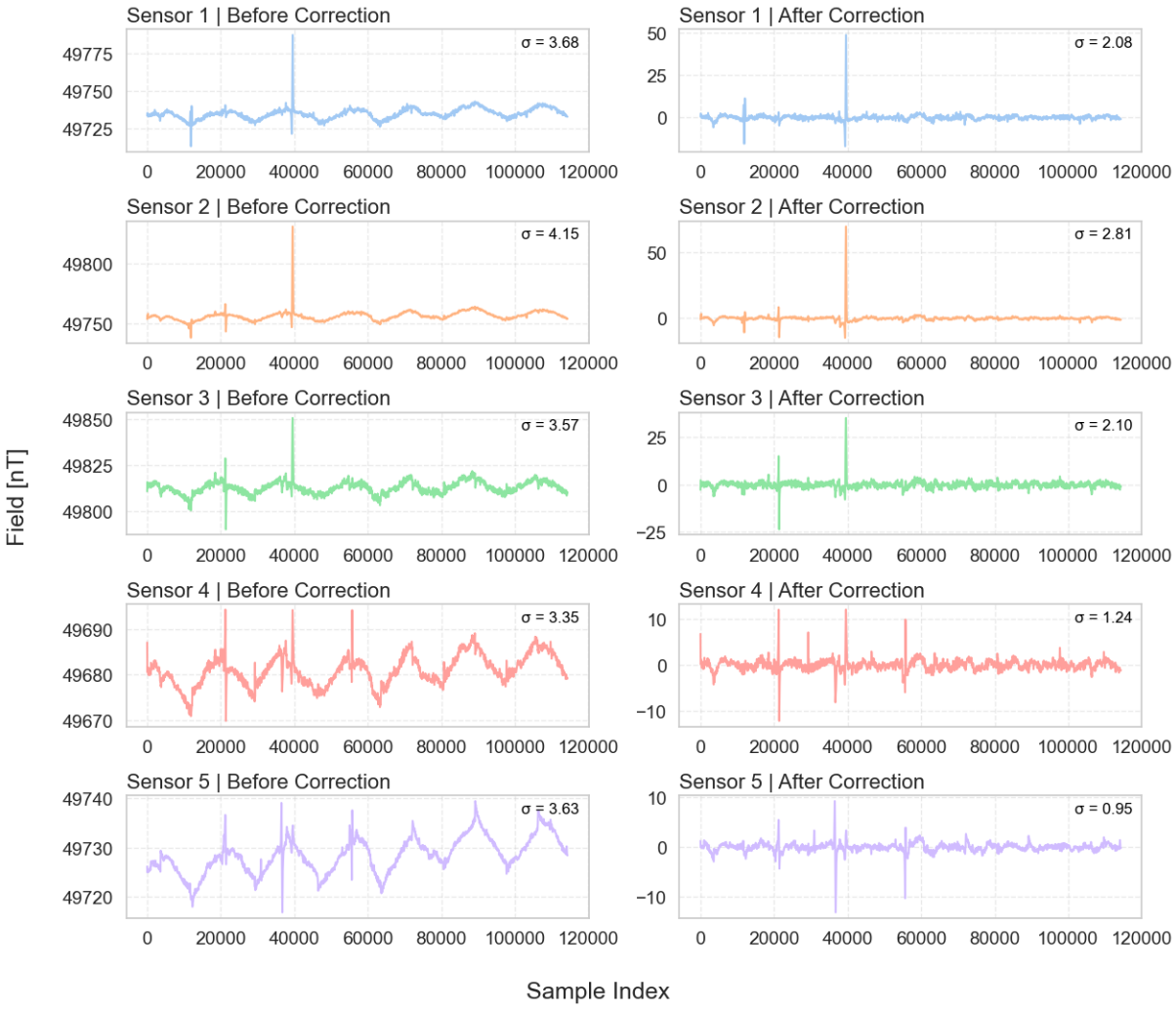

Figure A.32: This figure illustrates the 2 Hz lowpass filtered signal on the left and the compensated signal (2 Hz lowpass filtered, slope- and tieline-corrected) on the right for all sensors. The variance of the signal before and after compensation is shown in the top right. A decrease in variance is observed.